

# Tone Relative to Substance in European ESG Disclosure: A Continuous Corpus-Relative Score, 2018–2024

## 1 Introduction and scope

This paper measures one quantity: how the tone of a report’s ESG disclosure stands relative to the substance of that disclosure, within a defined corpus. The corpus is 343 reports of 49 listed firms in seven European countries over 2018–2024 (§2). For each report we compute a tone score, the Loughran–McDonald polarity of its ESG text (Loughran and McDonald, 2011), and a substance score, the density of mentions of a published ESG vocabulary, and we combine them as the difference of their standardised values,  $\text{ESGSI} = Z(\text{SEN}) - Z(\text{SUS})$  (§5). A report scores high when its tone ranks higher in the corpus than its substance does, and low in the opposite case.

**What the score is, and what it is not.** The score is continuous and corpus-relative. Both components are standardised over the pooled 343 reports, so a value is a position within this corpus and has no meaning outside it. Its sign says only on which side of the corpus mean a report falls. We use no threshold and we classify no report: the score does not separate reports into ESG-washing and genuine disclosure, and nothing in the paper should be read as such a classification. The reason is not caution for its own sake. A cut-off on a standardised difference is a statement about the location of two distributions, not about firm conduct (§5.8.4), and no external benchmark of ESG-washing exists against which a cut-off for these reports could be calibrated (§8). The motivating idea — positive language in excess of the substance behind it — tells us which direction of the score is of interest. It does not turn the score into a detector.

**The two scores.** The paper reports two scores. The first, ESGSI, takes its architecture and its name from Lagasio (2024): a tone component minus a substance component, each rescaled across the sample. We keep that architecture and change how its parts are specified, for reasons given in §5.8. ESGSI is the main score of the paper. The second, the Extended ESGSI ( $\text{ESGSI}^{\text{ext}}$ ), is a new metric that we propose. It adds two components, both computed on the raw extracted text: QUANT, the density of quantitative expressions, which counts as further evidence of substance, and HEDGE, the share of hedging and modal words, which counts on the side of tone. The two enter with weights of 0.5. That weighting is our choice and is not estimated, so we report how far the score moves over a grid of alternative weights (§6).

**Relation to the antecedent.** Lagasio (2024) introduced the ESGSI as a cross-section of 749 standalone English-language ESG reports for 2023. We depart from it at six points (Table 1).

- *Corpus.* The original retains only standalone ESG reports; “companies that integrated their ESG disclosure within their annual financial reports were excluded” (Lagasio, 2024, p. 3). Our corpus is a balanced panel, and a balanced panel requires one disclosure that every firm publishes every year. The annual report, and in France the universal registration document (URD), is such a filing. A standalone sustainability report is issued at the firm’s discretion. Of our 49 firms, one is represented by standalone sustainability reports, in all

seven years; a panel restricted to that document type would have held a single firm. The corpus therefore consists of 235 annual reports, 101 URD and 7 standalone sustainability reports, largely the population the original excludes.

- *Extraction.* The original vectorises whole reports. An annual report or URD embeds its ESG disclosure in a much larger financial and legal document, so we score lexically extracted ESG zones (§3).
- *Substance measure.* The original derives its substance score from TF-IDF values and states neither how per-term values are aggregated nor whether document vectors are normalised (§5.8.1). We measure substance as mentions per 100 tokens. Under L2 vector normalisation, which is the library default, a TF-IDF score cannot distinguish a report from the same report with twenty times the ESG content, and it tracks the breadth of the vocabulary a report uses rather than the amount of disclosure (§5.8.2). Density also agrees more closely with independently counted quantitative content, although on fully disjoint bases the difference is not significant once firms are resampled (§5.8.3).
- *Sentiment dictionary.* The original uses TextBlob polarity, a general-purpose tool. We use the Loughran–McDonald finance dictionary, which was built on disclosure language. The original’s own conclusion names “refining sentiment analysis techniques” (Lagasio, 2024, p. 10) as an extension.
- *Score normalisation.* The original prints a difference of  $z$ -scores but describes, and on the evidence of its own tables computes, a difference of min–max-rescaled scores (§5.8.1). We use  $z$ -scores (§5.8.4).
- *No threshold.* The original labels every report “Potential ESGwashing” if its index exceeds zero and “Likely Genuine” otherwise (Lagasio, 2024, pp. 4–5). We do not use this rule or any other.

Because corpus, vocabulary and sentiment dictionary all differ, our scores and those of the original are different measurements. No numerical comparison between them is attempted.

**Contributions.** The paper makes four contributions. First, it specifies ESGSI so that each of its parts is fully determined and justified: a density substance measure, chosen over TF-IDF with L2 vector normalisation on algebraic grounds and corroborated by a convergent-validity test with firm-clustered intervals;  $z$ -score normalisation; and no threshold (§5). Second, it proposes the Extended ESGSI and reports its sensitivity to its weights (§5.4, §6). Third, it documents and publishes the ESG vocabulary, its sources, admission criteria and composition, and the hedge lexicon (§4, §A), and it shows how far the score moves when that vocabulary is perturbed (§6). Fourth, it describes a seven-year balanced panel of European listed firms by year, industry, country and ESG pillar, with inference that treats the firm, not the report, as the independent unit (§7). The original itself calls for “longitudinal studies to examine how changes in corporate characteristics and regulatory environments affect ESGSI over time” (Lagasio, 2024, p. 10).

**Roadmap.** §2 describes the corpus, §3 the extraction of ESG zones, and §4 the vocabulary and the hedge lexicon. §5 defines every component, both scores and the panel inference, and explains why substance is measured by density and why the scores are  $z$ -standardised and not thresholded. §6 reports the sensitivity of the scores to the substance specification, to the weights of the Extended ESGSI and to the vocabulary. §7 gives the results by year, industry, country and pillar, §8 the limitations, and Appendix A the parameters, files and scripts that reproduce every reported number.

Table 1: Design of this study against its antecedent, Lagasio (2024).

|                     | Lagasio (2024)                                                      | This study                                                                                                |
|---------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Design              | Cross-sectional, 2023 only                                          | Balanced panel, 2018–2024                                                                                 |
| Documents           | 749 standalone ESG reports, one per firm                            | 343 unique reports: 235 annual reports, 101 URD, 7 standalone sustainability reports                      |
| Firms               | 749, worldwide, from a Refinitiv top-2,000 starting sample          | 49, seven European countries, each present in all seven years                                             |
| Inclusion rule      | English only; firms reporting ESG inside the annual report excluded | One annually recurring filing per firm-year, chiefly the annual report or URD                             |
| Text scored         | Whole report                                                        | Lexically extracted ESG zones (§3)                                                                        |
| ESG vocabulary      | GRI/SASB-derived; size not stated, list not published               | 434 source entries (402 terms, 32 patterns); 399 lemmatised scoring entries; all files published (§4, §A) |
| Tone                | TextBlob polarity                                                   | Loughran–McDonald polarity                                                                                |
| Substance           | TF-IDF; aggregation and vector normalisation not stated             | Mentions per 100 tokens; two TF-IDF variants as sensitivity (§5.2)                                        |
| Score normalisation | Prose says min–max; printed formula is a $z$ -score difference      | $z$ -score (§5.8.4)                                                                                       |
| Further components  | None                                                                | QUANT and HEDGE, in the Extended ESGSI only                                                               |
| Threshold           | Index > 0 labelled “Potential ESGwashing”                           | None; continuous score                                                                                    |

## 2 Data and corpus construction

The corpus comprises 343 unique corporate reports from 49 listed companies headquartered in seven European countries (Belgium, Finland, France, Germany, Italy, the Netherlands, Spain), one report per company per year from 2018 to 2024. The panel is exactly balanced: every year contributes 49 documents, and each country contributes the same number of documents in every year (its firm count in Table 2). Balance is temporal, not cross-national: countries are represented in very unequal numbers, but no year departs compositionally from any other. Firms enter the paper only through their country, their industry and anonymous positions; no firm is named.

Three document types are represented: annual reports, the Universal Registration Document (URD; French *document d’enregistrement universel*), and standalone sustainability reports. Document type was assigned per report from an external metadata file (`metadata/muestras_informes.xlsx`), joined to each PDF by country, publication year and a normalised company name — Unicode-folded, punctuation-stripped, and reconciled through a manual company-name map covering divergences between the directory tree and the metadata source. The join resolved all 343 documents; no case was left unmatched.

### 2.1 The country/type confound is a structural property of the corpus

Table 2 reports the corpus by country and document type. Document type is confounded with country almost completely: 99 of the corpus’s 101 URD filings are French, and all 98 German documents are annual reports. Both are properties of national reporting practice rather than sampling accidents — the URD is the standard annual regulatory filing for French listed companies and routinely carries the extended sustainability disclosure, whereas the German issuers in this sample report ESG content inside the ordinary annual report. We state this here as a design property of the corpus: within the pooled panel, a country effect and an annual-report/URD

Table 2: Corpus composition by country and document type.

| Country     | Companies | Annual report | URD | Sustainability report | Total |
|-------------|-----------|---------------|-----|-----------------------|-------|
| France      | 17        | 13            | 99  | 7                     | 119   |
| Germany     | 14        | 98            | 0   | 0                     | 98    |
| Netherlands | 6         | 40            | 2   | 0                     | 42    |
| Italy       | 5         | 35            | 0   | 0                     | 35    |
| Spain       | 4         | 28            | 0   | 0                     | 28    |
| Finland     | 2         | 14            | 0   | 0                     | 14    |
| Belgium     | 1         | 7             | 0   | 0                     | 7     |
| Total       | 49        | 235           | 101 | 7                     | 343   |

Table 3: Corpus composition by year.

| Year  | Annual report | URD | Sustainability report | Total |
|-------|---------------|-----|-----------------------|-------|
| 2018  | 36            | 12  | 1                     | 49    |
| 2019  | 33            | 15  | 1                     | 49    |
| 2020  | 33            | 15  | 1                     | 49    |
| 2021  | 33            | 15  | 1                     | 49    |
| 2022  | 34            | 14  | 1                     | 49    |
| 2023  | 33            | 15  | 1                     | 49    |
| 2024  | 33            | 15  | 1                     | 49    |
| Total | 235           | 101 | 7                     | 343   |

effect can be told apart only through 15 documents (the 13 French annual reports and the 2 Dutch URDs), and the seven standalone sustainability reports all come from one French firm. Germany, Italy, Spain, Finland and Belgium are each a single country and a single document type (all annual reports); Germany, with 14 firms and 98 documents, is the largest such slice, and §7 reports a within-Germany regression for that reason.

By year (Table 3), the document-type mix is nearly constant, so the temporal series (§7) is read against a fixed compositional backdrop. The small shifts come from six firms that file more than one document type over the panel (five French firms and one Dutch firm, each filing both annual reports and URDs).

## 2.2 Industry composition

Each firm carries one STOXX supersector of the Industry Classification Benchmark (ICB). The classification was supplied by the authors as a firm-level metadata file (`metadata/empresas_supersector.csv`), keyed on the same company names as the directory tree; it is not derived from the reports. The 49 firms fall into 16 supersectors. For analysis we also group supersectors into their ICB industries, the level above in the ICB hierarchy: 10 of ICB’s 11 industries are represented, Real Estate being the exception. Table 4 gives both levels. The counts in Tables 2, 3 and 4, and those quoted around them, are produced by `scripts/corpus_composition.py` from the document list of the adopted run; they do not depend on the vocabulary.

**What the sector dimension can identify.** Sector cuts across countries: every industry has firms in at least two countries, 13 of the 16 supersectors do as well, and the two countries with most firms each span seven industries (France’s 17 firms and Germany’s 14). Industry contrasts can therefore be drawn within a single country and document regime, which a country contrast cannot, and §7.6 draws them within France and within Germany for that reason.

Table 4: Corpus composition by ICB industry and, indented, STOXX/ICB supersector. Industries with a single supersector are shown on one row. AR: annual report; SR: standalone sustainability report. Countries: BE Belgium, DE Germany, ES Spain, FI Finland, FR France, IT Italy, NL Netherlands. Every firm contributes seven documents.

| Industry / supersector                 | Firms | Docs | AR  | URD | SR | Countries         |
|----------------------------------------|-------|------|-----|-----|----|-------------------|
| Financials                             | 11    | 77   | 70  | 7   | 0  | FI FR DE IT NL ES |
| Banks                                  | 7     | 49   | 42  | 7   | 0  | FI FR IT NL ES    |
| Insurance                              | 3     | 21   | 21  | 0   | 0  | FR DE             |
| Financial Services                     | 1     | 7    | 7   | 0   | 0  | DE                |
| Consumer Discretionary                 | 10    | 70   | 48  | 22  | 0  | FR DE IT NL ES    |
| Automobiles and Parts                  | 5     | 35   | 33  | 2   | 0  | DE IT NL          |
| Consumer Products and Services         | 4     | 28   | 8   | 20  | 0  | FR DE             |
| Retail                                 | 1     | 7    | 7   | 0   | 0  | ES                |
| Industrials                            | 9     | 63   | 32  | 31  | 0  | FR DE NL          |
| Industrial Goods and Services          | 7     | 49   | 30  | 19  | 0  | FR DE NL          |
| Construction and Materials             | 2     | 14   | 2   | 12  | 0  | FR                |
| Consumer Staples                       | 5     | 35   | 14  | 21  | 0  | BE FR NL          |
| Food, Beverage and Tobacco             | 3     | 21   | 7   | 14  | 0  | BE FR             |
| Personal Care, Drug and Grocery Stores | 2     | 14   | 7   | 7   | 0  | FR NL             |
| Health Care                            | 3     | 21   | 7   | 7   | 7  | FR DE             |
| Technology                             | 3     | 21   | 21  | 0   | 0  | DE NL             |
| Basic Materials (Chemicals)            | 2     | 14   | 8   | 6   | 0  | FR DE             |
| Energy                                 | 2     | 14   | 7   | 7   | 0  | FR IT             |
| Telecommunications                     | 2     | 14   | 14  | 0   | 0  | FI DE             |
| Utilities                              | 2     | 14   | 14  | 0   | 0  | IT ES             |
| Total                                  | 49    | 343  | 235 | 101 | 7  | 7 countries       |

**What it cannot identify.** Three limits bound it. First, cells are small: industries hold 2 to 11 firms and supersectors 1 to 7, and 8 of the 16 supersectors hold one or two firms, so a group-level interval rests on very few degrees of freedom, or none (§5.6). Second, sector is constant within a firm, so it is identified only from the 49 firms, never from within-firm variation, and it is absorbed by firm fixed effects. Third, sector does not break the country/type confound where the confound is sharpest: the 16 firms that file any URD are 15 of the 17 French firms and one Dutch firm, so an industry’s URD share is essentially its French share, and the seven standalone sustainability reports belong to a single Health Care firm. Some industries sit in a narrow set of countries (Utilities in Italy and Spain, Technology in Germany and the Netherlands), and a contrast between them is also a contrast between those countries.

Firms are assigned to countries as in the directory tree and in the metadata workbook. One assignment is open to argument: one firm is incorporated in the Netherlands but formed from predecessors headquartered elsewhere, and is classified under the Netherlands, consistently with both sources. It is also the Dutch firm whose filings include the two non-French URDs (5 annual reports and 2 URDs). Reclassifying it would move seven documents out of the Netherlands and would leave its industry unchanged.

### 2.3 Comparability across document types

The panel spans three document types with different structure and different length: annual reports in which ESG disclosure is one section among many financial and operational ones, URD filings that combine regulatory financial, governance and non-financial statements, and standalone sustainability reports that are ESG content throughout. Raw comparability across types cannot be assumed. Every measurement from §3 onward accordingly operates not on whole documents but on the ESG-relevant zones the lexical filter identifies within each one, which is

what lets a single score construction (§5) be applied uniformly across all three types. It does not make the types equivalent: the extraction is applied uniformly, not validated separately by type, and because type is sorted by country, no document-type effect on either score is identified (§7.6).

## 2.4 Deduplication

Corpus assembly began from 344 collected PDF filings, one more than the final corpus size. The extra file is one 2019 report present twice, the second time as a `_repaired` re-issue of the same filing under a different file name. A content-hash comparison identified this single duplicate pair. The hash is taken over the preprocessed (lemmatised) representation of §4 rather than the raw extraction, so the criterion is that two documents are indistinguishable to every downstream measure; here the two files are in fact identical at the extraction stage as well. The `_repaired` copy was removed as a data-quality control, leaving 343 unique reports. The balanced panel reported above — 49 documents in every year, and a fixed per-country count within every year — holds only after this step: before deduplication, 2019 carried 50 documents, with the Netherlands contributing 7 rather than 6, while every other year already stood at 49.

## 3 Document extraction: the lexical ESG-zone filter

The original study vectorises each report in its entirety: a standalone sustainability report is, by construction, already ESG text, so Lagasio (2024) needs no step to separate ESG content from anything else. Our corpus is different by design (§1, §2): annual reports and URDs in which ESG disclosure is embedded inside a much larger financial and legal document. Feeding the whole report to the downstream measures would let financial-statement bulk and document length dominate every score. Extraction is therefore a required stage with no counterpart in the original, and it fixes the text, and hence the denominators, on which every component of §5 is computed.

### 3.1 Paragraph segmentation

The extractor is the class `LexicalDocumentFilter` in `src/lexical_document_filter.py`; its full parameter and dependency listing is in §A. Each PDF is read with PyMuPDF’s block-level extractor (`page.get_text("blocks")`), text blocks only, sorted top to bottom and left to right. A block is accepted as a paragraph only after passing five filters, in order: (i) *margin strip* — dropped if entirely in the top or bottom margin band ( $y_1 < 0.08h$  or  $y_0 > 0.92h$ ,  $h$  = page height); (ii) *running header/footer suppression* — every margin-band block of at most 120 characters is normalised (digits  $\rightarrow \#$ ) and its occurrences counted over the document, and any normalised string whose occurrences exceed 30% of the page count is blacklisted and dropped wherever it occurs; (iii) *financial-table detection* — a block is dropped as tabular only if it contains *no* ESG keyword (the regex defined below) *and* either its digit-to-character ratio exceeds 0.35 with at most 6 lines, or every line is under 60 characters, there are at least two lines, and the first has no terminal punctuation (period, exclamation mark, question mark, or colon); the keyword veto keeps a table row that happens to carry ESG content rather than discarding it as noise; (iv) *section headers* — an all-caps block (ignoring non-letters), at least two letters, at most 10 tokens, fewer than 60% of them with one letter or none, and containing no filename-like string (a word, a period, then 2–6 word characters, e.g. `report.pdf`), is emitted as a § TITLE § marker that tags subsequent paragraphs with their section, rather than entering the paragraph stream; (v) *navigation* — a block that survives (i)–(iv) is cleaned (below) and then dropped if it reads as a table of contents, a GRI/ESRS correspondence table or a stray navigation line (§3.3). Cleaning strips URLs, rejoins hyphenated line breaks, drops non-ASCII characters, and collapses whitespace. Non-ASCII characters are deleted, not transliterated, so a term written

with a typographic hyphen fuses (*netzero*, which the regex below does not match) and a word set with a ligature loses letters; the matches lost this way are not counted.

### 3.2 Keyword matching

Paragraphs are scored against a single compiled regular expression, `_ESG_KW_RE`, built at import time from the four exported vocabulary files of §4. Under the adopted specification it combines 402 natural-form terms (367 from `esg_terms.txt` and 35 from `esg_terms_sectorial.txt`) with 32 patterns (31 from `esg_patterns.txt`, one from `esg_patterns_sectorial.txt`); without the sectoral files it would hold 367 and 31. Each term is converted mechanically: its words are joined by `[\s\-]+`, so spaces and hyphens are interchangeable (*net zero* matches *net-zero*); an optional `s` after the last word covers *-s* plurals only, so *data breaches* is missed and *fatalities* is caught only because a pattern (`fatalit\w+`) supplies it. Terms are ordered by number of words, then by length, both descending, because alternation is leftmost-first rather than longest-match: without the ordering *carbon* would shadow *carbon footprint*. The 32 patterns are read from file as written, not generated, and follow the terms in the alternation: morphological families (`sustainab\w+`, `recycl\w+`), digit-bearing forms (`co2e?`, `scope\s*[123]`, ISO codes), collocations that fix the ESG sense of a polysemous head (*social*, *water*, *energy*), and phrases that do not survive lemmatisation (*paris agreement*, *just transition*). The whole alternation is wrapped in word boundaries and matched case-insensitively. A match consumes its span, so *carbon footprint* counts once, not also as *carbon*.

### 3.3 Navigation filter

Tables of contents and GRI/ESRS correspondence tables are lists of section names, dense in ESG keywords, so the density rule below would select them as ESG text; §4.5 gives the reason for filtering the text rather than removing the section-naming terms from the vocabulary. The filter (`_is_navigation`) acts on the cleaned block and counts *navigation markers* among its whitespace tokens: a number of one to three digits preceded by a word and followed by nothing, a capitalised token or a section number (a page reference between titles); a multi-level section number (2.1, 1.3.1) followed by a capitalised token; a GRI-style (205-3) or ESRS-style (S1-14) disclosure code; or *p.*, *pp.*, *page* or *paragraph* followed by a digit. A block is dropped if it contains two or more dotted-leader page references (five or more dots, then a page number), or if it has at least four markers making up at least 8% of its tokens. A short block, of at most 15 tokens, is kept if it ends in sentence punctuation (.!?:;) and contains no leader (four or more underscores or dots); otherwise it is dropped if it opens with a multi-level section number or a Roman numeral and period, ends in a one- to three-digit number after a word, or contains a leader. The last rule also drops short unpunctuated labels such as *Scope 3* standing alone. Unnumbered headings and glossaries are not targeted.

The thresholds above were calibrated on a hand-labelled sample of keyword occurrences that is not part of the repository. The constants are therefore documented but their calibration cannot be reproduced, and the pipeline does not log how many blocks the filter drops, so its effect on the extraction is unmeasured.

### 3.4 Density scoring and zone construction

Each paragraph  $p$  is scored by keyword density,

$$\text{density}(p) = 100 \times \frac{\text{kw\_count}(p)}{\text{words}(p)}, \quad (1)$$

where `kw_count(p)` is the number of matches of `_ESG_KW_RE` on the cleaned, unlemmatised paragraph text and `words(p)` its whitespace-token count. A paragraph is *hot* if  $\text{density}(p) \geq 1.0$ .

Table 5: Extraction parameters.

| Parameter                                 | Value                             |
|-------------------------------------------|-----------------------------------|
| Keyword-density threshold (hot paragraph) | 1.0 / 100 words                   |
| Context window                            | $\pm 1$ paragraph                 |
| Minimum zone length                       | 150 characters                    |
| Header/footer repeat threshold            | > 30% of pages                    |
| Margin bands                              | top 8% / bottom 8% of page height |

Every hot paragraph is expanded to a window of one paragraph either side; overlapping or adjacent windows ( $lo \leq prev\_hi + 1$ ) are merged into one contiguous zone; merged zones under 150 characters are dropped. Zone statistics (keyword count, word count, density) are recomputed on each surviving zone’s concatenated text, not summed from its paragraphs; because the term regex allows any whitespace between words, a phrase can match across a paragraph break there. Table 5 lists the free parameters governing hot-paragraph detection and zone construction, none tuned by grid search; the thresholds internal to financial-table, section-header and navigation detection above are additional constants, given in prose rather than tabulated. Per PDF, zones are sorted by start offset and joined into one **relevant\_text** field, which is all that every later stage (§4 onward) reads.

### 3.5 Yield

Extraction runs before deduplication, on all 344 PDFs, and every PDF yielded at least one zone. The extra file is one 2019 report filed twice (a **\_repaired** copy), removed in §2; its extraction is identical to the original’s. Table 6 gives the totals on both bases; every per-document statistic, the yield and the by-year figures are on the 343 unique reports, as is everything downstream of preprocessing. Zone and word counts come from **scripts/analysis\_v3.py**; keyword matches and extraction-stage density from **scripts/extraction\_stats\_v3.py**, which applies the extractor’s own compiled regex to each report’s **relevant\_text**. Summing the per-zone counts instead gives 50 fewer matches (1,071,489 on 343), phrases that span two concatenated zones.

Yield is extracted words over the words of the full PDF text layer (**page.get\_text()** over every page), both counted as whitespace tokens (**str.split()**, on all 343 reports. The tokeniser is the same; the text is not. The numerator is cleaned text, and cleaning deletes standalone non-ASCII tokens (bullets, dashes, currency signs), fuses words across a non-ASCII hyphen and rejoins line-break hyphenation, all of which the uncleaned denominator counts. Yield is therefore biased slightly downward, by an amount the committed scripts do not measure. It is the share of a report’s text layer that survives into its zones, not a share of its ESG content. The mean report has 389 pages.

The aggregate yield is 28,101,967 extracted words out of 70,966,812. By year (Table 7), every volume measure rises and extracted words grow faster than the full text. Extraction-stage keyword density rises in every year on both bases, aggregated and as a mean of per-report values (3.21 in 2018 to 4.52 in 2024).

What these rises do not show is that reports disclose more ESG content. Yield and extraction density are both functions of the vocabulary: a paragraph is kept because it is dense in these terms, and density counts the same terms. The vocabulary was built on reports from across 2018–2024 (§4) and contains the names of reporting regimes that postdate 2018 (§7.4), so a shift in reporting language towards this vocabulary raises both measures whether or not the underlying disclosure changed, and so does a change in how reports are organised that makes ESG passages longer and more contiguous. Nor do they show that extraction behaves alike in every year: nothing in the paper measures, year by year, how often the vocabulary’s matches or the navigation filter’s decisions are wrong. §7 reads the trends with these qualifications; here

Table 6: Extraction volume. Totals on all 344 extracted PDFs and on the 343 unique reports; per-document figures on 343 only. Keyword density is Eq. (1) with the full regex on raw extracted text; it is not  $SUS^{dens}$ .

| Statistic                                     | 344 PDFs   | 343 reports    |
|-----------------------------------------------|------------|----------------|
| Total zones                                   | 66,377     | 66,171         |
| Total extracted words                         | 28,113,340 | 28,101,967     |
| Keyword matches                               | 1,072,127  | 1,071,539      |
| Keyword density, aggregate (per 100 words)    | 3.814      | 3.813          |
| Zones per document (mean, sample sd)          |            | 192.9, 146.0   |
| Zones per document (min, median, max)         |            | 9, 162, 1,085  |
| Extracted words per document (mean, median)   |            | 81,930, 74,802 |
| Words per zone (aggregate)                    |            | 424.7          |
| Words per zone (mean of per-document values)  |            | 487.2          |
| Keyword density (mean of per-document values) |            | 3.673          |
| Yield vs. full PDF text (mean, median)        |            | 39.6%, 38.3%   |
| Yield vs. full PDF text (min, max)            |            | 7.1%, 75.6%    |
| Yield vs. full PDF text (aggregate)           |            | 39.6%          |

Table 7: Extraction by year, 49 reports per year (343 in all). Means per report except the last column; yield is extracted over full-text words; keyword density is the aggregate per 100 extracted words, as in Table 6.

| Year | Zones | Extracted words | Words/zone | Full-text words | Yield |        | KW density |
|------|-------|-----------------|------------|-----------------|-------|--------|------------|
|      |       |                 |            |                 | mean  | median |            |
| 2018 | 157.7 | 56,484          | 405.3      | 176,519         | 31.6% | 28.7%  | 3.24       |
| 2019 | 157.6 | 60,594          | 444.5      | 179,803         | 34.6% | 32.6%  | 3.30       |
| 2020 | 181.2 | 70,143          | 473.7      | 197,275         | 36.5% | 35.5%  | 3.49       |
| 2021 | 189.8 | 80,845          | 483.3      | 206,427         | 40.1% | 37.9%  | 3.63       |
| 2022 | 195.8 | 89,382          | 521.2      | 216,282         | 42.1% | 39.3%  | 3.75       |
| 2023 | 226.1 | 96,162          | 521.5      | 224,549         | 43.6% | 41.2%  | 4.00       |
| 2024 | 242.1 | 119,901         | 561.0      | 247,447         | 48.4% | 48.4%  | 4.56       |
| All  | 192.9 | 81,930          | 487.2      | 206,900         | 39.6% | 38.3%  | 3.81       |

they are recorded as properties of the extraction.

### 3.6 Design rationale

The filter is a pure function of the PDF and the lexicon files: no network call, no external API, no random seed on this path. This matters directly for the determinism claimed in §A. A semantic filter, selecting text by its similarity to anchor queries in the embedding space of a hosted model, would not satisfy that claim: a paid, versioned, externally hosted model can change or be deprecated independently of this codebase, and the same PDF is then not guaranteed to yield the same selection on a later run. A lexical filter also ties extraction to the same vocabulary that SUS counts, which is the design choice argued in §4.4.

### 3.7 Recall is bounded by the lexicon

Extraction and SUS draw on the same vocabulary (§4.4), but not on the same entries. Every entry of the 399-entry lemmatised SUS vocabulary is the lemma of a term in the extraction

regex, and every one of the 402 terms reaches SUS: the eleven that preprocessing would empty or collapse to a generic word are protected as single tokens (§4.7), and three pairs of terms share a lemma (Table 8). The converse fails for the 32 patterns, which act only at extraction: they can make a paragraph hot and pull a zone in, but no pattern is itself a SUS entry. Two bounds follow. A term outside the combined regex triggers no hot paragraph and enters no zone, so it is invisible everywhere. A pattern can bring text into a zone, where it feeds SEN, QUANT, HEDGE and the length of the zone, without being counted by SUS. The two stages also match slightly different strings for the same term: the extraction regex requires a space or hyphen between words, the protection of §4.7 also accepts the fused form (*speakup*), and a lemmatised n-gram also matches across removed stopwords (*board director* in *board and directors*). For the heaviest multi-word entry this last difference is 0.3% (§4.7). The extraction side of the coupling is the larger one: a term in the regex brings further occurrences of itself into the zones it helps select (§4.7). §8 develops the consequences; they are noted here as properties of this step.

## 4 The ESG vocabulary

This section presents the ESG vocabulary, how it was built and how it is brought into the same representation as the corpus, together with the hedge lexicon used by HEDGE. The vocabulary has 434 entries: 402 natural-form terms and 32 regular-expression patterns. Of these, 36 (35 terms and one pattern) are designated sectoral; they are part of the vocabulary used throughout the paper, and excluding them is a sensitivity check (§6.3). The source of truth is `metadata/ESG_terms_v3.csv`, one row per entry with its type, pillar and sectoral flag; every count below is recomputed from committed files by `scripts/vocab_v3_stats.py`. Lagasio (2024) does not publish its keyword list or its size; ours is published in full so that every inclusion can be inspected.

### 4.1 Sources of candidate terms

Candidates came from three sources.

1. *Reporting frameworks and standards.* The disclosure topics and terminology of GRI, SASB, TCFD and the ESRS, including the names of the frameworks and regulatory regimes themselves (*csrd*, *tcfd*, *eu taxonomy*), which firms use to organise and label their ESG disclosure.
2. *Prior literature.* ESG terminology from dictionary-based studies of corporate disclosure, which supplied the general environmental, social and governance vocabulary (*emission*, *human rights*, *remuneration*).
3. *A structured review of sample reports.* The extracted zones of 17 reports from the corpus (13 firms, 7 countries) were read by three reviewers, working independently and without sight of the vocabulary, each of whom listed every ESG term found. After normalising case, plurals, hyphenation and spelling (`scripts/build_consensus.py`), the pooled list was given in random order to three further evaluators, also working independently and without knowing how many reviewers had listed each term (`scripts/build_consensus_v2.py`). Each evaluator judged, for every term, whether it is an ESG term, its pillar, whether it is sector-specific, and whether it needs a collocation to fix its sense. A term entered the candidate pool from this source only if all three evaluators approved it. Germany, 98 of the 343 reports, is represented in the sample by a single firm. **[Author: identify the reviewers of the candidate lists.]**

The review supplements the first two sources with the vocabulary firms actually use; its reading sample is small, and absence from it says nothing against a term.

## 4.2 Admission criteria

A candidate, from any source, was admitted only if it met the following criteria.

1. *ESG sense in context.* Its occurrences in the extracted text of this corpus, read in context, predominantly carry the ESG sense: a disclosure, target, data point or policy on an environmental, social or governance matter. A word whose occurrences are dominated by a non-ESG use in these reports is not admitted as a bare term: *materiality* (accounting and audit materiality), *fuel* (the verb), *ecosystem* (business or digital ecosystems), *footprint* (industrial or geographic footprint), *offset* (netting, *partially offset by*).
2. *Dispersion across firms.* A candidate from the review of sample reports had to occur at least 50 times in the corpus and in the reports of at least 5 of the 49 firms (constants `MIN_OCC_CANDIDATE` and `MIN_COMPANIES_CANDIDATE` in `scripts/build_kwic_sample.py`). The cut keeps out terms that are idiosyncratic to one or two firms; it is a practical choice, not a statistical test.
3. *Collocations for polysemous heads.* A head that carries the ESG sense only in some combinations is admitted only inside enumerated collocations: *carbon footprint* rather than *footprint*, the collocations of *social* and *water*, and `environmental\w*` and *environmentally* rather than every word beginning *environ-*. A count vocabulary cannot carry a negative lookahead (“*social*, except *social security*”), so a polysemous head is either admitted bare or admitted only in the collocations listed; the second option gives up the head’s other valid uses in exchange for excluding its invalid ones. A collocation inherits the pillar and sectoral designation of its head. The vocabulary holds 82 collocations.
4. *Patterns for morphological families and digit-bearing forms.* Where a family of forms shares a stem (`sustainab\w+`, `recycl\w+`), a term carries digits that preprocessing removes (`co2e?`, `scope\s*[123]`, ISO standard numbers), or a phrase does not survive lemmatisation (`paris\s+agreement`, *just transition*), the entry is written as a regular expression. Patterns act at extraction only (§4.4). There are 32.
5. *Sectoral designation.* An entry whose occurrences are concentrated in one industry is admitted but designated sectoral, because counting it would reward a firm for belonging to that industry rather than for what it discloses. The designation rests on two criteria: concentration of the entry’s occurrences across all 343 reports, including the share held by the single largest firm, and, for candidates from the review of sample reports, a majority of the evaluators who approved the term judging it sector-specific. The exact concentration thresholds of the first criterion are not in a committed script. Collocations inherit the designation of their head.

## 4.3 Pillar assignment

Every entry has exactly one pillar: E (environmental), S (social), G (governance), or TRANS (transversal) for umbrella terms that name ESG as a whole or its reporting regimes (*sustainability*, *esg*, *csrd*). Candidates from the review of sample reports take the pillar most often given by the evaluators who approved them; collocations take their head’s. The index pools the pillars (§5); the assignment matters only for the leave-one-pillar-out stability tests (§6.3) and the pillar breakdowns of §7. The EU taxonomy family (*eu taxonomy*, *taxonomy eligible*, *taxonomy alignment*) is assigned to TRANS, as a reporting regime, although the regulation’s six objectives are all environmental; that assignment is a choice and could reasonably be E.

## 4.4 One vocabulary for extraction and scoring

The vocabulary does two jobs. Matched as regular expressions on raw text, its entries decide which paragraphs are extracted (§3); lemmatised, its terms are what SUS counts (§5). The 32 patterns act only at the first stage. Changing the vocabulary therefore changes the corpus as well as the score: because the same entries select the zones, even SEN, which never consults the ESG vocabulary, is computed on different text when the vocabulary changes.

We use one vocabulary for both jobs rather than a broad extraction list paired with a narrower scoring list, because every component is computed on the extracted text. A zone selected by a term that fails the admission criteria adds non-disclosure prose to SEN, inflates the denominator of  $SUS^{dens}$ , and adds accounting figures to QUANT and modal verbs to HEDGE. The criteria of §4.2 therefore apply to extraction and scoring alike. The consequence for evaluation is that every stability test either freezes the extraction or simulates re-extraction (§6.3).

## 4.5 Section-naming terms and navigation text

Tables of contents and GRI/ESRS correspondence tables are lists of section names, and many section names are vocabulary entries (*governance*, *esrs*, *tcf*); they are dense in ESG keywords without being disclosure. Removing the section-naming terms would lose their valid uses in prose, so the vocabulary keeps them and the navigation text is removed instead by a filter in the extractor (§3.3). The noise that comes from the kind of text selected is handled at extraction; the vocabulary is judged on the sense of its terms in running text.

## 4.6 Preprocessing

Every zone identified in §3 passes through one **TextProcessor**: spaCy’s `en_core_web_md`, with named-entity recognition and parsing disabled. URLs are stripped; text is lowercased; eleven protected vocabulary phrases are replaced by single tokens (§4.7); the text is then tokenised and lemmatised. A token survives if it is not a stopword, punctuation or whitespace, is longer than two characters, its lemma is not one of 189 corpus-specific stopwords, and it is alphanumeric with at least two letters (keeping `co2` and `sf6`, dropping `2023` and `p12`). The lemmatised string (`clean_text`) feeds SUS, SEN and BREADTH; a whitespace-collapsed copy of the extracted text with digits and case intact (`raw_text`) is the input to QUANT and HEDGE, which need the numerals and the stopwords respectively.

## 4.7 Vocabulary and corpus in one representation

The scoring vocabulary is generated, not written by hand. `scripts/export_vocab_v3.py` writes the 434 entries of `metadata/ESG_terms_v3.csv` into four source files (terms or patterns, sectoral or not: 367, 35, 31 and 1 entries), and `scripts/build_lexicons.py` passes every term through the same **TextProcessor** as the corpus. A vocabulary entry and its occurrences in the corpus are therefore in one representation by construction: no entry can fail to match for want of a shared tokenisation or lemmatisation. The loader warns when a source file is newer than its generated file and stops when the generated file is missing.

**Protected phrases.** Preprocessing would destroy some phrases. *Well-being* would be emptied, as both of its words are spaCy stopwords; *iso 14001* would become *iso* and *say on pay* would become *pay*, a generic word that would count every unrelated use. The build therefore first lemmatises every term with a processor that has no protection; each term that comes out empty, and each multi-word term that comes out as a single word and is not on a reviewed allow-list, is written to `metadata/esg_protected_phrases.txt` as a pair: a regular expression and a single token. The regular expression accepts the words separated by spaces, hyphens or nothing, plus an optional final *s*. The token is the term with its non-alphanumeric characters removed and

Table 8: From natural-form terms to the lemmatised SUS vocabulary, recomputed with the pipeline’s **TextProcessor**; it matches the committed generated files.

| Step                            | Result                                                                                                                                                                                     |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Base terms                      | 367 → 365 lemmas                                                                                                                                                                           |
| protected, empty without it     | <i>well-being</i> (both words are stopwords)                                                                                                                                               |
| protected, collapsed without it | <i>iso 14001, iso 45001, iso 50001</i> → <i>iso</i> ; <i>first aid</i> → <i>aid</i> ; <i>say on pay</i> → <i>pay</i> ; <i>speak up</i> → <i>speak</i> ; <i>2030 agenda</i> → <i>agenda</i> |
| collisions                      | <i>conflict of interest, conflicts of interest</i> → <i>conflict interest</i> ; <i>well-being, wellbeing</i> → <i>wellbee</i>                                                              |
| allowed collapses               | <i>due diligence</i> → <i>diligence</i> ; <i>eu taxonomy</i> → <i>taxonomy</i>                                                                                                             |
| Sectoral terms                  | 35 → 34 lemmas; <i>b corp, cop 28, iso 37001</i> protected, collapsed without it; <i>cop 28</i> and <i>cop28</i> collide                                                                   |
| Union (scoring vocabulary)      | 399 entries: 140 unigrams, 224 bigrams, 34 trigrams, 1 four-gram                                                                                                                           |

any leading digits moved to the end (*well-being* → *wellbeing*, *2030 agenda* → *agenda2030*). **TextProcessor** applies the substitutions to the lowercased text before tokenising, for the corpus and for the vocabulary alike, so a protected term and its occurrences meet in the same token. Eleven phrases qualify (Table 8), and thirteen vocabulary terms are represented by the eleven protected tokens: the unhyphenated *wellbeing* and the fused *cop28* share the token of their spaced variants. spaCy may lemmatise the token itself (*wellbeing* → *wellbee*), which is harmless because both sides are lemmatised alike; the build stops if a token does not survive as one word. Two collapses are on the allow-list and stand without protection, *due diligence* → *diligence* and *eu taxonomy* → *taxonomy*, on the judgement that the surviving word occurs almost only inside the phrase in these reports.

**N-gram structure.** Every one of the 402 terms reaches SUS, as one of  $V = 399$  lemmatised entries. Multi-word entries are 259 of the 399 (64.9%); the longest has four tokens, so the vectoriser’s **ngram\_range** is (1,4), derived at run time from the vocabulary. Under scikit-learn’s unigram default every multi-word entry would score zero, without an error.

**Extraction and scoring, one entry.** Stopword and punctuation removal makes tokens adjacent that were not, so a lemmatised multi-word entry can match more than its phrase: *board director*, the lemma of *board of directors*, also matches *board and directors*. The excess is small. On the extracted text the lemma counts 59,912 occurrences in **clean\_text** and the exact phrase, *board(s) of director(s)* in **raw\_text**, 59,749, a difference of 0.3%. The same entry illustrates the coupling of §4.4 from the other side. Because the phrase is in the extraction regex, paragraphs that mention it are made hot, and the zones built around them bring further occurrences of it, and of the governance text around it, into the extracted text on which it is then counted. It is the heaviest entry of the scoring vocabulary (Table 10); part of that weight is a product of its own role in extraction. The sensitivity of the index to single entries is reported in §6.3.

## 4.8 Composition

Table 9 gives the composition of the 434 entries by pillar and type, and, for the 399 lemmatised scoring entries, their number and share of all vocabulary occurrences in the extracted text by pillar. E is the largest pillar on both counts; TRANS has 30 scoring entries and 14.2% of the

Table 9: Composition of the vocabulary by pillar and entry type. Terms and collocations are natural-form literals; a collocation is a literal admitted in place of a polysemous bare head. Sectoral cuts across the three type columns. The last two columns refer to the  $V = 399$  lemmatised scoring entries (terms only; patterns act at extraction) and their share of the 1,047,951 vocabulary occurrences in the extracted text of the 343 reports.

| Pillar              | Entries (434) |         |          |       |          | Scoring (399) |          |
|---------------------|---------------|---------|----------|-------|----------|---------------|----------|
|                     | Terms         | Colloc. | Patterns | Total | Sectoral | Entries       | Occ. (%) |
| E (environmental)   | 123           | 29      | 16       | 168   | 14       | 152           | 38.2     |
| S (social)          | 105           | 23      | 11       | 139   | 14       | 127           | 17.6     |
| G (governance)      | 65            | 27      | 3        | 95    | 8        | 90            | 30.0     |
| TRANS (transversal) | 27            | 3       | 2        | 32    | 0        | 30            | 14.2     |
| Total               | 320           | 82      | 32       | 434   | 36       | 399           | 100.0    |

Table 10: The ten heaviest scoring entries: occurrences in the lemmatised extracted text (`clean_text`) of the 343 reports, share of all 1,047,951 vocabulary occurrences, and number of reports in which the entry occurs.

| Entry (lemma)  | Pillar | Occurrences | Share (%) | Reports |
|----------------|--------|-------------|-----------|---------|
| board director | G      | 59,912      | 5.72      | 331     |
| remuneration   | G      | 59,209      | 5.65      | 335     |
| emission       | E      | 53,307      | 5.09      | 339     |
| sustainability | TRANS  | 45,050      | 4.30      | 342     |
| governance     | G      | 44,174      | 4.22      | 342     |
| climate        | E      | 44,026      | 4.20      | 340     |
| sustainable    | TRANS  | 34,395      | 3.28      | 343     |
| water          | E      | 26,449      | 2.52      | 320     |
| carbon         | E      | 23,940      | 2.28      | 338     |
| human right    | S      | 20,624      | 1.97      | 330     |
| Ten heaviest   |        |             | 39.2      |         |
| Fifty heaviest |        |             | 75.0      |         |

occurrences, most of them from *sustainability*, *sustainable*, *esg* and *taxonomy*. The 34 sectoral scoring entries hold 1.0% of the occurrences.

Occurrence mass is concentrated. Every one of the 399 entries occurs at least once; the ten heaviest hold 39.2% of all occurrences and the fifty heaviest 75.0% (Table 10). The two heaviest entries are governance terms (*board director*, *remuneration*), consistent with a corpus of annual reports and URDs that carry a corporate-governance report (§2). The concentration means that a handful of entries carry most of SUS; §6.3 removes entries one at a time for that reason.

## 4.9 The hedge lexicon

The HEDGE lexicon is generated by the same script from the Loughran–McDonald master dictionary (Loughran and McDonald, 2011), in the copy distributed with `pysentiment2`, the package that computes SEN. It keeps four categories, Uncertainty (297 words), Constraining (184), WeakModal (27) and StrongModal (19), for 495 unique forms (`metadata/lm_hedge.txt`). The forms are kept as written, without lemmatisation, because HEDGE is matched against the raw extracted text (§5.3). The lemmatised text is stopwords-filtered, and the modals that define two of the four categories (*may*, *might*, *could*, *must*, *will*), together with *always*, *never* and

*perhaps*, are spaCy stopwords, so on lemmatised text they could never be counted.

On the raw extracted text of the 343 reports, tokenised as HEDGE tokenises it (lowercase alphabetic words, hyphenated compounds kept whole), the lexicon matches 498,477 of 27,252,597 words (1.83 per 100), and 429 of its 495 forms occur at least once. The count is dominated by a few forms: *risk* (87,687), *risks* (59,412), *will* (40,489), *may* (38,628) and *requirements* (20,237), the five most frequent, are 49.4% of all hits, and *risk* and *risks* alone outnumber the next three together. HEDGE is therefore to a large extent a measure of how often a report names risk. The lexicon also counts the month: an occurrence of *May* capitalised and next to a number (*31 May*, *May 2023*) is at least 6,632 of the 38,628 hits on *may* (17.2%) and 1.3% of all hits. This is a lower bound, as it misses the month where no number is adjacent; the lexicon is used as published, without removing these.

#### 4.10 What the construction does and does not establish

The construction makes each entry traceable to a source, a criterion, a type, a pillar and a sectoral flag, and it guarantees that every term is counted in the same representation as the text it is counted on. It does not validate the vocabulary, for three reasons.

1. *Built on the evaluation corpus.* The review sample is drawn from the 343 reports, and the sense criterion and the dispersion cut were applied to occurrences in the same corpus the index is computed on. There is no hold-out, and the stability results of §6.3 inherit this dependence.
2. *Judgements, not measurements.* Admission, the choice between a bare head and its collocations, the pillar and the sectoral designation are judgements under stated criteria; the paper reports no error rate for the vocabulary’s matches, and the concentration thresholds behind the sectoral designation are not in a committed script.
3. *Extraction and scoring match differently.* Patterns select text but never enter SUS; the extraction regex and the lemmatised n-grams match slightly different strings for the same term (§3.2); and an entry’s count depends in part on the zones it helped select (§4.7).

The TRANS assignment of the taxonomy family and the sectoral criteria are open choices; §6.3 reports how much the index depends on them.

## 5 Specification of the ESGSI and the Extended ESGSI

This section defines every quantity the paper computes and then explains the three choices on which the specification rests: substance measured by density rather than by TF-IDF with vector normalisation, standardisation by *z*-scores, and no threshold (§5.8). Later sections test sensitivity to these choices (§6) and report the resulting scores (§7); none of them redefines anything.

### 5.1 Notation and the two text representations

Let  $d = 1, \dots, n$  index the  $n = 343$  reports of §2 and  $j = 1, \dots, V$  the entries of the lemmatised ESG vocabulary of §4, with  $V = 399$ : the 365 lemmas of the base terms and the 34 further lemmas contributed by the sectoral terms (Table 8). The union removes any sectoral lemma already among the base lemmas; here none is. The 32 regular-expression patterns of the vocabulary act at extraction only and are not among the  $V$  entries. Preprocessing emits two representations of each document, and the components are not all computed on the same one:

- *lemmatised text* — lowercased, stopword-filtered, punctuation- and numeral-stripped, lemmatised (§4); whitespace token count  $\ell_d$ , with  $L_d = \max(\ell_d, 1)$ ;

- *raw extracted text* — the ESG zones of §3 with NUL characters dropped, whitespace collapsed and nothing else removed, digits and stopwords intact; whitespace token count  $r_d$ , with  $R_d = \max(r_d, 1)$ .

QUANT and HEDGE are computed on the raw extracted text; SUS, SEN and BREADTH are computed on the lemmatised text. The split is not incidental. Preprocessing discards numeric tokens, so on lemmatised text the **percentages** and **large\_numbers** families of QUANT match nothing and return exactly zero for every document, without error; it also discards stopwords, among them the modal verbs that HEDGE counts. Every length-normalised component is divided by a token count of the representation it is computed on —  $R_d$  for QUANT, a word count  $W_d$  of the same raw text for HEDGE (below),  $L_d$  for the others — and the counts are not interchangeable.

Let  $c_{dj}$  be the raw count of vocabulary entry  $j$  in the lemmatised text of  $d$ , obtained with a fixed-vocabulary count vectoriser (scikit-learn’s default tokeniser, lowercased) over token sequences of length 1 to  $n_{\max}$ , where  $n_{\max}$  is the token length of the longest entry, derived from the vocabulary at run time; here  $n_{\max} = 4$  (*supply chain human right*). Entries are counted independently, so an occurrence of a multi-word entry also counts for any shorter vocabulary entry it contains. Let  $df_j = |\{d : c_{dj} > 0\}|$  and

$$\text{idf}_j = \ln \frac{1+n}{1+df_j} + 1, \quad (2)$$

scikit-learn’s smoothed inverse document frequency (Pedregosa et al., 2011). Supplying the vocabulary in advance disables document-frequency and feature-count filtering, so the **max\_df**, **min\_df** and **max\_features** settings the original discloses (§5.8.1) are inoperative here: all  $V$  entries are retained and scored in every document. Cross-sectional rescaling of a finished component score  $x$  is

$$Z(x)_d = \frac{x_d - \bar{x}}{\sigma_x}, \quad \sigma_x = \sqrt{\frac{1}{n} \sum_d (x_d - \bar{x})^2}, \quad (3)$$

with  $Z(x) \equiv 0$  when  $\sigma_x = 0$ . Both  $\bar{x}$  and  $\sigma_x$  are taken over the  $n$  documents analysed, pooled across the seven years, so every score below is a position within this corpus, not an absolute quantity, and yearly means of a score are comparable with one another.

## 5.2 The substance component: three specifications

We compute three substance specifications, differing in term weighting and in vector normalisation. The first is the one ESGSI uses; the other two are TF-IDF constructions, reported as alternatives because the antecedent derives its substance score from TF-IDF without determining which construction (§5.8.1):

$$\text{SUS}_d^{\text{dens}} = 100 \cdot \frac{\sum_j c_{dj}}{L_d}, \quad (4)$$

$$\text{SUS}_d^{\text{tfidf-}\ell} = 100 \cdot \frac{\sum_j c_{dj} \text{idf}_j}{L_d}, \quad (5)$$

$$\text{SUS}_d^{\text{tfidf-}L2} = \frac{1}{V} \sum_j \frac{c_{dj} \text{idf}_j}{\|\mathbf{c}_d \odot \mathbf{idf}\|_2}, \quad (6)$$

where  $\mathbf{c}_d \odot \mathbf{idf}$  is the elementwise product of the count vector with the idf vector, and  $\text{SUS}_d^{\text{tfidf-}L2} = 0$  by convention when a document contains no vocabulary entry. The subscripts name the scaling applied to the term vector:  $\ell$  for division by document length,  $L2$  for vector normalisation, that is scaling the term vector to unit  $L2$  norm. Equation (4) is ESG mentions per 100 lemmatised tokens under uniform term weights. Equation (5) applies idf weights to

Table 11: The four QUANT regular-expression families.

| Family        | Pattern                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------|
| percentages   | <code>\b\d+(?:[.,]\d+)?\s*%</code>                                                                        |
| large_numbers | <code>\b(?:19 20)\d{2}\b\d{4,}\b</code>                                                                   |
| units         | <code>\b(?:tonne ton mt ktco2 co2e? ghg kwh mwh <br/>gwh twh mw gw litre liter m3 cubic meter)\b</code>   |
| frameworks    | <code>\b(?:gri tcf sasb issb sdg ungc sfdr csrd <br/>un_global_compact paris_agreement taxonomy)\b</code> |

the same length-normalised construction. Equation (6) is the closest reading of the antecedent that its description allows: idf weights, L2 vector normalisation of the document’s term vector (the library default), mean over the vocabulary. The factors 100 and  $1/V$  are positive affine rescalings and are annihilated by (3); the sum-versus-mean choice in (6) is therefore immaterial, and the specifications differ only in weighting and in vector normalisation. Both idf specifications estimate  $df_j$  on the analysed sample and are corpus-dependent; (4) is not. We adopt  $SUS^{\text{dens}}$  as the substance component of both scores; §5.8 gives the reasons and §6 the consequences of the alternatives.

### 5.3 Tone, quantification, hedging and breadth

**Sentiment.** SEN is Loughran–McDonald polarity (Loughran and McDonald, 2011) as implemented in `pysentiment2`. Its tokeniser splits the lowercased lemmatised text into maximal runs of letters and reduces each to its NLTK Porter stem; the dictionary words are stemmed the same way, giving 140 positive and 893 negative stems, disjoint. Let  $s_d^+$  and  $s_d^-$  count the tokens of  $d$  whose stem is positive and negative respectively:

$$SEN_d = \frac{s_d^+ - s_d^-}{s_d^+ + s_d^- + \varepsilon}, \quad \varepsilon = 10^{-6}. \quad (7)$$

Here  $\varepsilon$  is the package’s constant and only prevents division by zero. This is a bounded ratio on  $[-1, 1]$ , not a density: up to  $\varepsilon$  it depends only on the balance of  $s_d^+$  and  $s_d^-$ , not on how many sentiment-bearing tokens the document contains, and it is 0 when there are none.

**Quantification.** QUANT is the density of regular-expression hits from four families (Table 11) over the *raw* extracted text, matched case-insensitively:

$$QUANT_d = \frac{1}{R_d} \sum_{g \in G} m_g(\text{raw}_d), \quad (8)$$

with  $G$  the four families and  $m_g$  the number of non-overlapping matches of family  $g$ ; a string matched by two families counts twice.

The `large_numbers` family excludes four-digit strings of the form `19xx` and `20xx`, so that citing dates does not register as quantification.

**Hedging.** HEDGE is the share of words of the raw extracted text that belong to the hedge lexicon  $H$  of §4 ( $|H| = 495$  forms: the Uncertainty, WeakModal, StrongModal and Constraining words of the master dictionary of Loughran and McDonald, 2011, as written). The raw text is lowercased and split into words, the maximal runs of letters with internal hyphens allowed (`[a-z]+(?:-[a-z])*`), so digits and punctuation separate words and *ill-defined* is one word. With  $\text{words}(\text{raw}_d)$  that sequence and  $W_d = \max(|\text{words}(\text{raw}_d)|, 1)$ ,

$$HEDGE_d = \frac{|\{t \in \text{words}(\text{raw}_d) : t \in H\}|}{W_d}. \quad (9)$$

On the lemmatised text, which is stopwords-filtered, the modals *may*, *might*, *could*, *must* and *will*, and *always*, *never* and *perhaps*, could never match; on the raw text every form of *H* can. Matching is by form, not by sense: the count is dominated by *risk* and *risks*, so in these reports HEDGE largely registers how often risk is discussed, and *May* the month also counts as a modal (§4.9 gives the counts). HEDGE enters only ESGSI<sup>ext</sup>.

**Breadth.** BREADTH is the effective number of distinct ESG terms in a document: the exponential of the Shannon entropy of the term-share distribution, i.e. a Hill number of order one (Hill, 1973). With  $p_{dj} = c_{dj} / \max(\sum_k c_{dk}, 1)$ ,

$$\text{BREADTH}_d = \exp\left(-\sum_{j: p_{dj} > 0} p_{dj} \ln p_{dj}\right) \in [1, V]. \quad (10)$$

It equals 1 when all mentions fall on one term — and, by the  $\max(\cdot, 1)$  convention in  $p_{dj}$ , also when a document contains no vocabulary entry at all — and  $V$  when mentions are spread uniformly across the vocabulary. BREADTH is not a component of either score. It is reported alongside them because it is the quantity  $\text{SUS}^{\text{tfidf-}L^2}$  turns out to measure (§5.8.2); reporting it as a separate column is what allows that claim to be checked against the published results rather than taken on assertion.

## 5.4 The two scores

$$\text{ESGSI}_d = Z(\text{SEN})_d - Z(\text{SUS}^{\text{dens}})_d, \quad (11)$$

$$\text{ESGSI}_d^{\text{ext}} = Z(\text{SEN})_d - Z(\text{SUS}^{\text{dens}})_d - w_q Z(\text{QUANT})_d + w_h Z(\text{HEDGE})_d, \quad w_q = w_h = 0.5. \quad (12)$$

**ESGSI.** Equation (11) is the main score of the paper. It keeps the architecture of Lagasio (2024) — a tone component minus a substance component, each rescaled across the sample — and reproduces the formula printed there (Lagasio, 2024, p. 4), with our substance and tone components in place of the original’s. Where the substance specification is varied (§6) we write  $\text{ESGSI}(\text{SUS}^{\text{tfidf-}\ell})$  and  $\text{ESGSI}(\text{SUS}^{\text{tfidf-}L^2})$  for the score with (5) or (6) in place of  $\text{SUS}^{\text{dens}}$ ; unqualified, ESGSI always means  $\text{ESGSI}(\text{SUS}^{\text{dens}})$ .

**Extended ESGSI.** Equation (12) is a new metric that this paper proposes. It adds to ESGSI two components that the tone and substance scores do not capture. Quantified disclosure — percentages, absolute figures, physical units, named reporting frameworks — is further evidence of substance, so  $Z(\text{QUANT})$  is subtracted, on the same side as  $Z(\text{SUS}^{\text{dens}})$ . Hedging and modal language qualifies whatever a report asserts, so  $Z(\text{HEDGE})$  is added, on the side of tone. Both directions are part of the proposal rather than findings: HEDGE counts word forms, and in these reports it largely registers how often risk is discussed (§5.3). The weights of 0.5 give each added component half the weight of each original one, so that together they weigh as much as one of the two original components and the tone–substance contrast remains the core of the score. They are a stated choice, not an estimate: no external criterion is available against which to calibrate them (§8). §6.2 therefore reports the score over the grid  $w_q, w_h \in \{0, 0.25, 0.5, 0.75, 1\}$ , whose corner  $w_q = w_h = 0$  is ESGSI itself: the ranking depends mostly on  $w_h$ , the within-firm trend of §5.6 almost only on  $w_q$ . The two scores are related but not interchangeable (§7).

**Reading the scores.** Both scores are continuous. Every term is a  $z$ -score over the pooled corpus, so each score has mean zero by construction, and its sign says only whether a report lies above or below the corpus mean of the combination. No threshold is applied and no report is assigned to a class. §5.8.4 explains why. The comparisons we report — across years, industries,

countries and specifications — are comparisons of positions within the corpus: differences of means, trends, rank correlations and the overlap of the top and bottom deciles.

## 5.5 Pillar decomposition

The results decompose  $\text{SUS}^{\text{dens}}$  by pillar. The decomposition is descriptive: it is not a component of either score, it enters neither (11) nor (12), and it changes no score. Let  $\pi(j) \in \{\text{E}, \text{S}, \text{G}, \text{TRANS}\}$  be the pillar of entry  $j$  (§4). An entry takes the pillar of the natural-form term that lemmatises to it; where several terms share one lemma the first in `ESG_terms_v3.csv` is used, and no two such terms carry different pillars. The map partitions the  $V = 399$  entries into E 152, S 127, G 90 and TRANS 30. These are not the counts of Table 9 (E 168, S 139, G 95, TRANS 32), which are over the 434 source entries: patterns are not among the  $V$  entries, and some terms collide at lemmatisation. For pillar  $p$ ,

$$\text{dens}_d^p = 100 \cdot \frac{\sum_{j: \pi(j)=p} c_{dj}}{L_d}, \quad \sum_p \text{dens}_d^p = \text{SUS}_d^{\text{dens}}, \quad (13)$$

is the number of mentions of pillar- $p$  entries per 100 lemmatised tokens. For a set  $A$  of documents (a year, an industry, a country, or the corpus), the pillar share of mentions pools counts over the set,

$$s_A^p = \frac{\sum_{d \in A} \sum_{j: \pi(j)=p} c_{dj}}{\sum_{d \in A} \sum_j c_{dj}}, \quad (14)$$

so that a document weighs in proportion to its number of mentions, while the pillar density of the set is the unweighted mean of  $\text{dens}_d^p$  over  $d \in A$ . The same two quantities are computed for the 34 entries contributed only by sectoral terms, a set that cuts across the pillars. Because entries are counted independently (§5.1), the shares partition vocabulary hits, not text: one occurrence of a multi-word entry adds to every contained entry, possibly in different pillars.

## 5.6 Inference on the panel

The regressions of §7 are estimated on the panel of §2. Write  $d = (i, t)$  for the report of firm  $i = 1, \dots, M$  in year  $t$ , with  $M = 49$ ,  $N = n = 343$  and  $\tau_t$  the calendar year. The panel is balanced, so every firm is observed in all seven years. For an outcome  $y$  (a score, a component or a density), the pooled regression is

$$y_{it} = \alpha + \beta \tau_t + u_{it}. \quad (15)$$

The firm fixed-effects regression removes firm means,  $\ddot{y}_{it} = y_{it} - \bar{y}_i$  and  $\ddot{\tau}_{it} = \tau_t - \bar{\tau}_i$ , and estimates

$$\ddot{y}_{it} = \beta \ddot{\tau}_{it} + \ddot{u}_{it} \quad (16)$$

by least squares. In the balanced panel  $\bar{\tau}_i = 2021$  for every firm, and  $\hat{\beta}_{\text{FE}}$  equals the mean of the 49 firm-level OLS slopes.

Every panel regression is reported with standard errors clustered by firm. For a regressor matrix  $X$  with  $K$  columns and residuals  $\hat{u}$ , the variance is

$$\widehat{\text{Var}}(\hat{\beta}) = \frac{M}{M-1} \frac{N-1}{N-K} (X^\top X)^{-1} \left( \sum_{i=1}^M X_i^\top \hat{u}_i \hat{u}_i^\top X_i \right) (X^\top X)^{-1}, \quad (17)$$

with  $X_i$  and  $\hat{u}_i$  the rows of firm  $i$  (CR1), and  $t$  statistics are referred to a  $t$  distribution with  $M-1$  degrees of freedom (48 on the full panel, 13 on the German subsample). In the fixed-effects regression (16) the firm means are removed before estimation, so  $K$  counts the slope only. The conventional (homoskedastic, independent) OLS standard error of (15) is shown alongside

where it is reported at all, and is labelled as such: seven reports of one firm are not independent observations. Clustering by firm allows any correlation among a firm’s seven reports but assumes independence across firms within a year. A shock common to all firms in one year, such as a new reporting mandate, is not covered, and with seven years it cannot be clustered on. The slope therefore summarises the seven yearly means, and its standard error reflects dispersion across firms around them, not uncertainty about which years were drawn.

With  $M = 49$  clusters, and with fewer in the subsamples, the asymptotic cluster approximation can mislead, so the fixed-effects slope is also tested with a restricted wild cluster bootstrap (Cameron et al., 2008). Under  $H_0: \beta = 0$  the restricted model of (16) contains only the firm effects, so its residuals are  $\ddot{y}_{it}$  itself. For each replication  $b = 1, \dots, B$  we draw one Rademacher weight per firm,  $v_{ib}^* \in \{-1, +1\}$  with equal probability, form  $\ddot{y}_{itb}^* = v_{ib}^* \ddot{y}_{it}$ , re-estimate (16) and compute its cluster-robust  $t_b^*$ . The  $p$ -value is

$$p_{\text{wild}} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{|t_b^*| \geq |\hat{t}|\}, \quad (18)$$

with  $\hat{t}$  the observed statistic. The small-sample factor of (17) multiplies  $\hat{t}$  and every  $t_b^*$  alike and is omitted. We use  $B = 9,999$  and seed 20260925 (NumPy `default_rng`); when no  $t_b^*$  reaches  $|\hat{t}|$  we report  $p_{\text{wild}} < 1/(B+1) = 0.0001$ .

Group-specific slopes interact the detrended year with indicators of a time-invariant firm attribute  $\gamma(i)$  (industry, supersector or country),

$$\ddot{y}_{it} = \sum_k \beta_k \mathbf{1}\{\gamma(i) = k\} \ddot{\tau}_{it} + \ddot{u}_{it}, \quad (19)$$

with firm-clustered standard errors as in (17). In the balanced panel  $\hat{\beta}_k$  is the mean of the firm-level OLS slopes of the  $M_k$  firms in group  $k$ , and we also report the  $t$  interval on those firm slopes, with  $M_k - 1$  degrees of freedom. The two intervals differ in their effective degrees of freedom. The normal equation for  $\beta_k$  makes the  $M_k$  firm scores of group  $k$  sum to zero, so the clustered variance of  $\hat{\beta}_k$  rests on  $M_k - 1$  free contributions. Up to the CR1 factor it is  $(M_k - 1)/M_k$  times the squared standard error of the firm-slope mean, yet it is referred to  $t_{M-1}$ . For  $M_k = 1$  it is exactly zero; for  $M_k = 2$  the firm-slope interval has one degree of freedom (§2.2). We therefore read group slopes through the firm-slope interval and treat the clustered one as too narrow for small groups. Equality of the  $\beta_k$  is tested by a Wald  $F$  test on the clustered variance (denominator degrees of freedom  $M - 1$ ). It treats the slope of a one-firm group as known without error, and is therefore anti-conservative whenever such a group exists (Belgium among countries, two supersectors). With two such groups the variance of the restrictions is also singular, and the test uses its rank. Document type is not used as a grouping because it varies within six firms. Yearly means are reported with 95% intervals from a regression of  $y$  on a full set of year indicators, without constant, again clustered by firm.

## 5.7 Parameters

Table 12 collects every constant and every representation choice on which the definitions above depend; the full lexicons, versions and seeds are in §A. That appendix reprints the same values in Table 32, adding the extraction and preprocessing stages, so that a reader has one consolidated listing. The two tables are not independent sources: the values are defined here, and the appendix table is a consolidation of this one together with §3 and §4. Any disagreement between them would be an error in the appendix.

## 5.8 Why density, $z$ -scores and no threshold

Three choices distinguish our specification from its antecedent: substance is measured by density rather than by TF-IDF with vector normalisation, the components are standardised by  $z$ -scores

Table 12: Parameters of the index specification and of the panel inference.

| Parameter                        | Value                                                         | Applies to                                          |
|----------------------------------|---------------------------------------------------------------|-----------------------------------------------------|
| Text representation              | lemmatised                                                    | SUS, SEN, BREADTH                                   |
|                                  | raw extracted                                                 | QUANT, HEDGE                                        |
| Vocabulary size $V$              | 399 lemmatised entries<br>(365 base + 34 sectoral)            | SUS, BREADTH                                        |
| $n$ -gram range                  | (1, 4), from the longest entry                                | SUS, BREADTH                                        |
| Count tokenisation               | scikit-learn default                                          | SUS, BREADTH                                        |
| SEN tokenisation                 | letter runs, Porter stems                                     | SEN                                                 |
| HEDGE tokenisation               | hyphenated letter runs                                        | HEDGE                                               |
| Document-frequency filtering     | none; vocabulary fixed in advance                             | all SUS, BREADTH                                    |
| Term-frequency transform         | raw counts, no sublinear damping                              | all SUS                                             |
| idf weights                      | smoothed, Eq. (2)                                             | $SUS^{\text{tfidf-}\ell}$ , $SUS^{\text{tfidf-L2}}$ |
| Vector normalisation             | none                                                          | $SUS^{\text{dens}}$ , $SUS^{\text{tfidf-}\ell}$     |
|                                  | L2, within document                                           | $SUS^{\text{tfidf-L2}}$                             |
| Aggregation over $j$             | sum                                                           | $SUS^{\text{dens}}$ , $SUS^{\text{tfidf-}\ell}$     |
|                                  | mean over $V$                                                 | $SUS^{\text{tfidf-L2}}$                             |
| Length denominator               | $L_d$ (lemmatised tokens)                                     | $SUS^{\text{dens}}$ , $SUS^{\text{tfidf-}\ell}$     |
|                                  | $R_d$ (raw tokens)                                            | QUANT                                               |
|                                  | $W_d$ (raw words)                                             | HEDGE                                               |
|                                  | none                                                          | $SUS^{\text{tfidf-L2}}$ , SEN, BREADTH              |
| Hedge lexicon size               | 495 forms (L-M master)                                        | HEDGE                                               |
| Polarity smoothing $\varepsilon$ | $10^{-6}$                                                     | SEN                                                 |
| Score normalisation              | z-score, population $\sigma$                                  | ESGSI, $ESGSI^{\text{ext}}$                         |
| Substance component              | $SUS^{\text{dens}}$                                           | ESGSI, $ESGSI^{\text{ext}}$                         |
| Extended-score weights           | $w_q = w_h = 0.5$                                             | $ESGSI^{\text{ext}}$                                |
| Threshold                        | none; continuous score                                        | ESGSI, $ESGSI^{\text{ext}}$                         |
| Pillar partition of $V$          | E 152, S 127, G 90, TRANS 30                                  | pillar shares only                                  |
| Clusters                         | firm, $M = 49$                                                | panel regressions                                   |
| Cluster variance                 | CR1, Eq. (17)                                                 | panel regressions                                   |
| Reference distribution           | $t$ with $M - 1$ df (48 full panel)                           | panel regressions                                   |
| Wild cluster bootstrap           | restricted, Rademacher weights<br>$B = 9,999$ , seed 20260925 | fixed-effects slopes                                |

rather than rescaled by min-max, and no threshold is applied. This subsection gives the reasons. It begins with what Lagasio (2024) specifies, because the first choice answers a gap in that specification rather than a stated alternative.

### 5.8.1 What the antecedent discloses and leaves unstated

Lagasio (2024) computes its sentiment component with TextBlob polarity and its substance component with scikit-learn’s `TfidfVectorizer`, version 0.24.2, over an author-constructed keyword list “based on established ESG reporting frameworks such as Global Reporting Initiative (GRI) Standards . . . and Sustainable Accounting Standards Board (SASB) Standards” (pp. 3–4). Four vectoriser settings are disclosed in prose — terms in more than 95 % of documents ignored, terms in fewer than two ignored, the vocabulary limited to 10,000 terms, unigrams and bigrams — that is, `max_df=0.95`, `min_df=2`, `max_features=10000` and `ngram_range=(1,2)` (p. 4). The index is labelled definitionally: a positive value is “Potential ESGwashing”, a non-positive one “Likely Genuine” (pp. 4–5).

Four things are not stated. The article never gives the `norm` setting, and never describes a rescaling of document vectors under any wording. It never states how per-term TF-IDF values become a document-level  $SUS_i$ , which is glossed only as “sustainability score for company i” (p. 4). It never publishes the keyword list or gives its size; three example terms appear. And it never says whether the raw or the preprocessed text reached TextBlob. The Cohen’s Kappa, five-fold cross-validation and TF-IDF sensitivity analyses announced on p. 5 are not reported, and the data availability statement reads “No” (p. 10). One further point is stated twice, inconsistently: the article prints the index as a difference of  $z$ -scores and says in the same subsection that “both sentiment and sustainability scores are normalized to a uniform scale using min-max normalization” (p. 4).

Of these, the vector normalisation can be supplied by convention. The four disclosed settings are exactly four non-default arguments, so a reader who assumes that only departures from library defaults are reported would infer that `norm='l2'`, the default, was in force. We make that inference, and label it as one:  $SUS^{\text{tfidf-}L2}$  in (6) is the construction it implies. The aggregation rule has no default to fall back on, but on a  $z$ -standardised score it is immaterial, since a sum and a mean over the vocabulary differ by the factor  $1/V$  (§5.2). What matters, then, is the vector normalisation, and the next subsection shows why.

### 5.8.2 Vector normalisation discards the amount of disclosure

$L2$  vector normalisation scales a document’s term vector to unit length. For any  $\mathbf{v} \neq \mathbf{0}$  and scalar  $c > 0$ ,

$$\frac{c\mathbf{v}}{\|c\mathbf{v}\|_2} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \quad (20)$$

so two documents whose ESG term counts are proportional receive identical unit vectors, and any aggregation of those vectors returns identical scores. The normalised vector keeps the mix of terms and discards how much there is. That suits retrieval, where TF-IDF originated (Spärck Jones, 1972; Manning et al., 2008) and a document’s scale is a nuisance. It does not suit a score whose purpose is to set tone against the amount of substance behind it.

We verified the identity with the pipeline’s own substance code, starting from the report at the median of  $SUS^{\text{dens}}$  (60,951 lemmatised tokens, 3,984 vocabulary hits on 214 distinct entries,  $SUS^{\text{dens}} = 6.5364$ ), and re-vectorising all 343 documents in every run (Table 13). *Amplification*: we synthesised a document from the report’s observed counts, writing out each vocabulary entry as many times as it occurs, separated by a filler token that belongs to no entry and padded with the same filler, and multiplied the written-out block by  $k \in \{1, 2, 5, 20\}$ . The count vector is exactly  $k$  times the  $k = 1$  vector, every  $df_j$  is unchanged, and  $SUS^{\text{tfidf-}L2}$  does not move: its largest absolute deviation across the four runs is  $3.5 \times 10^{-18}$ . The synthetic document registers

Table 13: Behaviour of the substance specifications when a report’s ESG content is multiplied or diluted, all 343 documents re-vectorised in every run. Amplification: a synthetic document built from the counts of the median-SUS<sup>dens</sup> report, with its ESG counts multiplied by  $k$ ; SUS<sup>tfidf-L2</sup> is shown relative to its value at  $k = 1$ . Dilution: the real report with 152,378 non-ESG tokens appended. Source: `scripts/critique_v3.py`.

|                      | Tokens  | ESG hits | SUS <sup>dens</sup> | SUS <sup>tfidf-<math>\ell</math></sup> | SUS <sup>tfidf-L2</sup> |
|----------------------|---------|----------|---------------------|----------------------------------------|-------------------------|
| <i>Amplification</i> |         |          |                     |                                        |                         |
| $k = 1$              | 60,951  | 4,131    | 6.7776              | 8.2286                                 | 1.000000                |
| $k = 2$              | 70,612  | 8,262    | 11.7006             | 14.2055                                | 1.000000                |
| $k = 5$              | 99,595  | 20,655   | 20.7390             | 25.1789                                | 1.000000                |
| $k = 20$             | 244,510 | 82,620   | 33.7900             | 41.0240                                | 1.000000                |
| <i>Dilution</i>      |         |          |                     |                                        |                         |
| Original report      | 60,951  | 3,984    | 6.5364              | 7.9703                                 | 0.0141953042            |
| With filler          | 213,329 | 3,984    | 1.8675              | 2.2772                                 | 0.0141953042            |

4,131 hits rather than 3,984 because entries are counted independently (§5.1). *Dilution*: we appended 152,378 filler tokens, cycled from the report’s 50 most frequent tokens outside the vocabulary, to the real report. Density falls 3.50-fold; SUS<sup>tfidf-L2</sup> is bit-for-bit unchanged.

If magnitude is discarded, what remains is dispersion across terms. For a non-negative unit vector  $\mathbf{u} \in \mathbb{R}^V$ ,

$$1 \leq \sum_{j=1}^V u_j \leq \sqrt{V}, \quad (21)$$

with the lower bound attained when all mass sits on one term and the upper bound when it is spread evenly over all  $V$ ; mass spread evenly over  $m$  terms gives  $\sqrt{m}$ . Aggregating an L2-normalised vector therefore rewards spreading mentions over many terms, not mentioning them more. On our corpus the unit sums  $V \cdot \text{SUS}_d^{\text{tfidf-L2}}$  run from 1.93 to 8.62 (mean 5.73) and correlate 0.9575 with  $\sqrt{\text{BREADTH}_d}$ . SUS<sup>tfidf-L2</sup> itself correlates 0.9546 with BREADTH (Spearman 0.951) and 0.4052 with SUS<sup>dens</sup> (Spearman 0.4524); Figure 1 plots both relationships. Under L2 vector normalisation the substance component is, to a correlation of 0.95, a measure of topical breadth.

The idf weighting is a smaller matter. In a corpus of ESG disclosure the terms present everywhere are the ones the genre is about, so idf moves weight away from them: the 49 entries with  $\text{idf} < 1.2$  (document frequency above 280) hold 69.26 % of the 1,047,951 ESG occurrences, while the 207 entries with  $\text{idf} > 2$  hold 6.70 %. One mention of *living income* (two occurrences in the corpus,  $\text{idf} = 5.7420$ ) weighs as much as 5.545 mentions of *board of directors*, the most frequent entry (59,912 occurrences,  $\text{idf} = 1.0355$ ). Because the promoted entries carry so little mass, the effect on the score is small: holding the length denominator fixed and varying only the weighting, SUS<sup>dens</sup> and SUS<sup>tfidf- $\ell$</sup>  correlate 0.9876 and the resulting scores have Spearman correlation 0.9926; varying the vector normalisation instead lowers it to 0.6421 (Table 16). We keep uniform weights because they need no document frequencies: a report’s density does not change when the corpus does.

### 5.8.3 Convergent validity with quantitative content

The argument above is algebraic. An empirical check needs a yardstick built by different means, and QUANT (§5.3) is one: a density of regular-expression matches over the raw text, with no vectoriser, no lemmatised vocabulary and no document-frequency weighting. A report that discloses substance should also quantify it, so a substance measure ought to covary with QUANT (Campbell and Fiske, 1959). Agreement does not establish validity, but a substance measure that does not covary with independently counted quantitative content is hard to defend.

The two instruments are not fully independent. Seventeen of the 399 vocabulary entries

Table 14: Pearson correlation of each substance specification with QUANT ( $n = 343$  reports, 49 firms). Disjoint bases: vocabulary reduced from 399 to 382 entries, QUANT restricted to **percentages** and **large\_numbers**. Intervals: 95 % percentile intervals from 9,999 bootstrap samples of firms drawn with replacement, each with its seven reports (seed 20260925). Last row: bootstrap samples in which  $SUS^{\text{dens}}$  does not correlate more strongly than  $SUS^{\text{tfidf-}L2}$ . Source: `scripts/validity_v3.py`.

|                                                             | Full bases |                | Disjoint bases |                 |
|-------------------------------------------------------------|------------|----------------|----------------|-----------------|
|                                                             | $r$        | 95 % interval  | $r$            | 95 % interval   |
| $SUS^{\text{dens}}$                                         | 0.6305     | [0.523, 0.732] | 0.3992         | [0.227, 0.539]  |
| $SUS^{\text{tfidf-}\ell}$                                   | 0.6392     | [0.550, 0.736] | 0.4007         | [0.256, 0.535]  |
| $SUS^{\text{tfidf-}L2}$                                     | 0.2984     | [0.041, 0.551] | 0.1850         | [-0.072, 0.483] |
| $SUS^{\text{dens}} - SUS^{\text{tfidf-}L2}$                 | 0.3321     | [0.075, 0.589] | 0.2142         | [-0.143, 0.519] |
| $SUS^{\text{tfidf-}\ell} - SUS^{\text{tfidf-}L2}$           | 0.3407     | [0.112, 0.578] | 0.2157         | [-0.103, 0.500] |
| Samples with $SUS^{\text{dens}} \leq SUS^{\text{tfidf-}L2}$ |            | 30/9,999       |                | 1,654/9,999     |

are also matched by a QUANT pattern (reporting-framework acronyms, the *taxonomy* family, *co2*, *co2e*, *ghg* and four ISO standards); they carry 5.39 % of the vocabulary’s occurrences, and strings of theirs make up 27.31 % of QUANT’s 186,984 hits. Both instruments also read the same extracted zones, so the test is conditional on the extraction stage (§3). We therefore report the comparison twice (Table 14): on the full bases, and on disjoint bases, with the 17 entries removed from the vocabulary (399 to 382) and QUANT restricted to its two purely numeric families, **percentages** and **large\_numbers**. The restricted QUANT correlates 0.8697 with the full one. Because the 343 reports are seven reports each of 49 firms, every interval resamples firms, not reports.

On the full bases the two length-normalised specifications correlate with QUANT about twice as strongly as  $SUS^{\text{tfidf-}L2}$  does (0.6305 against 0.2984, a ratio of 2.11; Spearman 0.6804 against 0.4521), and the firm-bootstrap interval of the difference excludes zero.  $SUS^{\text{tfidf-}L2}$  behaves like BREADTH, which correlates 0.3080 with QUANT. Length does not explain the ordering:  $\log L_d$  and  $\log R_d$  correlate 0.9992, but with both partialled out of both sides the correlations are 0.5904 for  $SUS^{\text{dens}}$  and 0.1894 for  $SUS^{\text{tfidf-}L2}$ .

**Disjoint bases.** Every correlation falls, by 37–38 %, so the full-basis levels are upper bounds and convergent validity on clean bases is moderate. The ratio does not fall: 2.16 for  $SUS^{\text{dens}}$  against  $SUS^{\text{tfidf-}L2}$ , against 2.11 on the full bases, so the shared strings did not produce the ordering. What changes is the width of the intervals. The difference is 0.2142 with an interval from -0.143 to 0.519, and one bootstrap sample in six reverses it; under Spearman the gap is 0.0821, interval [-0.117, 0.272]. The uncertainty sits on the  $SUS^{\text{tfidf-}L2}$  side. Across the 49 firm means  $SUS^{\text{dens}}$  correlates 0.5265 with the restricted QUANT and  $SUS^{\text{tfidf-}L2}$  0.3400; within firms, over time, 0.2071 against -0.0716. On clean bases the direction is the same in every cut, but a firm-clustered interval does not exclude equality: convergent validity corroborates the choice of a length-normalised specification without carrying it on its own. The case against  $SUS^{\text{tfidf-}L2}$  rests first on (20) and its demonstrations, which use no yardstick.

**Density rather than length-normalised TF-IDF.** Of the two length-normalised specifications,  $SUS^{\text{tfidf-}\ell}$  correlates slightly more strongly with QUANT, by 0.0087 on the full bases. The firm bootstrap does not distinguish that gap from zero (interval [-0.018, 0.044]; 2,748 of 9,999 samples reverse it), and on disjoint bases it shrinks to 0.0014. The criterion separates the length-normalised pair from  $SUS^{\text{tfidf-}L2}$ , not the two members of the pair. We adopt  $SUS^{\text{dens}}$  on a structural ground: it estimates nothing on the sample, so a report’s substance score does not

depend on which other reports are in the corpus. A further variant, sublinear term frequency divided by length, fails the test outright ( $r = 0.0500$  with QUANT), because once each term’s contribution is damped towards presence the length denominator dominates the ratio.

#### 5.8.4 $z$ -scores rather than min–max, and no threshold

**What the antecedent computed.** The printed formula of Lagasio (2024, p. 4) is a difference of  $z$ -scores, but its published numbers are min–max. Its Table 2 reports both components on  $[0, 1]$  and a mean index of  $-0.2179$ , which is exactly the difference of the two component means,  $0.3727 - 0.5906$  (Lagasio, 2024, p. 6); a difference of  $z$ -scores would have mean zero by construction. The article’s stated reason for min–max is to “address potential biases due to variations in document length and linguistic style” (p. 4). A min–max rescaling is monotone within each component, so any dependence on length present before it is present, in the same rank order, after it.

**What the choice changes.**  $Z(\cdot)$  and the min–max map  $M(\cdot)$  are both increasing affine maps, so within a component they preserve ranks. Once two components are differenced they are no longer equivalent, because they equalise the components’ spreads differently. On our corpus, with the adopted substance component, the difference is small in ranks: the  $z$ -score and min–max versions of ESGSI have Spearman correlation  $0.9971$  ( $0.9957$  with  $SUS^{\text{tfidf-}\ell}$ ,  $0.9888$  with  $SUS^{\text{tfidf-L2}}$ ). The difference is in location. Under  $Z$  the mean of the score is zero by construction. Under  $M$  it is  $\overline{M(\text{SEN})} - \overline{M(\text{SUS})}$ , which is set by where each component’s mass sits inside its own observed range. With  $SUS^{\text{dens}}$  the two rescaled means are almost equal ( $0.3645$  and  $0.3610$ ) and the score has mean  $+0.0035$ . With  $SUS^{\text{tfidf-L2}}$  the rescaled substance mean rises to  $0.5678$  and the score’s mean shifts to  $-0.2034$ , although nothing about the reports has changed. The antecedent’s published mean,  $-0.2179$ , has the same structure: a substance distribution sitting high in its range. The resemblance of the two figures is consistent with that mechanism; on a different corpus with a different vocabulary it is not evidence of what the antecedent ran.

**Consequences for our specification.** We standardise by  $z$ -scores because they fix the location of every component at the corpus mean and express each in units of its own standard deviation, so that the sign and size of a score describe a report’s position relative to the corpus and nothing else. Under min–max the location of zero would depend on the extremes of each component’s distribution, which a single outlying report can move. For the same reason we apply no threshold. Zero is the corpus mean of the combination, a property of the rescaling and of the corpus; it is not a calibrated boundary between kinds of disclosure, and there is no external benchmark against which a boundary could be calibrated (§8). The antecedent’s threshold at zero inherits its location from the rescaling, and we neither use it nor replace it with another.

## 6 Robustness: substance specification, weights and vocabulary

This section measures how far the two scores move when the substance specification (§6.1), the weights of the Extended ESGSI (§6.2) and the ESG vocabulary (§6.3) are varied, and closes with what these checks do and do not establish (§6.4). Every comparison is between a perturbed and a reference score over the same 343 reports, and uses three measures. The first is the Spearman rank correlation  $\rho$ , with the Pearson correlation  $r$  as a secondary figure. The second is the overlap of the extreme deciles: of the 34 reports with the highest reference score, how many are also among the 34 highest under the perturbed score, and likewise for the 34 lowest; we write it as top/bottom counts out of 34. The third is the within-firm trend, the fixed-effects slope of (16) with firm-clustered standard errors and, where stated, the wild cluster bootstrap  $p$ -value of (18) (§5.6). For the adopted scores the slope is  $-0.181$  per year for ESGSI (SE  $0.035$ ,  $p_{\text{cl}} = 3.7 \times 10^{-6}$ ,

Table 15: Pearson correlations among the three substance specifications and BREADTH ( $n = 343$ ). Spearman correlations, same cells:  $SUS^{\text{dens}}-SUS^{\text{tfidf-}\ell}$  0.9886,  $SUS^{\text{dens}}-SUS^{\text{tfidf-L2}}$  0.4524,  $SUS^{\text{dens}}-\text{BREADTH}$  0.4763,  $SUS^{\text{tfidf-}\ell}-SUS^{\text{tfidf-L2}}$  0.5234,  $SUS^{\text{tfidf-}\ell}-\text{BREADTH}$  0.5522,  $SUS^{\text{tfidf-L2}}-\text{BREADTH}$  0.9515. Source: `scripts/analysis_v3.py`.

|                           | $SUS^{\text{dens}}$ | $SUS^{\text{tfidf-}\ell}$ | $SUS^{\text{tfidf-L2}}$ | BREADTH |
|---------------------------|---------------------|---------------------------|-------------------------|---------|
| $SUS^{\text{dens}}$       | 1.0000              | 0.9876                    | 0.4052                  | 0.4173  |
| $SUS^{\text{tfidf-}\ell}$ | 0.9876              | 1.0000                    | 0.4823                  | 0.4985  |
| $SUS^{\text{tfidf-L2}}$   | 0.4052              | 0.4823                    | 1.0000                  | 0.9546  |
| BREADTH                   | 0.4173              | 0.4985                    | 0.9546                  | 1.0000  |

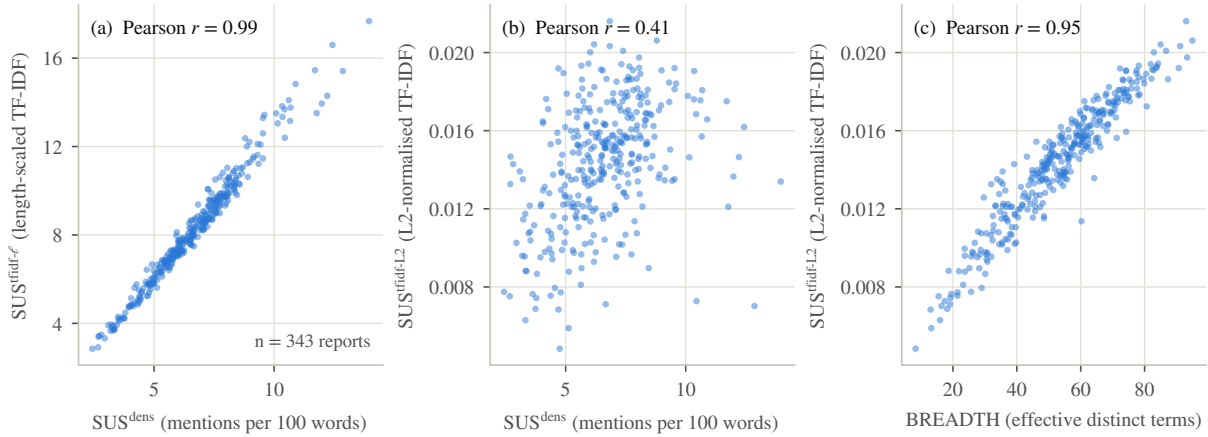


Figure 1: The substance specifications, one point per report ( $n = 343$ ). (a)  $SUS^{\text{tfidf-}\ell}$  against  $SUS^{\text{dens}}$ : the term-weighting contrast. (b)  $SUS^{\text{tfidf-L2}}$  against  $SUS^{\text{dens}}$ : both factors varied. (c)  $SUS^{\text{tfidf-L2}}$  against BREADTH. Panel correlations are the Pearson coefficients of Table 15, rounded. Source: `scripts/figures_v3.py`.

$p_{\text{wild}} < 0.0001$ ) and  $-0.260$  for  $\text{ESGSI}^{\text{ext}}$  (SE 0.040,  $p_{\text{cl}} = 3.7 \times 10^{-8}$ ,  $p_{\text{wild}} < 0.0001$ ; Table 22). The section does not choose the specification, which §5.8 does on other grounds, and agreement between two variants is not evidence that either measures substance well.

## 6.1 The substance specification

The three specifications of §5.2 vary one factor at a time.  $SUS^{\text{dens}}$  and  $SUS^{\text{tfidf-}\ell}$  share the length denominator and differ in term weighting (uniform or idf);  $SUS^{\text{tfidf-}\ell}$  and  $SUS^{\text{tfidf-L2}}$  share idf weights and differ in vector normalisation (division by document length or scaling to unit L2 norm). BREADTH is reported as a diagnostic, not as a component. Table 15 gives the component correlations and Figure 1 the three scatters that matter.

The components correlate at 0.9876 across the weighting contrast, at 0.4823 across the vector-normalisation contrast and at 0.4052 when both change.  $SUS^{\text{tfidf-L2}}$  correlates more closely with BREADTH (0.9546) than with either other substance measure, as §5.8.2 derives from (21). Table 16 carries the comparison through to the score.

Idf weighting leaves the score's ranking almost unchanged ( $\rho = 0.9926$ ) and keeps 33 and 32 of the 34 reports in each extreme decile. L2 vector normalisation reorders it:  $\rho = 0.6421$ , and only 18 of the top 34 and 21 of the bottom 34 remain. The trend survives the substitution but is weaker: under  $SUS^{\text{tfidf-L2}}$  the within-firm slope is  $-0.143$  per year (SE 0.035,  $p_{\text{cl}} = 1.9 \times 10^{-4}$ ,  $p_{\text{wild}} = 0.0004$ ), against  $-0.181$ . We did not estimate the slope under  $SUS^{\text{tfidf-}\ell}$ . Each index-level correlation exceeds its component-level counterpart ( $r$  of 0.9933 against 0.9876; 0.6764 against 0.4052), because the compared scores share the  $Z(\text{SEN})$  term; the component correlations are

Table 16: ESGSI under each substance specification against the adopted ESGSI(SUS<sup>dens</sup>) ( $n = 343$ , components standardised by  $z$ -scores throughout). Extreme deciles: reports of the adopted top (bottom) 34 that are also in the top (bottom) 34 under the alternative. Source: `scripts/analysis_v3.py`.

| Specification                          | Factors varied          | Spearman $\rho$ | Pearson $r$ | Top/bottom decile |
|----------------------------------------|-------------------------|-----------------|-------------|-------------------|
| SUS <sup>dens</sup> (adopted)          | —                       | 1.0000          | 1.0000      | 34/34             |
| SUS <sup>tfidf-<math>\ell</math></sup> | weighting               | 0.9926          | 0.9933      | 33/32             |
| SUS <sup>tfidf-L2</sup>                | weighting, vector norm. | 0.6421          | 0.6764      | 18/21             |

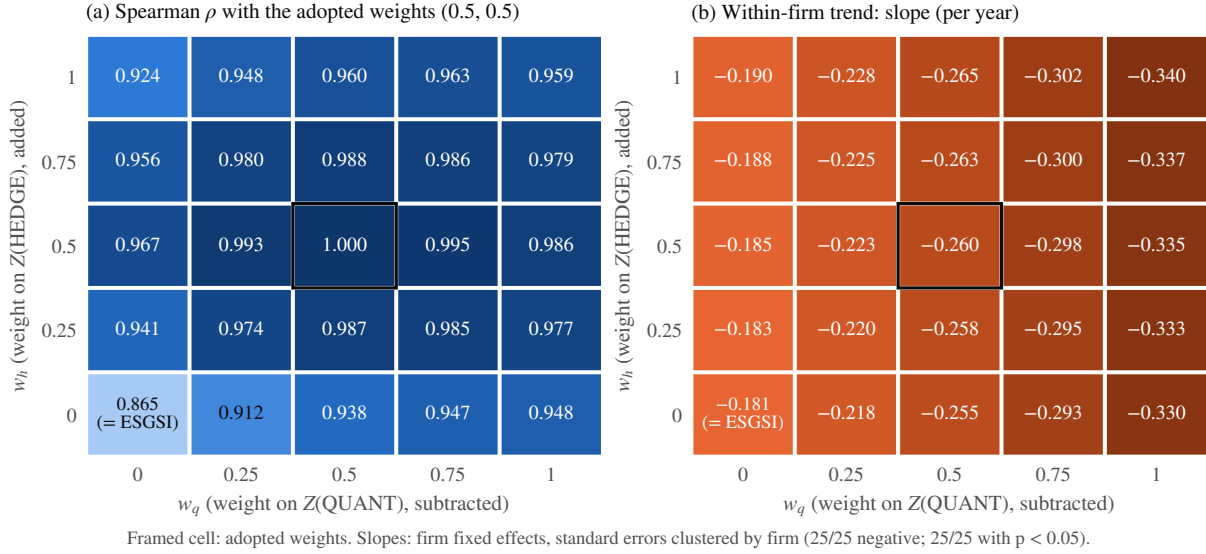


Figure 2: ESGSI<sup>ext</sup> over the weight grid ( $n = 343$ ). (a) Spearman  $\rho$  with the score at the adopted weights  $w_q = w_h = 0.5$  (framed); the cell  $w_q = w_h = 0$  is ESGSI. (b) Within-firm (fixed-effects) slope per year, standard errors clustered by firm. Source: `scripts/figures_v3.py`.

the stricter measure of how far the substance measurements agree. Term weighting is thus a second-order choice and vector normalisation a first-order one. The specifications are varied in ESGSI only; ESGSI<sup>ext</sup> is computed with SUS<sup>dens</sup> throughout.

## 6.2 The weights of the Extended ESGSI

Over the grid  $w_q, w_h \in \{0, 0.25, 0.5, 0.75, 1\}$  (§5.4), Figure 2 gives  $\rho$  with the adopted score and the within-firm slope, and Table 17 the extreme-decile overlaps.

Two things vary, and they depend on different weights. The *ranking* depends mostly on  $w_h$ . Holding  $w_q = 0.5$ ,  $\rho$  with the adopted score is 0.938 at  $w_h = 0$  and 0.960 at  $w_h = 1$ ; holding  $w_h = 0.5$ , it is 0.967 at  $w_q = 0$  and 0.986 at  $w_q = 1$ . The lowest value, 0.865, is at  $w_q = w_h = 0$ , where the score is ESGSI; the extreme deciles are least preserved there too (22/26); every other cell keeps at least 23 of the top and 27 of the bottom 34. Within 0.25 of the adopted weights on each axis  $\rho \geq 0.973$  and at least 28 of the top and 29 of the bottom 34 remain. The Spearman correlation with ESGSI falls from 1 at (0,0) to 0.701 at (1,1): at the far corner of the grid ESGSI<sup>ext</sup> orders the reports markedly differently from ESGSI.

The *trend* depends almost only on  $w_q$ . At  $w_h = 0.5$  the slope runs from  $-0.185$  ( $w_q = 0$ ) to  $-0.335$  ( $w_q = 1$ ), about  $-0.037$  per step of 0.25; at  $w_q = 0.5$  it runs from  $-0.255$  ( $w_h = 0$ ) to  $-0.265$  ( $w_h = 1$ ). QUANT rises over the panel and HEDGE does not (§7), so subtracting more of Z(QUANT) steepens the decline and adding more of Z(HEDGE) barely moves it. The

Table 17: Extreme-decile overlap of  $\text{ESGSI}^{\text{ext}}(w_q, w_h)$  with the adopted  $\text{ESGSI}^{\text{ext}}(0.5, 0.5)$ : reports of the adopted top/bottom 34 that remain in the top/bottom 34 ( $n = 343$ ). The cell (0, 0) is ESGSI. Source: `scripts/analysis_v3.py`.

| $w_q$ | $w_h$ |       |       |       |       |
|-------|-------|-------|-------|-------|-------|
|       | 0     | 0.25  | 0.5   | 0.75  | 1     |
| 0     | 22/26 | 25/28 | 30/29 | 27/28 | 24/28 |
| 0.25  | 23/28 | 28/30 | 32/31 | 30/30 | 23/28 |
| 0.5   | 26/29 | 31/32 | 34/34 | 30/30 | 24/29 |
| 0.75  | 28/28 | 31/31 | 33/32 | 31/29 | 25/27 |
| 1     | 29/28 | 30/29 | 31/28 | 30/29 | 26/27 |

slope is negative in all 25 cells, with firm-clustered standard errors between 0.032 and 0.052; the largest  $p_{\text{cl}}$ ,  $3.7 \times 10^{-6}$ , is that of ESGSI itself, and every 95 % interval excludes zero. The sign of the trend is therefore not a product of the weights; its magnitude in  $\text{ESGSI}^{\text{ext}}$  is, roughly in proportion to  $w_q$ .

### 6.3 Stability of the vocabulary

The standing objection to a dictionary measure is that the result belongs to the word list. We perturb our vocabulary of §4 in designed ways and at random, and read every designed deletion against random deletions of the same size.

**Frozen extraction and simulated re-extraction.** The vocabulary selects the text as well as scoring it (§4.4), so deleting entries changes both. Every perturbation is therefore run in two modes. In the *frozen* mode the extracted text is that of the adopted run and only the list that  $\text{SUS}^{\text{dens}}$  counts changes. This isolates the scoring effect, but paragraphs that were extracted only because of the deleted entries stay in the text, now unscored, and dilute the density of the reports that discuss those topics most. In the *re-extraction* mode the extraction rule of §3 is re-applied without the deleted entries, paragraphs that no longer qualify drop out, and SUS, SEN, QUANT and HEDGE are recomputed on what remains, as a real change of vocabulary would do.

Re-extraction is simulated from the extracted paragraphs, not from the PDFs. A subset of the vocabulary can only make fewer paragraphs hot, so the text it would extract is a subset of the adopted extraction, and re-applying the rule (hot paragraphs, one context paragraph on each side, merging, the 150-character minimum) to the extracted paragraphs reproduces it. Three approximations remain: each paragraph is lemmatised on its own rather than within its document; a shorter entry nested in a deleted one (*carbon* in *carbon footprint*) is not re-checked as a trigger, which slightly overstates the text lost; and QUANT and HEDGE are counted per paragraph. With the full vocabulary the simulation re-selects every extracted paragraph and reproduces the pipeline’s components at  $r = 0.99995$  (SUS), 0.999999 (SEN), 0.99994 (QUANT) and 0.99997 (HEDGE); ESGSI at  $r = 0.99997$  and  $\rho = 0.99991$  (largest absolute difference 0.091),  $\text{ESGSI}^{\text{ext}}$  at  $r = 0.99998$  and  $\rho = 0.99995$ ; its slopes are  $-0.180$  and  $-0.260$ . In both modes each perturbed score is compared with the simulation’s own full-vocabulary score.

`scripts/stability_v3_e2e.py` runs both modes over the 434 source entries (402 terms, 32 patterns; deleting a pattern changes nothing in the frozen mode). `scripts/stability_v3.py` runs the frozen mode over the 399 lemmatised entries against the pipeline’s own scores and adds single-entry deletion; where the two overlap they agree (without pillar E, frozen  $\rho$  0.7785 against 0.7761). `scripts/robustness_v3.py` adds firm-clustered and wild-bootstrap inference for the designed deletions.

Table 18: Designed deletions from the vocabulary (434 entries), against the full-vocabulary score of each mode. Text kept: share of extracted words that survive re-extraction. Top/bot: reports of the reference top/bottom 34 that remain. Slope: within-firm slope per year (full vocabulary:  $-0.180$  for ESGSI,  $-0.260$  for ESGSI<sup>ext</sup>). Null: median  $\rho$  over 1,000 random deletions of the same number of entries, and the percentile of the designed deletion’s  $\rho$  among them (re-extraction). Source: `scripts/stability_v3_e2e.py`.

| Deleted                             | Entries | Text kept | Re-extraction |         |        | Null   |       | Frozen |         |        |
|-------------------------------------|---------|-----------|---------------|---------|--------|--------|-------|--------|---------|--------|
|                                     |         |           | $\rho$        | Top/bot | Slope  | $\rho$ | Pct.  | $\rho$ | Top/bot | Slope  |
| <i>Panel A: ESGSI</i>               |         |           |               |         |        |        |       |        |         |        |
| Sectoral                            | 36      | 99.6      | 0.9995        | 33/33   | −0.176 | 0.9945 | 100.0 | 0.9987 | 34/34   | −0.184 |
| With caveat                         | 6       | 95.2      | 0.9912        | 32/32   | −0.180 | 0.9996 | 0.7   | 0.9908 | 33/34   | −0.195 |
| Pillar E                            | 168     | 81.1      | 0.8844        | 28/23   | −0.105 | 0.9513 | 0.2   | 0.7761 | 28/12   | −0.080 |
| Pillar S                            | 139     | 90.1      | 0.9836        | 32/30   | −0.168 | 0.9638 | 97.0  | 0.9839 | 32/30   | −0.190 |
| Pillar G                            | 95      | 68.6      | 0.9439        | 28/30   | −0.206 | 0.9804 | 0.4   | 0.9353 | 27/30   | −0.214 |
| Pillar TRANS                        | 32      | 94.8      | 0.9822        | 31/32   | −0.132 | 0.9952 | 3.1   | 0.9739 | 33/32   | −0.117 |
| <i>Board of directors</i>           | 1       | 93.6      | 0.9953        | 33/31   | −0.188 | 1.0000 | 0.5   | 0.9846 | 33/32   | −0.194 |
| <i>Panel B: ESGSI<sup>ext</sup></i> |         |           |               |         |        |        |       |        |         |        |
| Sectoral                            | 36      | 99.6      | 0.9994        | 33/32   | −0.256 | 0.9964 | 99.5  | 0.9991 | 34/33   | −0.264 |
| With caveat                         | 6       | 95.2      | 0.9940        | 30/31   | −0.255 | 0.9997 | 0.6   | 0.9950 | 33/31   | −0.275 |
| Pillar E                            | 168     | 81.1      | 0.9088        | 30/26   | −0.211 | 0.9633 | 0.5   | 0.8559 | 29/21   | −0.159 |
| Pillar S                            | 139     | 90.1      | 0.9886        | 31/31   | −0.253 | 0.9734 | 97.3  | 0.9866 | 31/31   | −0.269 |
| Pillar G                            | 95      | 68.6      | 0.9622        | 27/29   | −0.250 | 0.9860 | 0.3   | 0.9633 | 27/27   | −0.294 |
| Pillar TRANS                        | 32      | 94.8      | 0.9832        | 32/32   | −0.205 | 0.9969 | 0.6   | 0.9824 | 32/31   | −0.197 |
| <i>Board of directors</i>           | 1       | 93.6      | 0.9965        | 33/33   | −0.268 | 1.0000 | 0.6   | 0.9903 | 34/30   | −0.274 |

**Designed deletions.** Table 18 removes, in turn, the 36 sectoral entries, six entries admitted with a caveat, each pillar, and the single entry *board of directors*. The six are `sustainab\w+`, *sustainable*, *governance*, *audit committee*, *tcfd* and *hydrogen*: they meet the criteria of §4.2, but a visible part of their occurrences is navigation text the filter misses, legal templates, or the non-ESG sense of *sustainable* (*sustainable growth*). The null percentile is the share of the 1,000 matched random deletions whose  $\rho$  is no higher than the designed deletion’s: a low value means the designed deletion moves the score more than a random one of its size.

*Sectoral entries.* Excluding them, the sensitivity that §4 announces, leaves ESGSI at  $\rho = 0.9995$  under re-extraction (0.9987 frozen), keeps 33 of each extreme decile (all 34 frozen) and the slope at  $-0.176$  ( $-0.184$  frozen). No random deletion of 36 entries moves the score as little. That is a matter of mass, not of harmlessness: in the frozen programme the 34 sectoral lemmas hold 1.0 % of vocabulary hits, against 8.5 % on average for 34 lemmas drawn at random.

*Entries admitted with a caveat.* Removing the six gives  $\rho = 0.9912$ , keeps 32 of each extreme decile and leaves the slope at  $-0.180$  ( $-0.195$  frozen). They move the score more than 99 % of random six-entry deletions for the mirror reason: the five lemmatised ones hold 9.3 % of vocabulary hits, against 1.3 % for five random lemmas.

*Pillars.* The environmental pillar matters most. Without it  $\rho = 0.8844$  under re-extraction, 28 of the top 34 remain but only 23 of the bottom 34, and the slope falls to  $-0.105$  per year (SE 0.039,  $p_{cl} = 0.009$ ,  $p_{wild} = 0.011$ ), about 58 % of its full-vocabulary value; frozen,  $-0.080$  ( $p_{cl} = 0.027$ ,  $p_{wild} = 0.031$ ). Unlike the two previous sets, E is not heavy for its size (its lemmas hold 38.2 % of hits, against 38.0 % for 152 random lemmas), yet its  $\rho$  lies below all but 0.2 % of matched random deletions: the environmental entries order the reports differently from the rest of the vocabulary, and the reports they order are chiefly those at the bottom of the score, the ones whose substance is high relative to their tone. Without TRANS the slope is  $-0.132$  (SE 0.036,  $p_{cl} = 5.5 \times 10^{-4}$ ,  $p_{wild} = 0.0006$ ) at  $\rho = 0.9822$ ; these two pillars are the ones that change the trend’s magnitude. Pillar G removes the most text (31.4 % of extracted words) and lowers

Table 19: Random deletion of vocabulary entries, 1,000 draws per size, against the full-vocabulary score of each mode.  $\rho$ : median and 5th–95th percentiles. Top/bot: mean number of the reference top/bottom 34 that remain. Slope: median within-firm slope per year and 5th–95th percentiles (full vocabulary  $-0.180$  for ESGSI,  $-0.260$  for ESGSI<sup>ext</sup>). Text kept: mean share of extracted words surviving re-extraction (%). Source: `scripts/stability_v3_e2e.py`.

| Deleted                       | Entries | Text kept | Re-extraction |                |           |        |                  |        | Frozen         |           |        |                  |  |  |
|-------------------------------|---------|-----------|---------------|----------------|-----------|--------|------------------|--------|----------------|-----------|--------|------------------|--|--|
|                               |         |           | $\rho$        | Top/bot        |           | Slope  |                  | $\rho$ | Top/bot        |           | Slope  |                  |  |  |
| Panel A: ESGSI                |         |           |               |                |           |        |                  |        |                |           |        |                  |  |  |
| 10 %                          | 43      | 96.3      | 0.993         | [0.982, 0.998] | 32.4/31.8 | −0.174 | [−0.192, −0.150] | 0.994  | [0.971, 0.999] | 32.9/32.0 | −0.179 | [−0.200, −0.152] |  |  |
| 20 %                          | 87      | 91.8      | 0.982         | [0.964, 0.993] | 31.2/30.8 | −0.165 | [−0.194, −0.132] | 0.981  | [0.955, 0.994] | 31.9/30.7 | −0.175 | [−0.208, −0.137] |  |  |
| 30 %                          | 130     | 86.6      | 0.967         | [0.942, 0.985] | 29.8/29.7 | −0.154 | [−0.192, −0.107] | 0.966  | [0.939, 0.988] | 31.0/29.5 | −0.172 | [−0.217, −0.121] |  |  |
| 40 %                          | 174     | 80.8      | 0.950         | [0.912, 0.971] | 28.5/28.1 | −0.139 | [−0.181, −0.082] | 0.950  | [0.912, 0.977] | 30.0/28.0 | −0.167 | [−0.218, −0.102] |  |  |
| 50 %                          | 217     | 74.0      | 0.924         | [0.872, 0.956] | 27.0/26.0 | −0.118 | [−0.172, −0.057] | 0.932  | [0.878, 0.962] | 29.1/26.4 | −0.160 | [−0.222, −0.083] |  |  |
| Panel B: ESGSI <sup>ext</sup> |         |           |               |                |           |        |                  |        |                |           |        |                  |  |  |
| 10 %                          | 43      | 96.3      | 0.995         | [0.988, 0.999] | 32.4/31.8 | −0.253 | [−0.272, −0.226] | 0.996  | [0.985, 0.999] | 32.8/31.5 | −0.258 | [−0.279, −0.232] |  |  |
| 20 %                          | 87      | 91.8      | 0.987         | [0.974, 0.995] | 31.3/30.9 | −0.242 | [−0.273, −0.206] | 0.988  | [0.974, 0.996] | 31.9/30.4 | −0.255 | [−0.287, −0.216] |  |  |
| 30 %                          | 130     | 86.6      | 0.976         | [0.956, 0.989] | 30.2/30.0 | −0.231 | [−0.273, −0.180] | 0.979  | [0.962, 0.992] | 30.9/29.5 | −0.252 | [−0.297, −0.200] |  |  |
| 40 %                          | 174     | 80.8      | 0.961         | [0.932, 0.979] | 29.2/29.1 | −0.214 | [−0.264, −0.155] | 0.969  | [0.942, 0.984] | 30.1/29.0 | −0.246 | [−0.298, −0.182] |  |  |
| 50 %                          | 217     | 74.0      | 0.940         | [0.897, 0.967] | 27.9/28.0 | −0.192 | [−0.254, −0.122] | 0.957  | [0.914, 0.975] | 29.2/28.3 | −0.240 | [−0.302, −0.163] |  |  |

$\rho$  to 0.9439, below all but 0.4 % of the null, while steepening the slope to  $-0.206$ ; its lemmas hold 30.0 % of hits against 22.6 % for a random set of the same size. Pillar S is the exception in the other direction:  $\rho = 0.9836$ , above 97 % of the null, because it holds 17.6 % of hits against 31.7 %.

*Board of directors.* The heaviest entry (§4.7, Table 10) is also an extraction trigger: removing it cuts the extracted text by 6.4 %. Under re-extraction  $\rho = 0.9953$ , with 33 and 31 of the extreme deciles kept, and the slope steepens slightly to  $-0.188$ ; frozen,  $\rho = 0.9846$  and the slope is  $-0.194$  (SE 0.034,  $p_{cl} = 7.4 \times 10^{-7}$ ,  $p_{wild} < 0.0001$ ). One random entry holds 0.24 % of hits on average against its 5.7 %, so its percentile (0.5) says only that it is heavy.

Apart from pillars E and TRANS, every designed deletion leaves the ESGSI slope with  $p_{cl} < 10^{-4}$  in both modes; every designed deletion, E and TRANS included, leaves the ESGSI<sup>ext</sup> slope so. ESGSI<sup>ext</sup> moves less than ESGSI under each pillar deletion, as  $Z(\text{QUANT})$  and  $Z(\text{HEDGE})$  depend on the vocabulary only through the extracted text: without E its  $\rho$  is 0.9088 and its slope  $-0.211$  ( $p_{cl} = 6.5 \times 10^{-7}$ ).

*Frozen against re-extracted.* The modes agree except for pillar E, where freezing the text exaggerates the effect ( $\rho$  0.7761 against 0.8844; 12 of the bottom 34 kept against 23) through the dilution described above, and *board of directors*, where re-extraction also removes the governance text the entry brought into the zones.

**Random deletion.** Table 19 and Figure 3 give 1,000 random deletions each of 10–50 % of the 434 entries, in both modes.

The ranking degrades smoothly and slowly (Figure 3a). A random tenth of the entries costs a median  $\rho$  of about 0.007; half of them, about 0.08 under re-extraction (median 0.924) and 0.07 frozen (0.932), and even then the 5th percentile is 0.872. The extreme deciles degrade faster (panel b): at half the vocabulary about 27 of the top 34 and 26 of the bottom 34 remain under re-extraction, and for ESGSI the bottom decile is the less stable of the two at every size in both modes. Against the 5th–95th range of their matched nulls the designed deletions fall into three groups: the sectoral entries and pillar S above it; the entries with a caveat, *board of directors* and TRANS just below it; G and above all E well below it, E being the only designed deletion whose  $\rho$  and mean decile overlap are lower than the median for a random half of the vocabulary.

The slope is the least stable of the three measures (panel c), and its bands are wide. Under re-extraction the median drifts towards zero as more is deleted, from  $-0.174$  at 10 % to  $-0.118$  at 50 %, because the remaining text is less and less the text on which the full score was computed; frozen, the median stays between  $-0.179$  and  $-0.160$ . The 5th–95th band is already

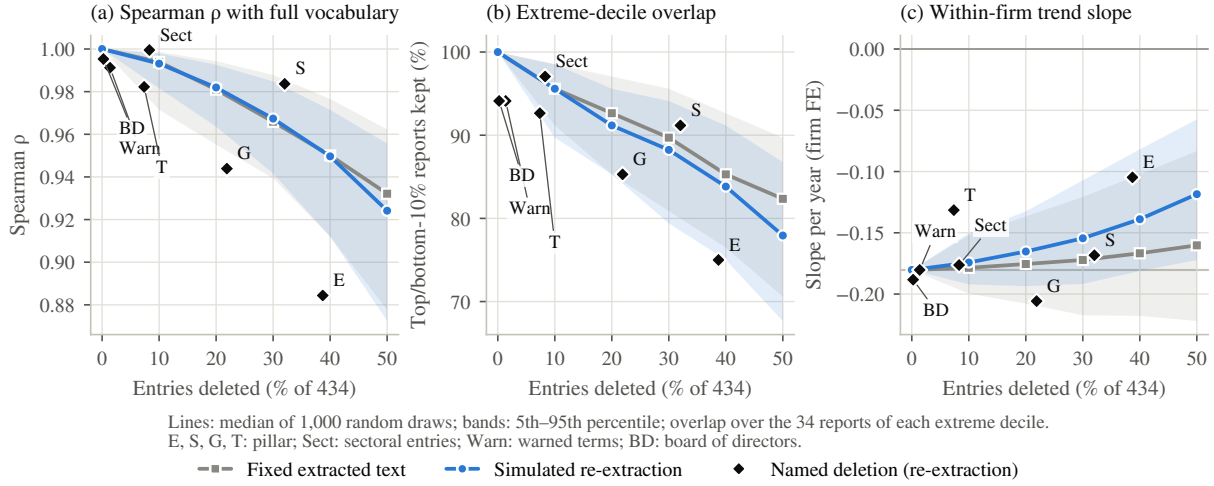


Figure 3: Stability of ESGSI under deletion of vocabulary entries ( $n = 343$ ). Lines: median over 1,000 random deletions at each size, simulated re-extraction (blue) and frozen extracted text (grey), both starting from the full vocabulary at 0 %; bands: 5th–95th percentiles. Diamonds: the designed deletions of Table 18 under re-extraction, placed at the share of the 434 entries they remove (Warn: entries admitted with a caveat; BD: *board of directors*). (a) Spearman  $\rho$  with the full-vocabulary score. (b) Mean of the top- and bottom-decile overlaps, as a percentage of the 34 reports in each. (c) Within-firm slope per year; the horizontal line is the full-vocabulary slope. Source: `scripts/figures_v3.py`.

$[-0.192, -0.150]$  for a random tenth and  $[-0.172, -0.057]$  for a random half under re-extraction ( $[-0.222, -0.083]$  frozen), a spread of a factor of three in magnitude. The designed deletions sit inside or near these bands, with TRANS above (less negative) and G below (more negative) the re-extraction band at their sizes. The sign is nonetheless stable: of the 5,000 random draws in each mode, the ESGSI slope is negative in all but one (a 50 % deletion under re-extraction), and the  $\text{ESGSI}^{\text{ext}}$  slope in all.  $\text{ESGSI}^{\text{ext}}$  keeps its ranking better at every size (Table 19, Panel B): at half the vocabulary its median  $\rho$  is 0.940 and its median slope  $-0.192$ .

**Single entries.** Deleting each of the 399 lemmatised entries in turn, frozen, gives  $\rho \geq 0.9756$  for ESGSI and  $\geq 0.9884$  for  $\text{ESGSI}^{\text{ext}}$ , keeps at least 31 of each extreme decile of ESGSI (29 of the top and 30 of the bottom of  $\text{ESGSI}^{\text{ext}}$ ), and leaves both extreme deciles of ESGSI unchanged for 366 of the 399 entries (319 for  $\text{ESGSI}^{\text{ext}}$ ). The within-firm slope of ESGSI stays between  $-0.194$  and  $-0.162$ , and that of  $\text{ESGSI}^{\text{ext}}$  between  $-0.274$  and  $-0.241$ . The largest movers by  $\rho$  are the heaviest entries: *remuneration* ( $\rho = 0.9756$ ; 31/33 of the extreme deciles kept; 5.6 % of hits), *board director*, the lemma of *board of directors* (0.9847; 33/31; 5.7 %), *sustainability* (0.9948; 34/34), *governance* (0.9955; 33/34) and *emission* (0.9961; 34/31). The slope moves most without *taxonomy* ( $-0.162$ ) and *sustainability* ( $-0.164$ ), and steepens most without *board director* ( $-0.194$ ).

## 6.4 What the checks establish

1. *The vocabulary was built on this corpus.* Candidates, admission criteria and the sectoral designation were all applied to the 343 reports (§4.10). The perturbations are of a vocabulary fitted to the sample, not of an independent one, and the stability they show is stability around that fit.
2. *High correlations are partly structural.* A deletion leaves the retained entries common to both scores, so the correlation has a floor. With  $t$  the ratio of the standard deviations

of the per-report densities of the deleted and the retained entries, the component-level floor is  $\sqrt{\max(0, 1 - t^2)}$ , carried to the index through the common  $Z(\text{SEN})$  term (frozen text). The sectoral deletion ( $t = 0.053$ ) gives  $r = 0.9993$  against a floor of 0.9992, *board of directors* ( $t = 0.192$ ) 0.9902 against 0.9900, pillar G 0.9456 against 0.9449; pillar E ( $t = 1.261$ ) 0.8064 against 0.4446. A high correlation after a small deletion records how little the deleted entries weigh, which is why each deletion is read against random deletions of the same size.

3. *Stability is not validity.* The checks bound how far the scores depend on the choices varied. Every designed deletion except pillar E leaves  $\rho \geq 0.935$  and at least 27 of each extreme decile; the sign of the within-firm trend survives every designed deletion, every weight pair and all but one of the 10,000 random draws. The trend’s magnitude is less secure: it depends on the environmental and TRANS entries, on how much of the vocabulary is kept, and in  $\text{ESGSI}^{\text{ext}}$  on  $w_q$ . None of this shows that the vocabulary measures substance; that question belongs to §4.10 and §5.8.3.

## 7 Results: time, firms, industries and countries

This section reports both scores on the panel of §2: ESGSI, the main score, and  $\text{ESGSI}^{\text{ext}}$ , the extended score this paper proposes (§5.4). Both are continuous and corpus-relative: each component is a population  $z$ -score over the 343 reports (3), so a value is a position relative to the corpus mean, in corpus standard deviations, whose sign says only on which side of that mean a report lies. No threshold is applied and no report is classified. Firms enter only as anonymous positions, and firm-level heterogeneity is reported as distributions by industry.

The specification is the adopted one:  $\text{SUS} = \text{SUS}^{\text{dens}}$ , our vocabulary with its sectoral entries ( $V = 399$ , §4), and  $w_q = w_h = 0.5$  in  $\text{ESGSI}^{\text{ext}}$ , whose weight sensitivity is in §6. Inference follows §5.6: firm-clustered standard errors (49 clusters,  $t_{48}$ ) and, for fixed-effects slopes, a restricted wild cluster bootstrap; conventional OLS  $p$ -values appear only beside them, labelled. Two scores are compared by Pearson  $r$ , Spearman  $\rho$  and the overlap of their extreme deciles (the 34 highest and the 34 lowest of 343 reports). Values come from `scripts/analysis_v3.py`, figures from `scripts/figures_v3.py`; industries are ordered by mean ESGSI throughout.

### 7.1 Components and scores in the cross-section

Table 20 describes the inputs and both scores, which have mean zero by construction. ESGSI has median  $-0.094$  and a longer right tail than left;  $\text{ESGSI}^{\text{ext}}$  is more dispersed (SD 1.708 against 1.364), with the longer left tail. The two agree closely but not exactly ( $r = 0.891$ ,  $\rho = 0.865$ ; 22 of the 34 highest and 26 of the 34 lowest reports shared): quantification and hedging move a sizeable minority of reports across the extremes.

### 7.2 The yearly series

The panel is balanced: the same 49 firms report in every year. Mean ESGSI falls in every year, from 0.502 in 2018 to  $-0.610$  in 2024. Mean  $\text{ESGSI}^{\text{ext}}$  also falls in every year, from 0.696 to  $-0.802$  (Table 21, Figure 4). The yearly intervals are wide, because firms differ far more than years (§7.5). For ESGSI, the 2018 interval is  $[0.073, 0.930]$  and the 2024 interval is  $[-1.039, -0.181]$ .

Table 22 estimates the trend. For ESGSI the firm fixed-effects slope is  $-0.181$  per year, equal to the pooled OLS slope as it must be in a balanced panel (§5.6); clustered by firm, its 95% interval is  $[-0.250, -0.111]$ ,  $p_{\text{cl}} = 3.7 \times 10^{-6}$ , and no wild bootstrap replication reached the observed statistic ( $p_{\text{wild}} < 0.0001$ ). For  $\text{ESGSI}^{\text{ext}}$  the slope is steeper,  $-0.260$  ( $[-0.340, -0.180]$ ),

Table 20: Distribution of the components and scores,  $n = 343$ .  $SUS^{dens}$  per 100 lemmatised tokens;  $BREADTH$  in effective distinct entries;  $QUANT$  and  $HEDGE$  as shares of raw tokens and words.  $SD$  is the sample standard deviation; the  $z$ -scores entering both scores use the population one (3).

|               | Mean   | SD     | Min    | P25    | Median | P75    | Max    |
|---------------|--------|--------|--------|--------|--------|--------|--------|
| $SUS^{dens}$  | 6.592  | 1.873  | 2.432  | 5.427  | 6.536  | 7.718  | 13.956 |
| $BREADTH$     | 53.24  | 17.11  | 8.32   | 41.19  | 54.27  | 65.04  | 94.96  |
| $SEN$         | -0.004 | 0.135  | -0.345 | -0.095 | -0.014 | 0.077  | 0.591  |
| $QUANT$       | 0.0061 | 0.0031 | 0.0002 | 0.0040 | 0.0060 | 0.0076 | 0.0332 |
| $HEDGE$       | 0.0187 | 0.0038 | 0.0083 | 0.0160 | 0.0179 | 0.0208 | 0.0336 |
| $ESGSI$       | 0.000  | 1.364  | -3.939 | -0.765 | -0.094 | 0.777  | 5.356  |
| $ESGSI^{ext}$ | 0.000  | 1.708  | -8.055 | -0.926 | -0.082 | 0.980  | 4.902  |

Table 21: Mean scores and standardised components by year; 49 reports per year. All  $z$ -scores are taken over the pooled 343 reports, so every row has mean zero over the window. The two alternative substance specifications (§5.2) are shown for reference (§7.8).

|                           | 2018   | 2019   | 2020   | 2021   | 2022   | 2023   | 2024   |
|---------------------------|--------|--------|--------|--------|--------|--------|--------|
| $ESGSI$                   | 0.502  | 0.443  | 0.110  | -0.018 | -0.128 | -0.298 | -0.610 |
| $ESGSI^{ext}$             | 0.696  | 0.585  | 0.258  | 0.038  | -0.187 | -0.587 | -0.802 |
| $Z(SEN)$                  | 0.052  | 0.005  | -0.103 | -0.090 | -0.043 | 0.011  | 0.169  |
| $Z(SUS^{dens})$           | -0.450 | -0.438 | -0.213 | -0.072 | 0.085  | 0.308  | 0.779  |
| $Z(QUANT)$                | -0.402 | -0.245 | -0.276 | -0.073 | 0.123  | 0.522  | 0.351  |
| $Z(HEDGE)$                | -0.012 | 0.040  | 0.019  | 0.039  | 0.005  | -0.057 | -0.034 |
| $ESGSI(SUS^{tfidf-\ell})$ | 0.522  | 0.455  | 0.137  | 0.004  | -0.118 | -0.302 | -0.697 |
| $ESGSI(SUS^{tfidf-L2})$   | 0.480  | 0.404  | 0.082  | -0.153 | -0.258 | -0.257 | -0.298 |

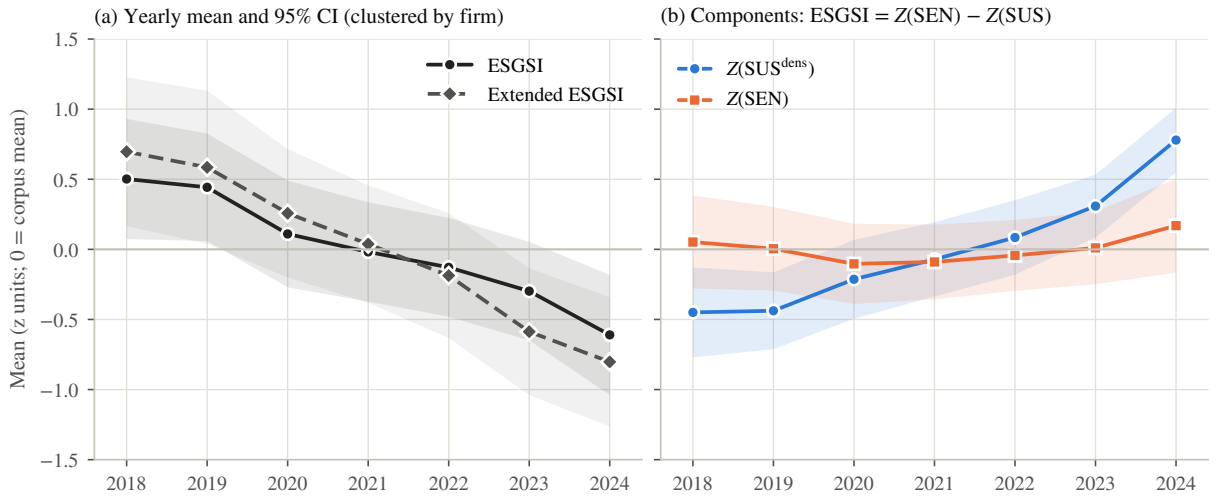


Figure 4: (a) Mean  $ESGSI$  and  $ESGSI^{ext}$  by year, 95% intervals from year indicators clustered by firm. (b) The same for  $Z(SUS^{dens})$  and  $Z(SEN)$ , whose difference is  $ESGSI$ .

Table 22: Trend regressions on calendar year, 343 reports of 49 firms (Germany: 98 of 14). Firm fixed effects, standard errors clustered by firm ( $t_{48}$ ;  $t_{13}$  for Germany) and restricted wild cluster bootstrap ( $B = 9,999$ ).  $p_{OLS}$ : conventional pooled OLS, which treats a firm’s seven reports as independent, for comparison only. QUANT and HEDGE in units of  $10^{-4}$  per year;  $SUS^{dens}$  in mentions per 100 tokens per year.

|                                | Slope  | SE <sub>cl</sub> | 95% interval      | $p_{cl}$              | $p_{wild}$           | $p_{OLS}$             |
|--------------------------------|--------|------------------|-------------------|-----------------------|----------------------|-----------------------|
| ESGSI                          | -0.181 | 0.035            | [-0.250, -0.111]  | $3.7 \times 10^{-6}$  | $< 0.0001$           | $6.3 \times 10^{-7}$  |
| ESGSI <sup>ext</sup>           | -0.260 | 0.040            | [-0.340, -0.180]  | $3.7 \times 10^{-8}$  | $< 0.0001$           | $8.0 \times 10^{-9}$  |
| SEN                            | 0.0020 | 0.0037           | [-0.0053, 0.0094] | 0.581                 | 0.577                | 0.577                 |
| $SUS^{dens}$                   | 0.366  | 0.041            | [0.283, 0.449]    | $1.1 \times 10^{-11}$ | $< 0.0001$           | $5.4 \times 10^{-14}$ |
| QUANT ( $\times 10^{-4}$ )     | 4.7    | 0.9              | [2.9, 6.5]        | $3.4 \times 10^{-6}$  | $< 0.0001$           | $1.6 \times 10^{-8}$  |
| HEDGE ( $\times 10^{-4}$ )     | -0.4   | 0.8              | [-1.9, 1.2]       | 0.633                 | 0.634                | 0.719                 |
| ESGSI, Germany                 | -0.268 | 0.070            | [-0.419, -0.116]  | 0.002                 | 0.003                | $1.6 \times 10^{-5}$  |
| ESGSI <sup>ext</sup> , Germany | -0.376 | 0.062            | [-0.511, -0.241]  | $4.3 \times 10^{-5}$  | $3.0 \times 10^{-4}$ | $7.7 \times 10^{-8}$  |

Table 23: Driver decomposition, 2018 against 2024; means over the 49 reports of each year. Extracted text is in lemmatised tokens, the denominator of  $SUS^{dens}$ . A mention is one vocabulary hit. Distinct entries are out of  $V = 399$ .

| Quantity                                          | 2018    | 2024    | Change  |
|---------------------------------------------------|---------|---------|---------|
| Extracted ESG text per report (lemmatised tokens) | 30,174  | 65,950  | +118.6% |
| ESG mentions per report                           | 1,760.7 | 5,371.8 | +205.1% |
| Distinct vocabulary entries present per report    | 129.2   | 205.1   | +58.7%  |
| $SUS^{dens}$ (mentions per 100 tokens)            | 5.75    | 8.05    | +40.0%  |
| BREADTH (effective distinct entries)              | 44.64   | 62.24   | +39.4%  |
| SEN                                               | 0.003   | 0.019   | +0.016  |
| QUANT                                             | 0.0048  | 0.0072  | +48.6%  |
| HEDGE                                             | 0.0186  | 0.0185  | -0.4%   |

with the same bootstrap result. Year explains little of either score: pooled  $R^2$  0.070 and 0.093, within-firm  $R^2$  0.211 and 0.295.

Three qualifications attach. The slope summarises an uneven path, not a constant annual rate. The zero crossing between 2020 and 2021 is not an event: both scores have mean zero over the window by construction. And a shock common to all firms in one year is not covered, since seven years cannot be clustered on; the standard error measures dispersion across firms around the yearly means, not uncertainty about which years were drawn (§5.6).

### 7.3 What moves, and what does not

Since  $ESGSI = Z(\text{SEN}) - Z(\text{SUS})$ , a decline means tone falling or substance rising, and it is substance (Figure 4b, Table 23). The first five quantities of the table rise in every year: mentions outgrow a denominator that more than doubles, and spread over more of the vocabulary. SEN has no detectable trend (+0.0020 per year,  $p_{cl} = 0.581$ ,  $p_{wild} = 0.577$ ); its yearly means dip in 2020 and rise in 2024, which works against the decline. A failure to reject is not a demonstrated null, and it answers one objection only, that a fixed sentiment dictionary scores 2024 language differently from 2018 language; for the ESG vocabulary, which carries the decline, the equivalent objection is open (§7.4).

$ESGSI^{ext}$  subtracts  $0.5 Z(\text{QUANT})$  and adds  $0.5 Z(\text{HEDGE})$ , and its steeper decline comes from quantification. QUANT rises by half over the window, not in every year (its standardised mean peaks in 2023), with a clearly positive slope ( $p_{cl} = 3.4 \times 10^{-6}$ ); HEDGE is flat ( $p_{cl} = 0.633$ ), its standardised yearly means within  $\pm 0.06$ . The **frameworks** family of QUANT (Table 11) names

Table 24: Trend with regime vocabulary removed from the scoring vocabulary, extraction held fixed. Mass: share of all mentions held by the deleted entries.  $SUS^{\text{dens}}$ : mean per report, 2018  $\rightarrow$  2024.  $\rho$  and extreme-decile overlap (of 34) are against ESGSI with the full vocabulary. Slopes: firm fixed effects per year;  $p$ -values as in Table 22. Removing both sets (368 entries remain, mass 14.3%) gives  $\rho = 0.973$  and slopes of  $-0.116$  (ESGSI) and  $-0.196$  (ESGSI<sup>ext</sup>).

| Vocabulary      | Entries | Mass  | $SUS^{\text{dens}}$     | $\rho$ | Top/bottom | ESGSI  |                      | ESGSI <sup>ext</sup> |                      |
|-----------------|---------|-------|-------------------------|--------|------------|--------|----------------------|----------------------|----------------------|
|                 |         |       |                         |        |            | Slope  | $p_{\text{cl}}$      | Slope                | $p_{\text{cl}}$      |
| Full            | 399     | —     | 5.75 $\rightarrow$ 8.05 | 1.000  | 34/34      | -0.181 | $3.7 \times 10^{-6}$ | -0.260               | $3.7 \times 10^{-8}$ |
| No regime names | 377     | 3.8%  | 5.65 $\rightarrow$ 7.50 | 0.992  | 34/34      | -0.149 | $8.9 \times 10^{-5}$ | -0.228               | $4.9 \times 10^{-7}$ |
| No TRANS pillar | 369     | 14.2% | 5.12 $\rightarrow$ 6.53 | 0.973  | 33/32      | -0.117 | 0.002                | -0.196               | $1.0 \times 10^{-5}$ |

regimes, so the rise in QUANT is exposed to the same chronology objection as SUS. Both scores register words, not activity.

## 7.4 Alternative explanations and the mandate confound

**Composition, length, extraction, country.** The balanced panel excludes drift in the set of firms and in the country mix by construction, and the document-type mix shifts only through the six firms that file more than one type (Table 3); it selects continuous reporters, which restricts external validity without generating an internal trend. A report that grows without adding ESG content lowers  $SUS^{\text{dens}}$ , so the rise is net of any such dilution. The extraction yield rises in every year, from 31.6% to 48.4% (Table 7), but context paragraphs enter the zones under no density requirement (§3), so marginal text should dilute  $SUS^{\text{dens}}$  rather than inflate it — an argument, not a measurement. Within Germany neither country nor document type varies, and the slope is steeper than the pooled one for both scores: a trend of this sign exists where the confound is absent, which does not show the confound absent from the pooled slope and says nothing about what acts on all countries at once.

**The confound this design cannot remove.** Over 2018–2024 European non-financial reporting moved from largely voluntary to substantially prescribed. A firm that adds a taxonomy-alignment table, a framework index and a standards-referenced sustainability statement adds ESG vocabulary in quantity without necessarily changing anything it does, and our numerator cannot tell the two apart. The vocabulary names regimes that did not exist or were not in force in 2018 (*csrd*, *esrs*, the *taxonomy* family, *sfd*, *issb*), and the transversal pillar that holds them nearly doubles its share of all mentions, from 9.9% to 18.6% (Figure 8b).

The purely definitional part can be bounded by deleting entries and rescoreing, with the extraction held fixed (Table 24). The first deletion removes an explicit list of 22 regime, standard and rating names (the *csrd*, *esrs*, *taxonomy*, *sfd* and *double materiality* family, reporting standards and initiatives from *global reporting initiative* to *tnfd*, and rating providers), deliberately over-inclusive (GRI, SASB, CDP and the Global Compact predate the window) so that it bounds the definitional component from above. The second removes the whole TRANS pillar, 30 entries, which holds 21 of the 22 names (*science based target* is filed under E) and also *sustainability*, *sustainable* and *esg*.

Striking the regime names leaves an ESGSI slope of  $-0.149$ , about four fifths of the original; striking the whole pillar leaves  $-0.117$ , about two thirds ( $p_{\text{wild}} = 1.0 \times 10^{-4}$  and 0.003). ESGSI<sup>ext</sup> keeps three quarters or more of its slope in both cases. The trend is not an artefact of counting regime names.

The check does *not* dispose of the mandate confound. A regime elicits substantive disclosure as well as its own acronyms — emissions, transition plans, most of the environmental vocabulary — in the words a firm that had changed its conduct would also use, and the environmental

pillar is where the rise sits (§7.7). The deletions bound the definitional component only; nothing here measures how much of the remaining slope the mandates produced. Breadth is a weak counterweight: a compliance template predicts more mentions on few entries, and we observe broadening instead, but the mandates prescribe reporting across all three pillars at once. A mandate is a common time shock, collinear with calendar year and outside what firm-clustered inference covers; the balanced panel, the German subsample and the fixed effects remove cross-sectional composition, not anything that moves with time. Identification would need variation in exposure (firms outside scope, staggered application), which 49 large European issuers, all in scope, do not provide. The confound is therefore left explicitly unresolved.

What the series identifies is that ESG-lexical content per unit of extracted text rose while tone did not. Whether the added text reflects changed conduct, compliance with a new regime, or unchanged conduct newly described in a prescribed vocabulary is not identified, and this design cannot rank the three. One property belongs to the architecture: both scores fall when substance rises, so a firm that adds a mandated taxonomy table moves down the scale for having complied, in any use of this index family across a period of regulatory change.

## 7.5 Firms

Most of what the scores record is a difference between firms, not between years. For ESGSI, 66.7% of the total sum of squares lies between firm means and 7.2% between year means, and the intraclass correlation of a firm’s seven reports, net of year, is 0.676 (ESGSI<sup>ext</sup>: 68.5%, 9.4%, 0.720). The standard deviation of the 49 firm means of ESGSI is 1.123, against a mean within-firm standard deviation of 0.753; a firm’s yearly values span 2.09 on average (2.53 for ESGSI<sup>ext</sup>), more than the distance between the 2018 and 2024 yearly means. The trend is real in the sense of Table 22, and small beside this.

Positions persist. The Spearman correlation of firms’ ESGSI between consecutive years runs from 0.728 (2023–2024) to 0.915 (2021–2022), mean 0.849, but is 0.456 between 2018 and 2024; for ESGSI<sup>ext</sup>, 0.672 to 0.933 (mean 0.823) and 0.393. Ranks are stable from one year to the next and reshuffle substantially over the window. Across firms the two scores agree about as well as across reports: firm means correlate  $r = 0.887$  ( $\rho = 0.848$ ), firm slopes  $r = 0.876$  ( $\rho = 0.860$ ).

The decline is general but not uniform: 39 of the 49 firm slopes are negative for ESGSI and 43 for ESGSI<sup>ext</sup>, with medians  $-0.177$  and  $-0.272$  (their means are the fixed-effects slopes); the standard deviation of the ESGSI firm slopes is 0.242. Table 25 and Figure 5 give the distribution by industry. Dispersion within an industry is often as large as the distance between industries: Financials (11 firms) spans  $-1.41$  to  $2.09$  in firm means of ESGSI, Consumer Discretionary (10)  $-1.93$  to  $1.61$ . Utilities, two firms, is the only industry with no negative firm slope, under either score. Nothing in this subsection concerns a firm’s conduct.

## 7.6 Industries and countries

Industry and country are constant within firm, so every comparison here rests on the 49 firms, not the 343 reports, in cells of 1 to 17 firms. Industry cuts across countries and is partly informative about the country/document-type confound (§2.2); country is not. The document types are almost perfectly sorted by country (99 of the 101 URDs are French, all 98 German reports are annual reports, the seven sustainability reports are one French firm’s), so every country contrast is also a document-type contrast. Table 26 gives both cuts for both scores; Figures 6 and 7 the trajectories.

**Levels.** Industry means of ESGSI run from 1.125 (Technology, 3 firms) to  $-1.919$  (Energy, 2 firms); Utilities has the highest mean SUS<sup>dens</sup>, 9.86, and the second-lowest score. Both ends rest on two or three firms, and the ordering moves within the window (Figure 6): Basic Materials and Telecommunications go from well above the corpus mean in 2018 to below it in 2024. ESGSI<sup>ext</sup>

Table 25: Distribution of firm means (2018–2024) and firm-level OLS slopes, by ICB industry. Min, median and max are over the industry’s firm means; “Slope < 0” counts firms with a negative slope. Firms are anonymous.

| Industry               | Firms | ESGSI |        |       |           | ESGSI <sup>ext</sup> |        |       |           |
|------------------------|-------|-------|--------|-------|-----------|----------------------|--------|-------|-----------|
|                        |       | Min   | Median | Max   | Slope < 0 | Min                  | Median | Max   | Slope < 0 |
| Technology             | 3     | −0.01 | 0.43   | 2.95  | 3/3       | −0.19                | 0.23   | 3.76  | 3/3       |
| Telecommunications     | 2     | 0.03  | 0.94   | 1.84  | 2/2       | −0.18                | 0.75   | 1.68  | 2/2       |
| Industrials            | 9     | −0.36 | 0.23   | 1.53  | 7/9       | −1.06                | 0.22   | 1.10  | 9/9       |
| Consumer Discretionary | 10    | −1.93 | 0.06   | 1.61  | 9/10      | −1.41                | −0.02  | 1.29  | 9/10      |
| Financials             | 11    | −1.41 | −0.51  | 2.09  | 7/11      | −0.92                | 0.62   | 2.68  | 9/11      |
| Health Care            | 3     | −1.79 | 0.52   | 0.96  | 2/3       | −1.68                | 0.77   | 0.94  | 2/3       |
| Basic Materials        | 2     | −1.52 | −0.19  | 1.14  | 2/2       | −1.95                | −0.73  | 0.49  | 2/2       |
| Consumer Staples       | 5     | −1.15 | −0.39  | −0.20 | 5/5       | −2.29                | −0.83  | −0.04 | 5/5       |
| Utilities              | 2     | −1.99 | −1.17  | −0.35 | 0/2       | −4.04                | −2.57  | −1.10 | 0/2       |
| Energy                 | 2     | −2.60 | −1.92  | −1.24 | 2/2       | −2.95                | −2.64  | −2.33 | 2/2       |

Table 26: Scores by ICB industry and by country; means over reports. SUS<sup>dens</sup> per 100 tokens. Slope: firm fixed effects with group-specific slopes (19), equal to the mean of the group’s firm-level slopes. Interval: 95%  $t$  interval on those firm slopes, with  $M_k - 1$  degrees of freedom (§5.6); none for a one-firm group.

|                        | Firms | SEN    | SUS <sup>dens</sup> | ESGSI  |        |                  | ESGSI <sup>ext</sup> |        |                  |
|------------------------|-------|--------|---------------------|--------|--------|------------------|----------------------|--------|------------------|
|                        |       |        |                     | Mean   | Slope  | 95% interval     | Mean                 | Slope  | 95% interval     |
| <i>Industry (ICB)</i>  |       |        |                     |        |        |                  |                      |        |                  |
| Technology             | 3     | 0.117  | 6.17                | 1.125  | −0.328 | [−0.827, 0.171]  | 1.268                | −0.399 | [−1.136, 0.338]  |
| Telecommunications     | 2     | 0.071  | 5.89                | 0.935  | −0.404 | [−1.816, 1.007]  | 0.750                | −0.495 | [−1.882, 0.892]  |
| Industrials            | 9     | 0.023  | 6.41                | 0.297  | −0.137 | [−0.280, 0.006]  | 0.200                | −0.276 | [−0.365, −0.188] |
| Consumer Discretionary | 10    | 0.012  | 6.56                | 0.138  | −0.287 | [−0.471, −0.103] | 0.078                | −0.334 | [−0.539, −0.128] |
| Financials             | 11    | −0.047 | 5.92                | 0.041  | −0.097 | [−0.221, 0.027]  | 0.791                | −0.244 | [−0.383, −0.104] |
| Health Care            | 3     | −0.012 | 6.67                | −0.104 | 0.066  | [−0.819, 0.952]  | 0.013                | −0.019 | [−0.892, 0.855]  |
| Basic Materials        | 2     | 0.005  | 7.08                | −0.190 | −0.437 | [−1.236, 0.361]  | −0.731               | −0.448 | [−1.999, 1.103]  |
| Consumer Staples       | 5     | −0.040 | 7.17                | −0.575 | −0.239 | [−0.374, −0.104] | −0.946               | −0.240 | [−0.382, −0.098] |
| Utilities              | 2     | 0.074  | 9.86                | −1.171 | 0.282  | [−0.933, 1.497]  | −2.573               | 0.499  | [−3.595, 4.594]  |
| Energy                 | 2     | −0.210 | 7.33                | −1.919 | −0.286 | [−0.953, 0.381]  | −2.642               | −0.455 | [−0.716, −0.195] |
| <i>Country</i>         |       |        |                     |        |        |                  |                      |        |                  |
| Germany                | 14    | 0.059  | 5.60                | 1.003  | −0.268 | [−0.419, −0.116] | 1.107                | −0.376 | [−0.511, −0.241] |
| Belgium                | 1     | −0.047 | 6.37                | −0.203 | −0.372 | —                | −0.043               | −0.401 | —                |
| Finland                | 2     | −0.060 | 6.26                | −0.242 | −0.123 | [−2.287, 2.041]  | 0.216                | −0.399 | [−0.565, −0.233] |
| France                 | 17    | −0.031 | 6.84                | −0.335 | −0.182 | [−0.296, −0.068] | −0.575               | −0.250 | [−0.367, −0.132] |
| Spain                  | 4     | 0.086  | 8.51                | −0.361 | −0.055 | [−0.336, 0.226]  | −0.460               | −0.074 | [−0.361, 0.214]  |
| Netherlands            | 6     | −0.020 | 7.04                | −0.363 | −0.181 | [−0.446, 0.085]  | −0.031               | −0.302 | [−0.570, −0.034] |
| Italy                  | 5     | −0.109 | 6.65                | −0.808 | −0.017 | [−0.360, 0.325]  | −0.817               | 0.013  | [−0.602, 0.628]  |

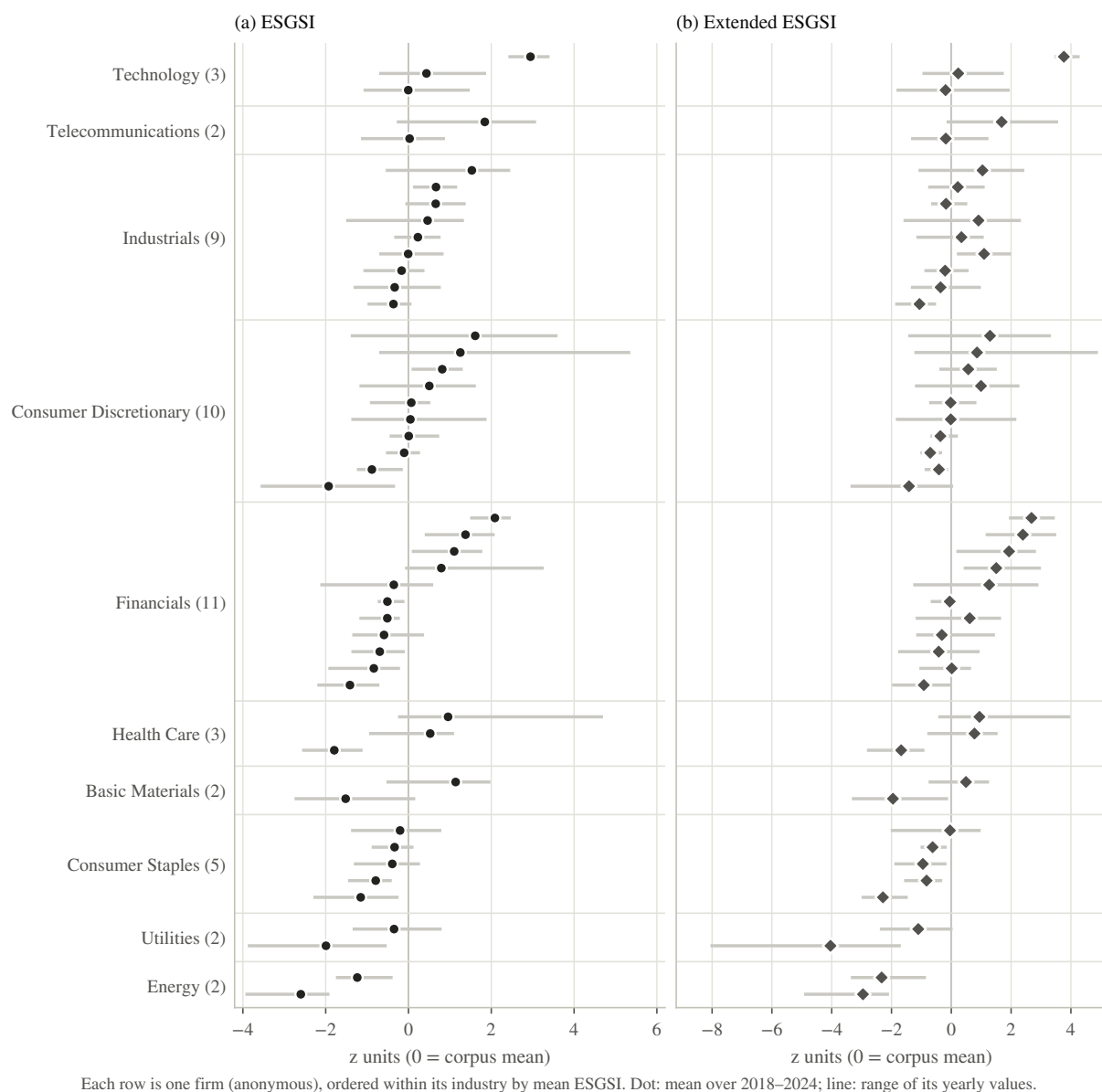


Figure 5: (a) ESGSI and (b)  $\text{ESGSI}^{\text{ext}}$  by firm, one anonymous row per firm, grouped by ICB industry and ordered within it by mean ESGSI. Dot: 2018–2024 mean; line: range of yearly values.

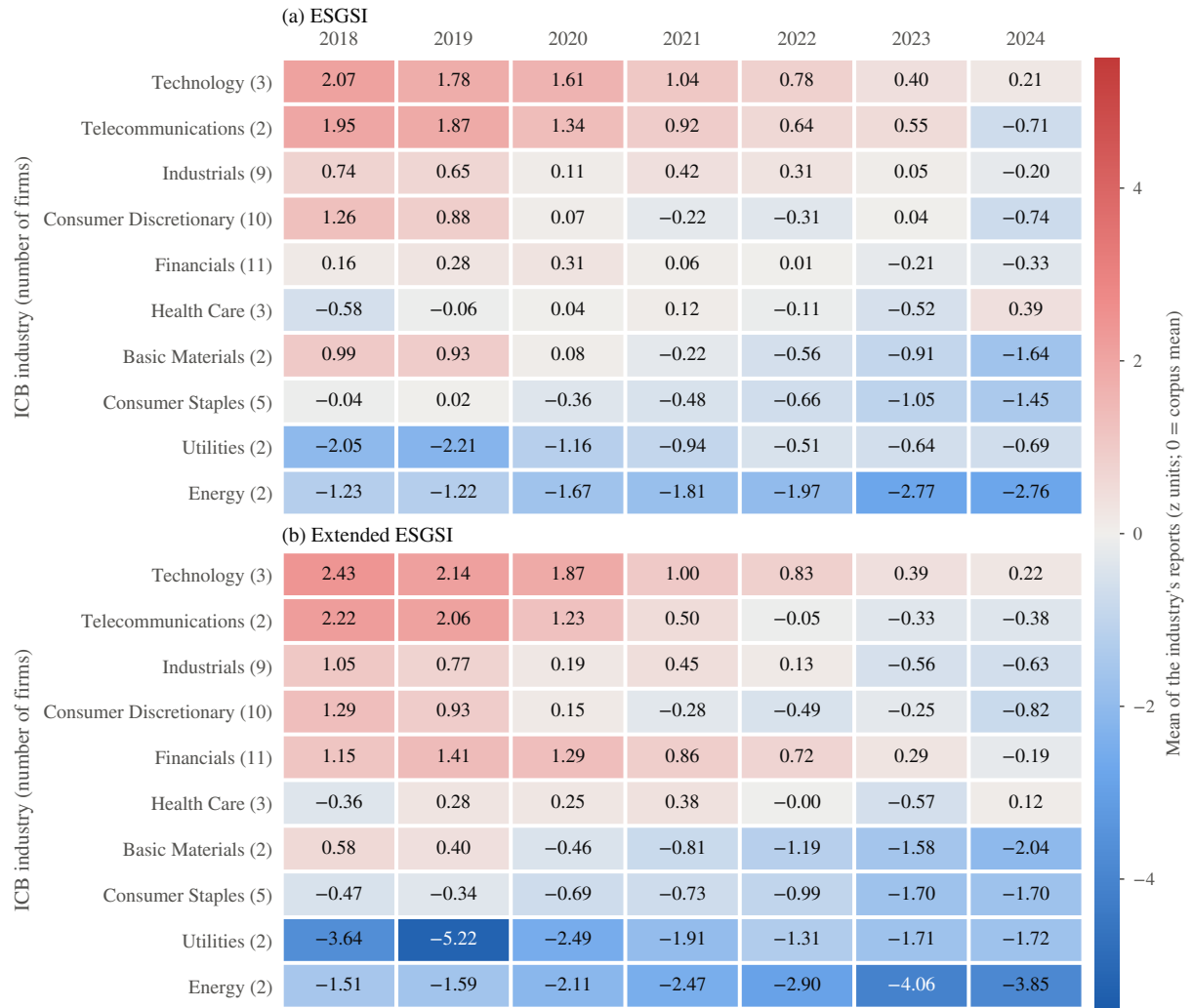


Figure 6: Mean (a) ESGSI and (b)  $\text{ESGSI}^{\text{ext}}$  by ICB industry and year, industries ordered by mean ESGSI; diverging colour scale centred on the corpus mean. Number of firms in parentheses, which is also the number of reports averaged in each cell (2 to 11).

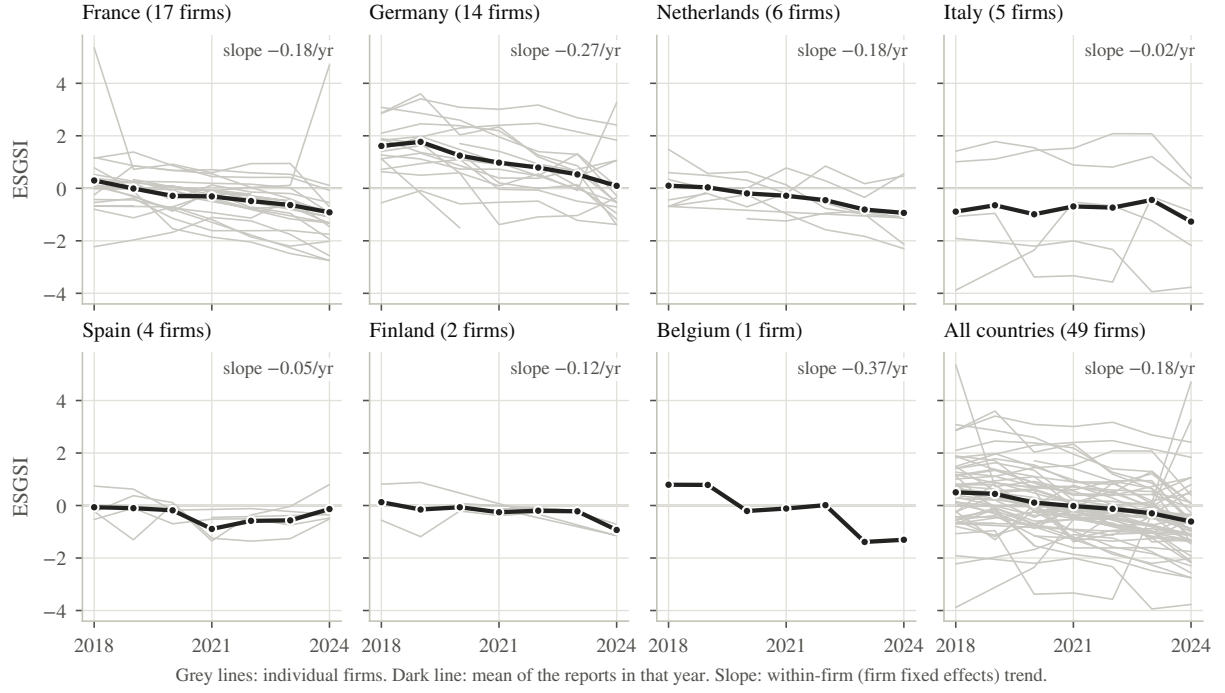


Figure 7: ESGSI by country: firms (grey, anonymous) and the country’s yearly mean (dark line), with its fixed-effects slope printed in each panel (Table 26). The last panel pools all 49 firms.

reorders the industries where quantification and hedging differ most: Financials, with the highest mean HEDGE and among the lowest QUANT, moves from 0.041 to 0.791; Utilities, with the highest QUANT, from  $-1.171$  to  $-2.573$ . Industry therefore structures  $\text{ESGSI}^{\text{ext}}$  more than ESGSI (Table 27).

Among countries, Germany stands apart (mean ESGSI 1.003 against  $-0.335$  for France), but its reports are all annual reports and France’s mostly URDs, so the gap is not a country effect we can identify. By document type, annual reports average 0.167, URDs  $-0.263$  and the seven sustainability reports, one firm’s,  $-1.792$ : no document-type effect is identified either. Within France, industry accounts for an adjusted  $R^2$  of 0.191 of the variance of firm means (17 firms, seven industries); within Germany,  $-0.464$  (14 firms, seven industries). Neither supports an industry ordering of ESGSI. In each of the five industries present in both countries the German mean exceeds the French one, so the country gap survives holding industry fixed, and so does the document-type confound it carries.

**Slopes.** Group slopes are read through the firm-slope intervals of Table 26: the clustered interval of a group slope rests on  $M_k - 1$  free contributions but is referred to  $t_{48}$ , and is too narrow for small groups (§5.6). Utilities shows the difference: its positive ESGSI slope has a clustered interval of  $[0.143, 0.421]$  and a firm-slope interval of  $[-0.933, 1.497]$ . Read this way, the ESGSI slope is distinguishable from zero in Consumer Discretionary (10 firms), Consumer Staples (5), Germany (14) and France (17). Financials (11) and Industrials (9) have the flattest declines among the larger industries, with intervals that just include zero; Italy and Spain are nearly flat. Under  $\text{ESGSI}^{\text{ext}}$ , whose slopes are steeper, Financials, Industrials and the Netherlands also decline distinguishably. No group of one to three firms, including both industries with positive slopes, has an ESGSI interval narrow enough to say anything; the  $\text{ESGSI}^{\text{ext}}$  exceptions (Energy, Finland) are two-firm intervals on one degree of freedom.

**Variance decomposition.** At report level (Table 27, Panel A), industry and country each add more than year does, and firm identity adds 0.301 of  $R^2$  ( $\text{ESGSI}^{\text{ext}}$ : 0.259) after all four

Table 27: Variance of ESGSI and ESGSI<sup>ext</sup>. A: report-level regressions ( $n = 343$ ), effects added in the order shown; firm effects nest industry, country and most of document type. B: regressions of the 49 firm means (the year effect cancels in the balanced panel);  $p$  from the overall  $F$  test, descriptive on cells this small.

|                      |        | ESGSI      |              | ESGSI <sup>ext</sup> |              |
|----------------------|--------|------------|--------------|----------------------|--------------|
|                      |        | $R^2$      | $\Delta R^2$ | $R^2$                | $\Delta R^2$ |
| <i>A. Reports</i>    |        |            |              |                      |              |
| Year (7)             |        | 0.072      | 0.072        | 0.094                | 0.094        |
| + Industry (10)      |        | 0.275      | 0.203        | 0.417                | 0.323        |
| + Country (7)        |        | 0.412      | 0.137        | 0.506                | 0.089        |
| + Document type (3)  |        | 0.443      | 0.031        | 0.525                | 0.018        |
| + Firm (49)          |        | 0.744      | 0.301        | 0.784                | 0.259        |
| <i>B. Firm means</i> | Params | Adj. $R^2$ | $p$ ( $F$ )  | Adj. $R^2$           | $p$ ( $F$ )  |
| Industry             | 9      | 0.143      | 0.082        | 0.349                | 0.001        |
| Country              | 6      | 0.248      | 0.005        | 0.174                | 0.027        |
| Document type        | 2      | 0.028      | 0.193        | 0.049                | 0.119        |
| Industry + country   | 15     | 0.287      | 0.023        | 0.421                | 0.002        |

attributes. Between firms (Panel B), industry and country together reach an adjusted  $R^2$  of 0.287 for ESGSI, each adding about as much to the other (0.168 and 0.206, unadjusted); for ESGSI<sup>ext</sup> industry weighs more (0.349 alone, 0.421 with country). On 49 observations these figures are fragile, and the country figure is also a document-type figure. They support one statement: neither industry nor country accounts for most of the difference between firms; the larger part is firm-specific.

## 7.7 Pillar composition and the mass of the vocabulary

The mentions that SUS<sup>dens</sup> pools do not grow evenly across pillars (§5.5, Figure 8b): the rise is environmental and transversal, and governance loses ground. In mean pillar density per 100 tokens (13), the environmental pillar rises in every year, from 1.73 to 3.25; the transversal one more than doubles, from 0.63 to 1.52, though not in every year; the social one is roughly level until 2024 (1.05 to 1.38); governance *falls*, from 2.34 to 1.90. These are descriptive means; we estimate no pillar-level trend regressions. Across industries the composition follows the business (Figure 8a): the environmental share runs from 26.2% (Telecommunications) to 63.2% (Utilities), whose environmental density of 6.04, against 2.33 over the corpus, gives it the highest SUS<sup>dens</sup> in Table 26; Financials and Telecommunications are the governance-heavy industries.

The mass is concentrated: 15 of the 399 entries hold 48.1% of the 1,047,951 mentions, each occurring in all 49 firms. The heaviest are *board director* (5.72%), *remuneration* (5.65%) and *emission* (5.09%), and three of the five heaviest are governance entries, so any statement about the governance pillar inherits the single-entry sensitivity reported in §6; we make none beyond the descriptive one above.

Distinctive entries by industry are sector vocabulary. Ranked by lift (the entry’s share of the industry’s mentions over its share of all mentions, among entries with at least 50 mentions in the industry and present in two or more of its firms), they include *hydroelectric* (lift 19.1) and *smart meter* (17.3) for Utilities, *oil spill* (12.2) and *methane* (8.9) for Energy, *sedex* and *food safety* (8.4) for Consumer Staples, and *access healthcare* (14.7) for Health Care; Financials’ are governance terms (*line defence*, *esg risk management*). A firm thus scores substance partly for describing its own industry. The sectoral designation of §4 covers some of these entries (*smart meter*, *oil spill*, *sedex*, *food safety*, *access healthcare*) and not others (*hydroelectric*, *methane*). The 34 sectoral entries hold 1.0% of all mentions, at most 2.7% in any industry (Consumer Staples) and 1.7% in Utilities, so they cannot account for the distance between Utilities or Energy and

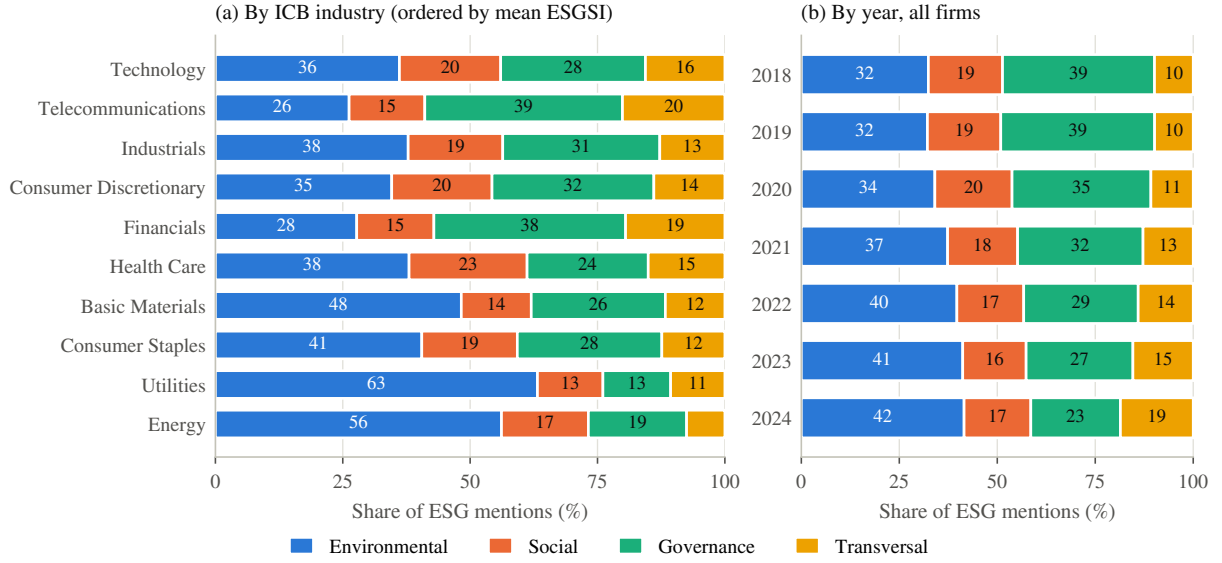


Figure 8: Share of ESG mentions by pillar, pooled over the group’s reports (14), (a) by ICB industry, ordered by mean ESGSI as in Table 26, and (b) by year.

the rest; environmental density at large carries it. Scoring without them is a sensitivity reported in §6.

## 7.8 Sensitivity of the series

**Substance specification.**  $\text{ESGSI}(\text{SUS}^{\text{tfidf-}\ell})$  is nearly identical to the adopted score (Table 16) and falls in every year (Table 21); we ran no clustered trend test on it.  $\text{ESGSI}(\text{SUS}^{\text{tfidf-}L^2})$  differs more and declines at  $-0.143$  per year (§6.1), but its yearly mean is nearly flat after 2022. The direction is common to all three specifications, its significance tested for two, and monotonicity holds for  $\text{SUS}^{\text{dens}}$  and  $\text{SUS}^{\text{tfidf-}\ell}$  only, as expected since  $\text{SUS}^{\text{tfidf-}L^2}$  tracks BREADTH (§6).

**Weights and vocabulary.** The weights of  $\text{ESGSI}^{\text{ext}}$  are a choice, not an estimate; §6 varies them over a grid and perturbs the vocabulary (§6.3). Neither check bears on the mandate confound, which concerns chronology, not our discretion over the list; no perturbation of a list built on these documents (§4.10) moves the bound of Table 24.

## 7.9 What the results license

Every component is score-normalised across the sample, so neither score has cardinal meaning. A yearly mean of  $-0.610$ , or a firm mean of  $2.947$ , is a position within this corpus, not a quantity of anything; no level here is comparable with Lagasio (2024) or any other application of the architecture, and slopes are in pooled standard deviations of this sample. The sign of a score says only on which side of the corpus mean a report lies. Only orderings within the sample and the direction of change are interpretable.

Within those bounds, both scores decline in every year from 2018 to 2024, at  $-0.181$  ( $\text{ESGSI}$ ) and  $-0.260$  ( $\text{ESGSI}^{\text{ext}}$ ) corpus standard deviations per year. The decline is carried by substance, and in  $\text{ESGSI}^{\text{ext}}$  also by quantification; tone and hedging show no detectable movement. It is not a composition artefact, appears where the country/document-type confound is absent, holds in direction under all three substance specifications, and survives, at four fifths and two thirds of its magnitude, the removal of regime names and of the whole transversal pillar. It is the smaller part of what the scores record: firms differ far more than years, their positions persist from year to year, and neither industry nor country, on cells this small, accounts for most of the difference

Table 28: Limitations, ranked by how much of what we claim they constrain, with the sections that treat each and their status.

|    | Limitation                                                                                                       | Treated in            | Status                             |
|----|------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------|
| 1  | Scores corpus-relative; no cardinal meaning; sign is position only                                               | §5.4, §5.8.4, §7      | Inherent to the design             |
| 2  | No external criterion; QUANT is internal and reads the same zones                                                | §5.8.3                | Conceded                           |
| 3  | Mandate/chronology confound in the trend                                                                         | §7.4                  | Bounded, unresolved                |
| 4  | Vocabulary built on this corpus; admission by the authors; error rate not reported; reviewers not yet identified | §4.1–4.2, §4.10, §6.4 | Conceded                           |
| 5  | Extraction–scoring coupling; additions not testable                                                              | §4.4, §6.3            | Measured for deletions             |
| 6  | Trend magnitude depends on the E and TRANS entries and on vocabulary size                                        | §6.3, §6.4            | Measured                           |
| 7  | Extended ESGSI weights are a choice; slope scales with $w_q$                                                     | §5.4, §6.2            | Grid reported                      |
| 8  | Country/document-type confound; small sector cells                                                               | §2, §7.6              | Conceded                           |
| 9  | 49 clusters; year-common shocks not covered                                                                      | §5.6                  | Partly addressed                   |
| 10 | Recall bounded by the lexicon; patterns extract but never score                                                  | §3, §4.2              | Unmeasured                         |
| 11 | Extraction parameters never varied; navigation filter unmeasured                                                 | §3, §3.3              | Untested                           |
| 12 | QUANT: overlap with the vocabulary, <code>large_numbers</code> noise, regime names                               | §5.8.3, here          | Overlap removed; noise unvalidated |
| 13 | HEDGE dominated by <i>risk</i> ; <i>may</i> as a month                                                           | §4.9, §5.3            | Conceded                           |
| 14 | SEN: finance dictionary, no negation, volume-insensitive                                                         | §5.3                  | Conceded                           |
| 15 | Protected phrases cover collapses only; lemma artefacts                                                          | §4.7                  | Partly repaired                    |
| 16 | PDF text-layer artefacts (non-ASCII deletion, NUL characters)                                                    | §3, §A                | Partly repaired                    |
| 17 | Adopted run and navigation-filter calibration not committed                                                      | §A.10                 | Open                               |

between firms. None of this is evidence about conduct: the reporting mandates of the same window could move the scores as they move with conduct held fixed, and this design cannot exclude them (§7.4).

## 8 Limitations

Most limitations of this paper are stated where the choice concerned is made, with their magnitudes. This section collects them, ranks them by how much of what we claim they constrain, points to where each is treated, and states in full the few that no earlier section owns. Table 28 is the index.

### 8.1 What bounds every result

**The scores are corpus-relative and have no cardinal meaning.** Every component of ESGSI and  $\text{ESGSI}^{\text{ext}}$  is a  $z$ -score over the 343 pooled reports (3), so a score is a position within this corpus, in corpus standard deviations, and a slope is in pooled standard deviations of this

sample. The sign says only on which side of the corpus mean a report lies. No level is comparable across corpora, vocabularies or studies, and every score, including the yearly series, would move if the panel were extended or reduced (§5.4, §7). The adopted  $SUS^{\text{dens}}$  estimates nothing on the sample, but the score normalisation is cross-sectional under every specification, and the two TF-IDF alternatives also estimate their idf weights on it (§5.2). We use no threshold and classify no report (§5.8.4); this limitation is the reason, and no calibration could remove it without an external criterion.

**No external criterion.** Neither score is tested against an external benchmark — regulatory findings, controversy data or hand-coded cases — and no figure in this paper is evidence about any firm’s conduct. The one empirical check on the substance component is convergent validity with QUANT (§5.8.3), and QUANT is internal: our regular expressions over our own extracted zones. §5.8.3 removes the strings the two instruments share, after which the correlations fall by more than a third and the firm-bootstrap interval of the difference between  $SUS^{\text{dens}}$  and  $SUS^{\text{tfidf-}L2}$  no longer excludes zero. The shared extraction cannot be removed within this design, since both instruments see only the text that §3 admits, so the test is conditional on that stage. The rejection of  $SUS^{\text{tfidf-}L2}$  rests first on the algebra of §5.8.2, which uses no criterion, and the choice of  $SUS^{\text{dens}}$  over  $SUS^{\text{tfidf-}\ell}$  on a structural argument, not on evidence (§5.8.3).

**The mandate and chronology confound is unresolved.** The 2018–2024 window is the one over which European non-financial reporting became prescribed, and a firm that adds mandated tables adds ESG vocabulary without necessarily changing anything it does. §7.4 bounds the purely definitional part — with the extraction held fixed, striking 22 regime names leaves an ESGSI slope of  $-0.149$  and striking the transversal pillar  $-0.117$ , against  $-0.181$  — and states why no design here does better: exposure is common to all 49 firms and collinear with calendar time. Whether the rise in substance reflects changed conduct, compliance or unchanged conduct described in a prescribed vocabulary is not identified. The architecture adds a further property: both scores fall when substance rises, so a firm that complies with a new disclosure regime moves down the scale for having complied.

**The vocabulary was built on this corpus, and its error rate is not reported.** Candidate terms came in part from a reading of 17 of the 343 reports, in which Germany, 98 of the reports, is represented by one firm; the admission criteria, the collocation choices, the pillar assignment and the sectoral designation were applied by the authors to occurrences in the same corpus on which the scores are computed, and no part of the corpus was held out (§4.1, §4.2). The admission criteria are judgements under stated rules, not measurements: the paper reports no error rate for the vocabulary’s matches, and the concentration thresholds behind the sectoral designation are not in a committed script (§4.10). The reviewers and evaluators of the candidate lists are not yet identified in the text; §4.1 carries a marker for that information. The extraction parameters, the QUANT families and the set of specifications were likewise fixed with the corpus visible. The stability checks of §6.3 perturb this fitted vocabulary, so the stability they show is stability around the fit (§6.4). Every choice is disclosed (§A); disclosure is not pre-commitment, and we claim only the first.

## 8.2 The vocabulary and the extraction

**One vocabulary selects the text and scores it.** The regular expression that decides which paragraphs are extracted is built from the same terms whose lemmas  $SUS$  counts (§4.4), so admitting a term changes the text it is then counted on, and changes the text on which  $SEN$ ,  $QUANT$  and  $HEDGE$  are computed. *Board of directors* is the measured case: the heaviest scoring entry, it also brings governance text into the zones, and removing it cuts the extracted

text by 6.4% (§4.7, §6.3). The coupling also limits the stability programme. Re-extraction is simulated from the paragraphs the adopted vocabulary already extracted, which is exact in principle for deletions only; an addition to the vocabulary, which would pull in text that was never extracted, cannot be simulated at all, and the single-entry deletions were run with the extraction frozen (§6.3).

**The magnitude of the trend depends on parts of the vocabulary.** The sign of the within-firm trend survives every designed deletion, every weight pair and all but one of 10,000 random deletions; its magnitude does not. Without the environmental pillar the ESGSI slope falls to  $-0.105$  under re-extraction, and without the transversal pillar to  $-0.132$ . The environmental pillar is also the designed deletion that falls furthest outside its matched random null: its  $\rho$  lies below all but 0.2% of random deletions of the same size, because its entries order the reports differently from the rest of the vocabulary. Under random deletion of half the vocabulary the 5th–95th band of the slope spans a factor of three (§6.3, §6.4). A statement about the size of the decline is therefore a statement about this vocabulary.

**Recall is bounded by the lexicon.** A term outside the extraction regex triggers no hot paragraph and enters no zone, so it is invisible to every component, not under-weighted (§3). The 32 patterns pull text into zones, where it feeds SEN, QUANT, HEDGE and the length denominators, without ever being counted by SUS. A count vocabulary cannot carry a negative lookahead, so a polysemous head is admitted bare or only inside enumerated collocations, losing its other valid uses (§4.2); paraphrase is invisible throughout. No recall figure exists for the vocabulary.

**The extraction stage is untested at its free parameters.** The density threshold (1.0 keyword per 100 words), the  $\pm 1$ -paragraph window and the 150-character minimum zone were set by judgement and never varied (§3). The stage keeps under two fifths of the corpus’s full text layer (Table 6) and sits upstream of every number here. The navigation filter is documented by a calibration whose script is not committed, and the pipeline does not log what the filter removes, so its effect on the adopted extraction is unmeasured (§3.3). Neither the sense of the vocabulary’s matches nor the filter’s error rates were measured by year, so a drift in extraction accuracy over the window is not excluded (§3.5).

**Protected phrases are a narrow repair.** Protection covers exactly the terms that preprocessing would empty or collapse to one word, eleven phrases in all (§4.7). It does not cover the other ways preprocessing reshapes the vocabulary: two collapses (*due diligence*, *eu taxonomy*) are allowed by judgement; stopword removal lets a lemmatised n-gram match across removed words (*board director* in *board and directors*, 0.3% of that entry’s count); and some entries are lemmatisation artefacts rather than words (*esr*, *ecovadi*, *sustainalytic*). The shared representation makes these harmless for matching, since corpus and vocabulary pass through the same transformation, but not for weighting: under  $SUS^{\text{tfidf-}\ell}$  and  $SUS^{\text{tfidf-L2}}$  an artefact entry carries the idf of its artefactual form. The adopted  $SUS^{\text{dens}}$  weights every entry equally and is insensitive to that.

**PDF text-layer artefacts.** Cleaning deletes non-ASCII characters rather than transliterating them, so a term written with a typographic hyphen fuses (*netzero*) and a word set with a ligature loses letters; the matches lost this way are not counted, and the yield of §3.5 is biased downward by an unmeasured amount (§3). Two extracted texts carried NUL characters that truncated them whenever the stored text was read back; the defect is repaired and the adopted run is unaffected (Table 34), but it shows that the text layer can carry characters no stage was designed for.

### 8.3 The components and the Extended ESGSI

**The weights of the Extended ESGSI are a choice.**  $w_q = w_h = 0.5$  is stated, not estimated, and no external criterion exists to calibrate it (§5.4). Over the grid  $w_q, w_h \in \{0, 0.25, 0.5, 0.75, 1\}$  the Spearman correlation with the adopted score is at least 0.865, and at least 0.973 within 0.25 of the adopted weights on each axis; the within-firm slope is negative in all 25 cells, but its magnitude grows roughly in proportion to  $w_q$ , from  $-0.185$  at  $w_q = 0$  to  $-0.335$  at  $w_q = 1$  with  $w_h = 0.5$  (§6.2). The level and the rate of decline of  $\text{ESGSI}^{\text{ext}}$  are therefore properties of the chosen weights as well as of the reports.

**QUANT is crude.** Seventeen vocabulary entries are also matched by a QUANT pattern; the overlap is measured and removed in §5.8.3, but it matters for  $\text{ESGSI}^{\text{ext}}$  too, which subtracts both  $Z(\text{QUANT})$  and  $Z(\text{SUS}^{\text{dens}})$ , so the shared strings reinforce rather than cancel, and their share of QUANT hits rises from 16.29 % in 2018 to 37.62 % in 2024. The **frameworks** family names reporting regimes, so the rise of QUANT carries the chronology objection of §7.4. The **large\_numbers** family matches any run of four or more digits other than a calendar year, so it counts page numbers, note references, identifiers, share counts and monetary figures alongside quantified ESG disclosure; it is 13,380 of the 186,984 QUANT hits (7.2 %), and no family was validated against ESG content. No two components share a token base (SUS and SEN on lemmatised tokens, QUANT on raw whitespace tokens, HEDGE on raw words; §5.1), so it is score normalisation, not anything about the quantities, that makes them addable.

**HEDGE largely counts risk.** The Loughran–McDonald categories are matched by form, not sense: *risk* and *risks* alone are 147,099 of the 498,477 hits, so in these reports HEDGE largely registers how often risk is discussed, and at least 6,632 of the 38,628 matches of *may* are the month (§4.9, §5.3). The lexicon is used as published, without removing these. HEDGE enters only  $\text{ESGSI}^{\text{ext}}$ , and it has no detectable trend (§7).

**SEN inherits the limits of its dictionary.** Loughran–McDonald polarity is a finance word list applied to non-financial narrative: it has no negation or intensifier handling, an asymmetric list of 140 positive and 893 negative stems, and a bounded ratio insensitive to the volume of matched terms — a report with 4 positive and 2 negative matches scores exactly as one with 400 and 200 (§5.3). Its input is lemmatised text that the package’s tokeniser stems again. The dictionary replaces the antecedent’s TextBlob by design (§1), which is one of the reasons our scores and the antecedent’s are different measurements. The absence of a trend in SEN answers only the objection that a fixed dictionary scores later language differently (§7).

### 8.4 Design and inference

**Country and document type are confounded, and sector cells are small.** 99 of the 101 URDs are French and all 98 German documents are annual reports; the two effects can be separated only through 15 documents, and the seven standalone sustainability reports belong to one firm (§2). No country or document-type contrast is interpretable as such, and one score construction is applied across three document types whose comparability rests on the extraction stage. The German subsample holds both fixed but establishes only that a trend of the same sign exists there (§7). Sector cuts across countries and is partly informative, but industries hold 2 to 11 firms and supersectors 1 to 7, 8 of the 16 supersectors hold one or two, and sector is identified only between firms (§2.2); no group of one to three firms supports a statement about its slope (§7.6).

**Inference rests on 49 clusters and seven years.** Every panel regression clusters by firm, with  $t_{48}$  references and a restricted wild cluster bootstrap for the fixed-effects slopes (§5.6); with

49 clusters, and 14 in the German subsample, the asymptotic approximation is not relied on alone. Firm clustering does not cover a shock common to all firms in one year, and seven years cannot be clustered on, so a trend standard error measures dispersion across firms around seven yearly means, not uncertainty about which years were drawn. Group-slope intervals clustered by firm are too narrow for small groups and the Wald test of equal slopes is anti-conservative when a one-firm group is present; we read group slopes through firm-slope intervals instead (§5.6). The panel requires seven consecutive reports, so it selects large, continuously reporting European issuers; that restricts external validity without generating an internal trend (§7.4).

**The adopted run is not yet committed.** Every number is regenerated by a script, but the adopted run depends on pipeline changes that exist only in the working tree (rows 5–8 of Table 34), and most analysis scripts are not yet in the repository history; the last commit does not reproduce the adopted run (§A.10). The navigation filter’s calibration script is not committed at all (§A.5). Until the working tree is committed, the commit hash is a necessary, not a sufficient, condition for reproduction.

## 8.5 Checks not run

Several checks that would bear on the results were not run, and the sections that raise them say so: variation of the extraction parameters (§3); the effect of the navigation filter on the adopted extraction, and the accuracy of extraction by year (§3.5); re-extraction from the PDFs rather than simulated from extracted paragraphs, single-entry deletions under re-extraction, and additions to the vocabulary (§6.3); a clustered trend test for ESGSI( $SUS^{\text{tfidf-}\ell}$ ) (§6.1); pillar-level trend regressions (§7.7); and any comparison with an external criterion (above).

# A Reproducibility

This appendix discloses every element of our pipeline at the level of detail whose absence §5.8.1 records for the antecedent: the code, the files each stage reads and writes, the parameters, the seeds, the output schema and the software versions. Everything below describes the adopted specification: our vocabulary with its sectoral entries,  $SUS = SUS^{\text{dens}}$  and  $w_q = w_h = 0.5$  (§4, §5). The paper reports one run of the pipeline, and every number in it comes from that run or from the analysis scripts that read it.

## A.1 What is in the repository and what must be regenerated

The repository holds the code (`main.py`, `src/`, `scripts/`), every lexicon and metadata file (`metadata/`) and this paper. The directories `data/` and `results/` are excluded by `.gitignore`: no PDF, no extracted or preprocessed text and no output file is versioned. A reader who wants to regenerate the paper’s numbers needs three things from outside the repository. First, the 344 PDFs, placed as `data/pdf/<COUNTRY>/<Company>/<file>.pdf`; the two folder names become the `País` and `Compañía` fields and are the keys on which `metadata/muestras_informes.xlsx` (document type) and `metadata/empresas_supersector.csv` (supersector) are joined, so the folder names must match those files exactly. Second, the environment of Table 31. Third, two environment variables: `ESG_RUN_TAG=v3` and `ESG_INCLUDE_SECTORAL=1`. Both `src/config.py` and `run_suffix()` in `src/lexical_document_filter.py` turn them into the suffix `_v3_sec` on every pipeline path; the second also adds the sectoral term and pattern files to the extraction regex and the sectoral lemmas to the scoring vocabulary. Without them the pipeline scores the base vocabulary only and writes to the unsuffixed directories. We ran every script with `PYTHONIOENCODING=utf-8` as well, which only affects console output on Windows.

Everything else is regenerated by the code: `main.py` produces the pipeline outputs (Table 29) and the scripts of Table 30 produce every number and figure the paper reports.

## A.2 Pipeline and intermediate artefacts

The pipeline is a fixed sequence of stages, each reading the previous stage’s output and writing its own; no stage reads back from a later one. Table 29 gives the code, input and output of each. The two lexicon stages run offline, once per change of vocabulary; the three pipeline stages are invoked together as `python main.py`, with the flags `run_extraction`, `run_preproc` and `run_esgsi_analysis` set to `True` and `run_lda` set to `False` (§A.5). With `run_extraction=False` the stored extraction JSONs are reused, and with `run_preproc=False` the stored `processed_texts.csv`. Runtimes are from the pipeline logs of the adopted run on one Windows 10 desktop (Intel x86-64), single process.

Table 29: Pipeline stages and intermediate artefacts of the adopted run. All paths are relative to the repository root.

| Stage                                       | Code                                                                                                                          | Input                                                                                                                                                                            | Output                                                                                                                                                                                                                                            |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Vocabulary export (offline)                 | <code>scripts/export_vocab_v3.py</code>                                                                                       | <code>metadata/ESG_terms_v3.csv</code> (434 entries)                                                                                                                             | <code>metadata/esg_terms.txt</code> (367 terms),<br><code>esg_terms_sectorial.txt</code> (35),<br><code>esg_patterns.txt</code> (31 patterns),<br><code>esg_patterns_sectorial.txt</code> (1)                                                     |
| Lexicon build (offline)                     | <code>scripts/build_lexicons.py</code> , using <code>TextProcessor</code>                                                     | the two term files; <code>metadata/personal_stopwords.txt</code> ; the Loughran–McDonald master dictionary bundled with <code>pysentiment2</code> ( <code>static/LM.csv</code> ) | <code>metadata/esg_protected_phrases.txt</code> (11 phrases),<br><code>esg_terms_lemmatized.txt</code> (365 lemmas), <code>esg_terms_sectorial_lemmatized.txt</code> (34),<br><code>lm_hedge.txt</code> (495 forms)                               |
| Extraction (≈37 min)                        | <code>src/lexical_document_filter.py</code> ( <code>LexicalDocumentFilter</code> )                                            | <code>data/pdf/</code> (344 PDFs); the four term and pattern files                                                                                                               | <code>data/chunks_lexical_v3_sec/</code> (344 JSON files mirroring the PDF tree; fields <code>file</code> , <code>path</code> , <code>zones</code> , <code>relevant_text</code> )                                                                 |
| Preprocessing (≈34 min)                     | <code>src/text_processor.py</code> ( <code>TextProcessor</code> ), called from <code>main.py</code>                           | <code>relevant_text</code> of each JSON; protected phrases; stopword list                                                                                                        | <code>data/clean_v3_sec/processed_texts.csv</code> (344 rows, ;-separated; <code>Documento</code> , <code>País</code> , <code>Compañía</code> , <code>Año</code> , <code>TipoDocumento</code> , <code>clean_text</code> , <code>raw_text</code> ) |
| Deduplication, measurement, scores (≈5 min) | MD5 of <code>clean_text</code> in <code>main.py</code> ;<br><code>src/esgsi_analyzer.py</code> ( <code>ESGSIAnalyzer</code> ) | <code>processed_texts.csv</code> ; the lemmatised vocabulary and hedge lexicon via <code>src/config.py</code> ; <code>metadata/muestras_informes.xlsx</code>                     | <code>results/metrics_v3_sec/results.csv</code> (343 rows, 17 columns; Table 33)                                                                                                                                                                  |

`src/config.py` refuses to start if a generated lexicon is missing and logs a warning if its source file is newer, so a vocabulary edit that is not followed by `scripts/build_lexicons.py` is reported rather than silently scored with the old lemmas. The adopted run was executed in three calls: extraction; then preprocessing on that extraction (`run_extraction=False`), with the protected phrases of row 5 of Table 34; then measurement on the stored preprocessed text (`run_preproc=False`), with rows 6 and 7 in place and the stored file rewritten under the NUL rule of row 8. None of rows 5–8 alters the extraction regex, and each call reads only what the previous one wrote.

## A.3 Analysis layer

No number in the paper is read off the pipeline by hand. The scripts of Table 30 read the pipeline outputs, write every quotable quantity to `results/analysis_v3/` or `results/stability_v3/`, and draw the figures into `paper/figures/`. Because `results/` is not versioned, the paper never

\inputs a file from it; every table was transcribed from these outputs. The table gives a valid running order. Three dependencies fix it: `robustness_v3.py` reads the outputs of both stability scripts, `extraction_stats_v3.py` reads a per-document file written by `analysis_v3.py`, and `figures_v3.py` reads the outputs of `analysis_v3.py` and `stability_v3_e2e.py`. The other scripts depend only on the pipeline outputs; `corpus_composition.py`, `vocab_v3_stats.py`, `validity_v3.py`, `critique_v3.py` and `robustness_v3.py` import functions from `analysis_v3.py`, so that corpus loading, deduplication, firm grouping, the pillar map and the trend estimator are defined once.

Table 30: Analysis scripts (`scripts/`), in a valid running order. All read the adopted `results.csv`; “preprocessed text” and “extraction JSONs” are the files of Table 29. Outputs without a directory are in `results/analysis_v3/`. Runtimes on the machine of Table 29, as recorded by the scripts themselves; “—” where none was recorded.

| Script                              | Further inputs                                                                                                                                                                                                     | Outputs                                                                                                                                                                 | Time   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <code>stability_v3.py</code>        | preprocessed text; <code>ESG_terms_v3.csv</code>                                                                                                                                                                   | <code>results/stability_v3/</code> :<br><code>summary.json</code> , <code>variantes.csv</code> ,<br><code>borrado_aleatorio.csv</code> ,<br><code>quitar_uno.csv</code> | 99 s   |
| <code>stability_v3_e2e.py</code>    | extraction JSONs (zones);<br><code>ESG_terms_v3.csv</code>                                                                                                                                                         | <code>results/stability_v3/</code> :<br><code>e2e_cache.pkl</code> , <code>e2e_summary.json</code> ,<br><code>e2e_borrado_aleatorio.csv</code>                          | 19 min |
| <code>robustness_v3.py</code>       | stability outputs, including<br><code>e2e_cache.pkl</code> ; preprocessed text                                                                                                                                     | <code>robustness_v3.json</code>                                                                                                                                         | 33 s   |
| <code>analysis_v3.py</code>         | preprocessed text; extraction JSONs;<br><code>data/pdf/</code> (full-text word counts,<br>cached in <code>pdf_wordcounts.csv</code> );<br><code>empresas_supersector.csv</code> ;<br><code>ESG_terms_v3.csv</code> | <code>summary.json</code> (every quotable<br>scalar), CSV tables, <code>tex/*.tex</code><br>fragments                                                                   | 74 s   |
| <code>extraction_stats_v3.py</code> | extraction JSONs;<br><code>extraction_by_doc_internal.csv</code>                                                                                                                                                   | <code>extraction_stats.json</code>                                                                                                                                      | —      |
| <code>corpus_composition.py</code>  | <code>empresas_supersector.csv</code>                                                                                                                                                                              | <code>corpus_composition.csv</code> ,<br><code>corpus_composition_internal.csv</code><br>(firm rows),<br><code>corpus_industry_x_country.csv</code>                     | —      |
| <code>vocab_v3_stats.py</code>      | vocabulary, lexicon and stopword files<br>of <code>metadata/</code> ; preprocessed text                                                                                                                            | <code>vocab_stats.json</code>                                                                                                                                           | —      |
| <code>validity_v3.py</code>         | preprocessed text; extraction JSONs;<br><code>ESG_terms_v3.csv</code>                                                                                                                                              | <code>validity_v3.json</code>                                                                                                                                           | 285 s  |
| <code>critique_v3.py</code>         | preprocessed text;<br><code>empresas_supersector.csv</code>                                                                                                                                                        | <code>critique_v3.json</code>                                                                                                                                           | 854 s  |
| <code>figures_v3.py</code>          | analysis and stability outputs                                                                                                                                                                                     | <code>paper/figures/fig*.pdf</code> (8<br>figures)                                                                                                                      | —      |

Three runtimes need a gloss. `stability_v3_e2e.py` ran in 19 minutes with its cache of per-paragraph lemmatised counts already built; it reuses the cache unless called with `-rebuild`, and by the script’s own estimate rebuilding it adds about 40 minutes. `analysis_v3.py` took 74 seconds with the PDF word counts already cached; a first run reads the full text layer of all 344 PDFs. And `validity_v3.py` and `critique_v3.py` recompute the components from the text with `ESGSIAnalyzer` and check them against `results.csv` before reporting anything, which is what their runtime buys.

**Other scripts and outputs not reproduced.** No script of the analysis layer compares the adopted run with any other run. The repository still holds the scripts that produced candidate lists during the construction of the vocabulary (`build_consensus*.py`, `build_vocab_v3*.py`, `build_kwic_sample.py`); no number in this paper comes from them except the two dispersion constants that §4.2 quotes from the last. The analysis outputs identify firms only by any-

mous positions (`<Industry>-NN`). The keys that link a position to a company folder are written to files whose names end in `_internal` (`doc_level_internal.csv`, `firm_ids_internal.csv`, `extraction_by_doc_internal.csv`, `extraction_internal.json`, `corpus_composition_internal.csv`); the company folder names also appear in `results.csv` itself and in the PDF word-count cache. None of these is reproduced in the paper.

## A.4 Dependencies

`pyproject.toml` declares dependency names without version constraints, so it is not itself a reproducibility record. Table 31 gives the resolved environment (conda environment `esgwashing`) in which the adopted run and every analysis script were executed; it is the binding version record for this paper.

Table 31: Runtime dependencies and versions.

| Package                     | Version | Used for                                    |
|-----------------------------|---------|---------------------------------------------|
| Python                      | 3.11.15 | everything                                  |
| spaCy                       | 3.8.11  | lemmatisation                               |
| <code>en_core_web_md</code> | 3.8.0   | spaCy model                                 |
| PyMuPDF                     | 1.27.2  | PDF text blocks                             |
| scikit-learn                | 1.8.0   | count and TF-IDF vectorisers                |
| pysentiment2                | 0.1.1   | Loughran–McDonald dictionary, SEN           |
| nlTK                        | 3.9.3   | Porter stemmer inside pysentiment2          |
| numpy                       | 2.4.3   | scores, random draws                        |
| pandas                      | 3.0.1   | tables, CSV input and output                |
| scipy                       | 1.17.1  | analysis scripts                            |
| statsmodels                 | 0.15.0  | <code>analysis_v3.py</code>                 |
| matplotlib                  | 3.10.8  | figures                                     |
| openpyxl                    | 3.1.5   | reading <code>muestras_informes.xlsx</code> |
| loguru                      | 0.7.3   | logging                                     |
| gensim                      | 4.4.0   | LDA only (not run)                          |

## A.5 Determinism

Extraction, preprocessing and measurement together are a pure function of the PDFs, the lexicon files of Table 29 and the pinned versions of Table 31: no network call, no external API, and no random draw anywhere on this path (§3). The one non-trivial control-flow step, the deduplication that takes the 344 preprocessed reports to 343, keeps the first of any two reports whose `clean_text` has the same MD5 hash (§2). Which of the two identical files of the duplicated report counts as first follows the order in which the file system lists the JSON files; on our machine the file without the `_repaired` suffix is kept. The choice changes the `Documento` string of one row and nothing else, since the two texts are identical. Row order in `results.csv` follows the same listing; no score depends on it.

Randomness enters only in the analysis layer, and every draw is seeded:

- *Wild cluster bootstrap* (§5.6): `analysis_v3.py`,  $B = 9,999$ , seed 20260925 (`SEED`, `N_BOOT`), NumPy `default_rng`, one Rademacher weight per firm and replication. `robustness_v3.py` calls the same function with the same seed for its trend  $p$ -values.
- *Firm bootstrap of the convergent-validity correlations* (§5.8.3): `validity_v3.py`, 9,999 samples of firms with replacement, seed 20260925, percentile 95% intervals.
- *Random vocabulary deletions, frozen extraction* (§6.3): `stability_v3.py`, seed 20260918, 2,000 draws at each deletion share (10, 20, 30, 40 and 50% of the entries) and 5,000 size-matched draws for each designed deletion.

- *Random vocabulary deletions, both modes* (§6.3): `stability_v3_e2e.py`, seed 20260919, 1,000 draws at each deletion share and 1,000 size-matched draws for each designed deletion.

In both stability scripts all draws come from one generator in a fixed sequence, so a given draw is reproducible only by running the whole script, not in isolation. The amplification and dilution demonstrations of `critique_v3.py` (§5.8.2) involve no draw: the source report is the one at the median of  $SUS^{\text{dens}}$ , and the dilution filler is taken deterministically from its 50 most frequent non-vocabulary tokens.

One documented quantity cannot be regenerated from the repository. The calibration of the navigation filter (§3.3) is recorded in the docstring of `_is_navigation`, but the script that computed it is not committed; its rates are recorded there but not reproducible, and the paper reports neither. The yield figures of §3.5, by contrast, are computed on all 343 reports by `analysis_v3.py`; no sample is involved.

The LDA topic model (`src/topic_modeler.py`) is the only stochastic component of the pipeline code, and it is switched off (`run_lda=False`). Its `fit()` builds the model with `random_state=None`, so it is unseeded; its stability analysis is gated behind `LDA_N_SEEDS>0` and never invoked at the repository’s setting `LDA_N_SEEDS=0`. LDA feeds none of  $SUS$ ,  $SEN$ ,  $QUANT$ ,  $HEDGE$ ,  $ESGSI$  or  $ESGSI^{\text{ext}}$ , and no output of it is used in this paper.

## A.6 Parameters

Table 32 consolidates, in one place, every free parameter from extraction through score assembly and panel inference, with the identifier the code uses. Each value is defined and justified where the corresponding stage is described (§3, §4, §5); nothing here is derived independently of those sections. The index-stage and inference rows repeat Table 12, which is the definitional source for them; the extraction rows repeat Table 5 and the rules of §3. Where this table and those disagree, they govern and this one is in error.  $Z$  denotes score normalisation throughout; it is never vector normalisation (§5).

## A.7 Output schema

The file `results/metrics_v3_sec/results.csv` is ;-separated, UTF-8, 343 rows by 17 columns, listed in Table 33. The row count is already net of the duplicated report that deduplication removes (§2); nothing downstream needs to drop a row by hand. The column names and the country values are Spanish (`ALEMANIA`, `FRANCIA`, ...); `Compañía` is the company folder name and equals the `Empresa` column of `metadata/empresas_supersector.csv` exactly.

The last two columns, `Etiqueta` and `Etiqueta_ext`, are a remnant of the antecedent’s classification rule: the pipeline still writes, for each report, a label derived from the sign of each score. **They are not used anywhere in this paper:** no script of Table 30 reads them, no figure or table reports them, and no threshold is applied to either score (§5.8.4). A reader regenerating the results should ignore them.

The score columns are rounded to four decimals, so a score recomputed from the rounded component columns does not match the stored one exactly.  $QUANT$  and  $HEDGE$  have standard deviations of the order of  $10^{-3}$ , so recomputing  $ESGSI^{\text{ext}}$  from their rounded columns reproduces it only to about  $10^{-2}$ . The analysis scripts therefore recompute the components from the text at full precision wherever precision matters, and check the recomputation against the stored columns.

## A.8 Vocabulary and lexicons as published artefacts

Every lexicon that feeds a score in this paper is a plain-text or CSV file in the repository, loaded at run time rather than transcribed into the paper:

Table 32: Parameters, extraction through panel inference, under the adopted specification.

| Stage                 | Parameter                    | Value (code identifier)                                                                                                                                                                                                                                                                                                     |
|-----------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extraction            | Keyword regex                | 402 terms (367 base, 35 sectoral), then 32 patterns (31, 1), case-insensitive, in word boundaries; term words joined by <code>[\\s\\-]+</code> , optional final <code>s</code> ; longest terms first ( <code>_ESG_KW_RE</code> )                                                                                            |
| Extraction            | Zones                        | hot paragraph if keyword density $\geq 1.0$ per 100 whitespace tokens ( <code>LEXICAL_KW_THRESHOLD</code> ); window of $\pm 1$ paragraph ( <code>LEXICAL_CONTEXT_PARAS</code> ), windows merged if <code>lo <math>\leq</math> prev_hi + 1</code> ; zones under 150 characters dropped ( <code>LEXICAL_MIN_ZONE_LEN</code> ) |
| Extraction            | Margin bands                 | block dropped if $y_1 < 0.08h$ or $y_0 > 0.92h$ ( <code>_HEADER_RATIO</code> , <code>_FOOTER_RATIO</code> )                                                                                                                                                                                                                 |
| Extraction            | Running header/footer        | margin blocks of $\leq 120$ characters, digits normalised to <code>#</code> , blacklisted if on $> 30\%$ of pages ( <code>_MAX_HF_LEN</code> , <code>_REPEAT_THRESH</code> )                                                                                                                                                |
| Extraction            | Financial table              | no ESG keyword, and either digit share $> 0.35$ with $\leq 6$ lines, or $\geq 2$ lines all $< 60$ characters with the first not ending in <code>.!?:</code>                                                                                                                                                                 |
| Extraction            | Section header               | all-caps, $\geq 2$ letters, $\leq 10$ tokens, $< 60\%$ of them with $\leq 1$ letter, no file name; becomes a section marker                                                                                                                                                                                                 |
| Extraction            | Cleaning                     | URLs removed, line-break hyphenation rejoined, non-ASCII deleted, whitespace collapsed ( <code>_clean_block</code> )                                                                                                                                                                                                        |
| Extraction            | Navigation filter            | cleaned block dropped if $\geq 2$ dotted-leader page references, or $\geq 4$ navigation markers forming $\geq 8\%$ of its tokens; separate rules for blocks of $\leq 15$ tokens, §3.3 ( <code>_is_navigation</code> )                                                                                                       |
| Preprocessing         | Before spaCy                 | URLs removed, lowercased, 11 protected phrases replaced by one token each ( <code>esg_protected_phrases.txt</code> )                                                                                                                                                                                                        |
| Preprocessing         | spaCy pipeline               | <code>en_core_web_md</code> , <code>ner</code> and <code>parser</code> disabled, <code>nlp.max_length</code> 5,000,000                                                                                                                                                                                                      |
| Preprocessing         | Token filter                 | alphanumeric with $\geq 2$ letters, length $> 2$ , not stop, punctuation or space, lemma not in the 189-entry list ( <code>personal_stopwords.txt</code> )                                                                                                                                                                  |
| Preprocessing         | Raw text                     | <code>relevant_text</code> with NUL characters removed and whitespace collapsed ( <code>main.py</code> )                                                                                                                                                                                                                    |
| Preprocessing         | Deduplication                | MD5 of <code>clean_text</code> , first occurrence kept; 344 $\rightarrow$ 343                                                                                                                                                                                                                                               |
| Scores                | Text representation          | lemmatised: SUS, SEN, BREADTH; raw extracted: QUANT, HEDGE                                                                                                                                                                                                                                                                  |
| Scores                | Vocabulary size $V$          | 399 lemmatised entries, 365 base + 34 sectoral ( <code>config.ESG_KEYWORDS</code> )                                                                                                                                                                                                                                         |
| Scores                | $n$ -gram range              | (1, 4), from the longest entry at run time                                                                                                                                                                                                                                                                                  |
| Scores                | Count tokenisation           | scikit-learn default, lowercased; vocabulary fixed in advance, no document-frequency filtering                                                                                                                                                                                                                              |
| Scores                | Term frequency, idf          | raw counts, <code>sublinear_tf=False</code> ; smoothed idf, <code>smooth_idf=True</code> , for $SUS^{tfidf-\ell}$ and $SUS^{tfidf-L2}$                                                                                                                                                                                      |
| Scores                | Vector normalisation         | none for $SUS^{dens}$ and $SUS^{tfidf-\ell}$ ; L2 within document for $SUS^{tfidf-L2}$ ( <code>norm='l2'</code> )                                                                                                                                                                                                           |
| Scores                | Aggregation                  | sum over the vocabulary divided by $L_d$ , per 100 tokens, for $SUS^{dens}$ and $SUS^{tfidf-\ell}$ ; mean over $V$ for $SUS^{tfidf-L2}$                                                                                                                                                                                     |
| Scores                | Length denominators          | $L_d$ lemmatised tokens ( $SUS^{dens}$ , $SUS^{tfidf-\ell}$ ); $R_d$ raw whitespace tokens (QUANT); $W_d$ raw words (HEDGE); none ( $SUS^{tfidf-L2}$ , SEN, BREADTH)                                                                                                                                                        |
| Scores                | SUS fed to the scores        | $SUS^{dens}$ ( <code>config.SUS_MODE='density'</code> ); all three written to <code>results.csv</code>                                                                                                                                                                                                                      |
| Scores                | SEN                          | pysentiment2 Loughran-McDonald list (Loughran and McDonald, 2011), Porter stems (140 positive, 893 negative); polarity with $\varepsilon = 10^{-6}$                                                                                                                                                                         |
| Scores                | QUANT                        | four regex families, Table 11, over raw text, divided by $R_d$ ( <code>config.QUANT_PATTERNS</code> )                                                                                                                                                                                                                       |
| Scores                | HEDGE                        | raw text lowercased and split into words <code>[a-z]+(?:-[a-z]+)*</code> ; hits in the hedge lexicon ( <code>lm_hedge.txt</code> , 495 forms) divided by $W_d$                                                                                                                                                              |
| Scores                | $Z$ (score norm.)            | population sd, ddof = 0; zero vector if sd = 0                                                                                                                                                                                                                                                                              |
| Scores                | ESGSI <sup>ext</sup> weights | $w_q = w_h = 0.5$ , a stated choice ( <code>config.ESGSI_EXT_WEIGHTS</code> ); grid in §6.2                                                                                                                                                                                                                                 |
| Scores                | Threshold                    | none; both scores continuous                                                                                                                                                                                                                                                                                                |
| Descriptive Inference | Pillar partition of $V$      | E 152, S 127, G 90, TRANS 30; pillar shares only                                                                                                                                                                                                                                                                            |
| Inference             | Panel regressions            | firm clusters, $M = 49$ ; CR1 variance; $t$ with $M - 1$ df                                                                                                                                                                                                                                                                 |
| Inference             | Wild cluster bootstrap       | restricted, Rademacher weights, $B = 9,999$ , seed 20260925                                                                                                                                                                                                                                                                 |

Table 33: Column schema of **results.csv**.

| Column(s)                                     | Content                                                                                                      | Rounding    |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------|
| Documento, País, Compañía, Año, TipoDocumento | JSON file name, country, company folder, year, document type (Annual report, URD, Informe de sostenibilidad) | none        |
| SUS_Score                                     | the specification selected by <code>config.SUS_MODE</code> , currently $SUS^{\text{dens}}$                   | 6 dp        |
| SUS_density, SUS_tfidf_length, SUS_lagasio    | $SUS^{\text{dens}}$ , $SUS^{\text{tfidf-}\ell}$ , $SUS^{\text{tfidf-L2}}$                                    | 6 dp        |
| Breadth, SEN_Score, QUANT_Score, HEDGE_Score  | remaining components                                                                                         | 4 dp        |
| ESGSI, ESGSI_ext                              | the two scores                                                                                               | 4 dp        |
| Etiqueta, Etiqueta_ext                        | sign-based label written by the pipeline;<br><i>not used in this paper</i>                                   | categorical |

- the v3 vocabulary, `metadata/ESG_terms_v3.csv`: 434 entries, each with its type (term or pattern), pillar and sectoral flag, plus bookkeeping columns from its construction that no score reads (§4);
- its four exported files, the input of the extraction regex: `esg_terms.txt`, `esg_terms_sectorial.txt`, `esg_patterns.txt`, `esg_patterns_sectorial.txt`;
- the scoring vocabulary that SUS and BREADTH actually count, `esg_terms_lemmatized.txt` and `esg_terms_sectorial_lemmatized.txt`:  $V = 399$  entries, of which 140 unigrams, 224 bigrams, 34 trigrams and one four-gram;
- the 11 protected phrases, `esg_protected_phrases.txt`, as regular expression and replacement token;
- the hedge lexicon, `lm_hedge.txt`: the 495 distinct forms of the Uncertainty, Constraining, WeakModal and StrongModal words of the Loughran–McDonald master dictionary (Loughran and McDonald, 2011) as bundled with `pysentiment2`, unlemmatised;
- the 189-entry supplementary stopword list, `personal_stopwords.txt`;
- the document-type and supersector maps, `muestras_informes.xlsx` and `empresas_supersector.csv`.

The files a third party would need to recompute every number in this paper are the same files that produced them.

## A.9 Defects found and repaired

Table 34 lists the defects that we found in the pipeline and repaired before the adopted run, in two groups. The first four were fixed in commit **44114d0**; the other four were found while preparing the adopted run and are fixed in the working tree. The adopted run includes all eight. We report them because most produced a plausible-looking number instead of an error, and that failure mode is the one most likely to survive review in dictionary-based text measurement.

The general lesson is not specific to this codebase. In dictionary-based measurement, a mis-specified component returns zero, or a plausible non-zero number, rather than raising an error: a token filter, a lemmatisation mismatch, a vectoriser default or a stopwords list can each silently remove an entire category of matches while the pipeline runs to completion and writes a number that looks exactly like a low-incidence finding. None of the defects in Table 34 raised an exception or an out-of-range value; each was found only by checking a component against text known in

Table 34: Defects found and repaired. Rows 1–4: commit 44114d0. Rows 5–8: working tree (§A.10).

| Component                  | Failure                                                                                                                                                                                                                                                                            | Fix                                                                                                                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 QUANT                    | <code>percentages/large_numbers</code> ran on lemmatised <code>clean_text</code> , which preprocessing had already stripped of digit-bearing tokens; both families matched nothing in any document                                                                                 | Routed onto the untouched <code>raw_text</code> column                                                                                                                                                                                 |
| 2 SUS vocabulary           | <code>TextProcessor</code> kept only <code>token.is_alpha</code> tokens, silently deleting alphanumeric ESG terms ( <code>co2</code> , <code>co2e</code> , <code>sf6</code> , <code>iso14001</code> ) from <code>clean_text</code> , so those vocabulary entries could never match | Filter replaced by an alphanumeric test requiring $\geq 2$ letters (still excludes pure figures and page references)                                                                                                                   |
| 3 HEDGE                    | The L&M dictionary is distributed in inflected form; matched directly against lemmatised <code>clean_text</code> , it missed most of its own entries                                                                                                                               | Lexicon extended with each form's spaCy lemma (superseded by rows 6 and 7)                                                                                                                                                             |
| 4 SUS (multi-word entries) | The vectorisers used scikit-learn's default <code>ngram_range=(1,1)</code> ; every multi-word entry scored zero in every document, with no warning                                                                                                                                 | $n_{\max}$ derived at run time from the vocabulary and passed as <code>ngram_range=(1, n_max)</code> ; 4 for our vocabulary                                                                                                            |
| 5 SUS vocabulary           | 11 vocabulary terms vanished or collapsed under preprocessing ( <i>well-being</i> $\rightarrow$ empty, <i>iso 14001</i> $\rightarrow$ <i>iso</i> , <i>say on pay</i> $\rightarrow$ <i>pay</i> ); the lexicon build logged them and excluded them from the scoring vocabulary       | <code>build_lexicons.py</code> writes <code>esg_protected_phrases.txt</code> ; <code>TextProcessor</code> replaces each phrase by one token before tokenising, in corpus and vocabulary alike; $V = 399$                               |
| 6 HEDGE                    | Computed on lemmatised, stopwords-filtered text, where the modals ( <i>may</i> , <i>might</i> , <i>could</i> , <i>must</i> , <i>will</i> ) and <i>always</i> , <i>never</i> , <i>perhaps</i> can never occur; no error                                                             | Computed on the raw extracted text, split into words <code>[a-z]+(?:-[a-z]+)*</code> , divided by that word count $W_d$                                                                                                                |
| 7 HEDGE lexicon            | On raw text the lemma forms added in row 3 matched function words that are not Loughran–McDonald words ( <i>and</i> , <i>for</i> , <i>one</i> , <i>up</i> ); the source file was an extended word list, not the master dictionary                                                  | Original forms only, from the master dictionary bundled with pysentiment2: 495 forms in <code>lm_hedge.txt</code>                                                                                                                      |
| 8 <code>raw_text</code>    | Two extracted texts contain NUL characters; pandas' C parser truncates a field at the first NUL when the CSV is read back (one text to 99,091 of its 795,471 characters). Scores computed in memory were correct; every script re-reading the CSV was not                          | <code>main.py</code> removes NUL characters before writing; the stored file was rewritten with the same rule before the measurement call of the adopted run, and QUANT and HEDGE recomputed from it reproduce <code>results.csv</code> |

advance to contain the pattern it was supposed to detect, or by recomputing a stored quantity from its inputs.

## A.10 Provenance

The last commit on the branch is [54a419d](#). It contains the vocabulary and its exported files, the navigation filter, the three SUS specifications with `config.SUS_MODE`, the `Breadth` column, the deduplication step and earlier versions of the two stability scripts. It is not, on its own, sufficient to regenerate the adopted `results.csv` or the paper’s numbers:

- rows 5–8 of Table 34 exist only as uncommitted changes to `main.py`, `src/config.py`, `src/text_processor.py`, `src/esgsi_analyzer.py` and `scripts/build_lexicons.py`, together with the regenerated lemmatised lexicons, the deletion of the former lemmatised hedge lexicon, and the new files `esg_protected_phrases.txt`, `lm_hedge.txt` and `empresas_supersector.csv`;
- `stability_v3.py` and `stability_v3_e2e.py` carry uncommitted changes, and the other eight scripts of Table 30 and the figures are not yet in the repository history.

A reader who checks out [54a419d](#) exactly gets the scoring vocabulary without the 11 protected phrases and HEDGE on lemmatised text, not the adopted run. Closing this gap requires a commit capturing the working tree; until it exists, the commit hash names a necessary but not sufficient condition for reproduction.

## References

- Cameron, A.C., Gelbach, J.B., Miller, D.L., 2008. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics* 90, 414–427.
- Campbell, D.T., Fiske, D.W., 1959. Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin* 56, 81–105. doi:10.1037/h0046016.
- Hill, M.O., 1973. Diversity and evenness: A unifying notation and its consequences. *Ecology* 54, 427–432. doi:10.2307/1934352.
- Lagasio, V., 2024. ESG-washing detection in corporate sustainability reports. *International Review of Financial Analysis* 96, 103742. doi:10.1016/j.irfa.2024.103742. open access, CC BY 4.0.
- Loughran, T., McDonald, B., 2011. When is a liability not a liability? textual analysis, dictionaries, and 10-ks. *The Journal of Finance* 66, 35–65. doi:10.1111/j.1540-6261.2010.01625.x.
- Manning, C.D., Raghavan, P., Schütze, H., 2008. *Introduction to Information Retrieval*. Cambridge University Press.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, É., 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* 12, 2825–2830.
- Spärck Jones, K., 1972. A statistical interpretation of term specificity and its application in retrieval. *Journal of Documentation* 28, 11–21. doi:10.1108/eb026526.